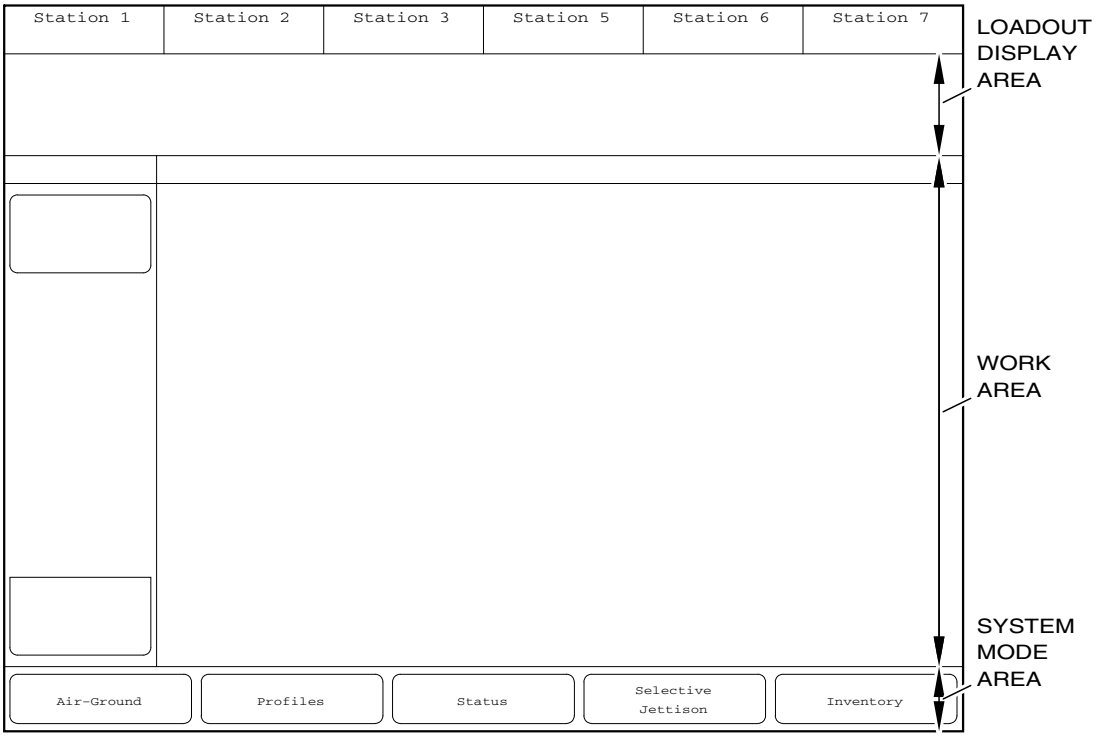


**Figure 1-2. GSMS Weapons Display.**

1.3.1.1 Loadout Display Area. Extending across the top of the Weapons Display (Figure 1-2) is the Loadout Display area. This display area provides visual indication of what weapon is loaded on each station and its current status with the use of Station Boxes and Store Information Boxes.

1.3.1.1.1 Station Box. Station Boxes provide store identification using icons to represent different weapons depending on what is currently configured. There are six Station Boxes located across the upper portion of the Loadout Display area. When the corresponding station is selected the Station Box becomes outlined by an animated dashed border. When the station store is released the icon will disappear from inside the Station Box.

Station Boxes are labeled as follows, from left to right:

- Station 1 (left outboard)
- Station 2 (left mid station)
- Station 3 (left inboard)
- Station 5 (right inboard)
- Station 6 (right mid station)
- Station 7 (right outboard)

Station 4 is reserved for future centerline indication and does not appear.

ER Stations 3 and 5 are capable of being configured with external fuel tanks; when configured, the respective station will appear grayed out and will not be selectable.

Stations 2 and/or 6 are capable of being fitted with the M310 or 2411.0.0 the M299 missile launcher. When a launcher is installed, each rail is represented as a substation (Figure 1-3). Substations are labelled clockwise starting at upper left when viewed from aft of the aircraft (2411.0.0 Figure 1-234). The number assigned also indicates the order by which the missile information will be displayed in the Store Information Box from the Station Box down. Block 5 Stations 3 and 5 are capable of being fitted with M310 launchers and AGM-114 missiles.

[2411.0.0 Block 5 ] Up to 12 AGM-114 missiles can be loaded onto the aircraft utilizing two M299 launchers on Stations 2 and 6, and two M310 launchers on Stations 3 and 5.

Station 2, 3, 5, and 6 are capable of being fitted with the BRU-78/A Dual Carriage bomb rack.

The below list provides the number listing for each substation:

- Left rail – Substation 1, first Store Information Box
- Right rail – Substation 2, second Store Information Box

1.3.1.1.2 Store Information Box. The Store Information Box provides a visual of the current store status and weapon type indicator (Figure 1-3). Each Station Information Box contains three fields for information display: Store Power, Store Status, and Store Type. When the corresponding store is selected the store information box becomes outlined by an animated dashed border. When the store has been deselected or released the outline will disappear.



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Store Power (Figure 1-3) indicates the current store power status and only applies to stores that require a powered state prior to release. The indicator will turn green when the store is powered up, and will remain dark if the store is not, or cannot be, powered up. The following list provides the possible indication for each weapon.

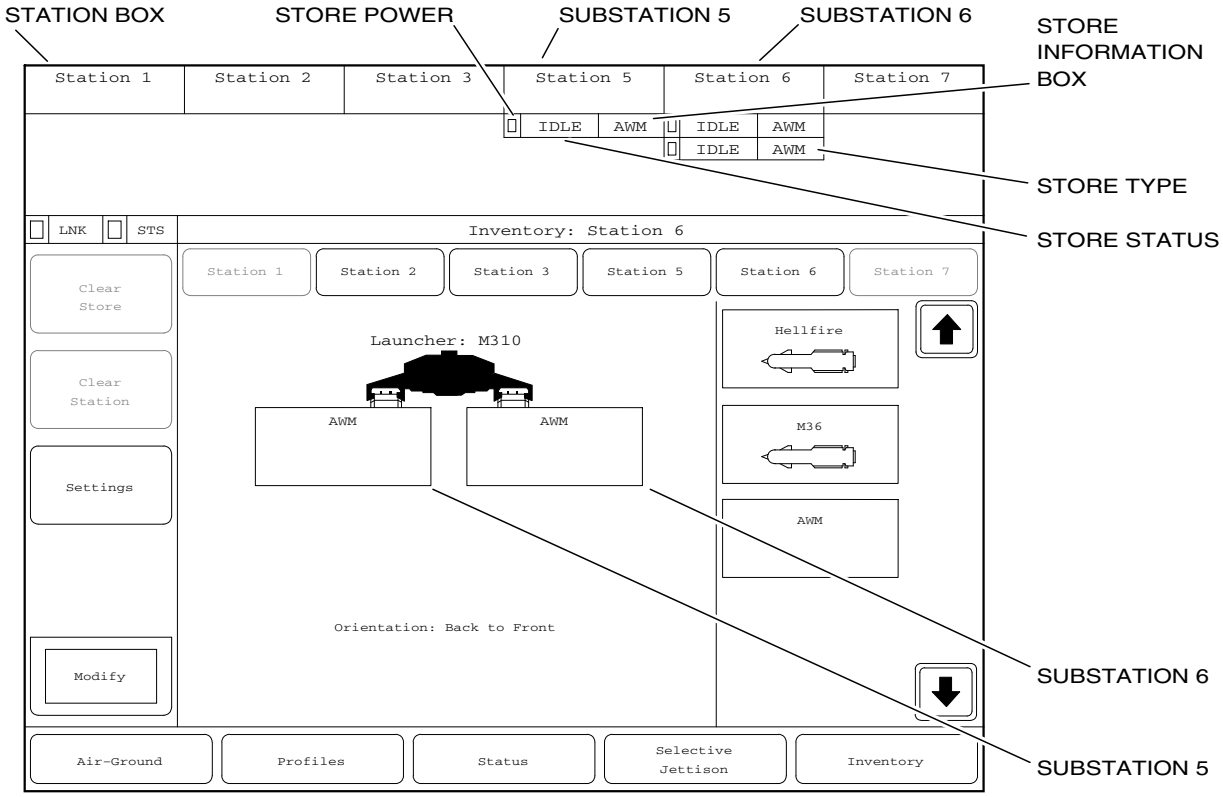


**Figure 1-3. Loadout Display Area.**

AGM-114

- Dark – missile is not powered up
- Green – missile is powered up

GBU-12/51/GBU-12T

- Dark – weapon cannot be powered up by aircraft

JDAM and GBU-49/48

- Dark – weapon is not powered up
- Green – weapon is powered up

LSDB/SDB

- Dark – missile is not powered up
- Green – missile is powered up

Store Status (Figure 1-3) provides an indication of the current status for the store and is displayed against a colored background. The possible status with the respective background color and the meaning are provided below.

Applicable to all weapons:

- UNVRFD (yellow background) – Store is unverified.
- INVLD (red background) – Verification process is complete and Store Type selected does not match Store Type recognized by SCC.
- REL (green background) – Master Arm is armed, launch enable button and trigger have been depressed, aircraft is indicating it is in the weapon release process.
- AWAY (green background) – Store released.
- WARN (yellow background) – A degraded condition exists that does not preclude a normal release of a store.
- ERROR (red background) – A fatal error occurred which precludes a normal release of a store.

Applicable to AGM-114 and AWM during missile system test:

- BIT (green background) – Missile is running Built-In Test (BIT).
- FAILED (red background) – Missile has failed to release.
- READY (green background) – Missile has passed BIT, power up sequence is complete, and PRF code has been accepted by launcher.
- IDLE (green background) – Missile is verified but not powered.



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Applicable to GBU-12/51/GBU-12T:

- READY (green background) – Weapon is ready for release.
- JET (green background) – Store is in the process of being jettisoned.
- JETTED (black background) – Store jettisoned.
- HUNG (red background) – Store hung.

Applicable to JDAM and GBU-49:

- IDLE (green background) – Weapon is verified but not powered.
- JET (green background) – Store is in the process of being jettisoned.
- JETTED (green background) – Store jettisoned.
- HUNG (red background) – Store hung.
- IBIT (green background) – Initiated Built-In Test (IBIT) in process.
- WARMUP (green background) – Weapon IBIT was successful, waiting for weapon to warm up and preparing to transfer align.
- Non-UAI DEGRD (yellow background) – Weapon is powered up and warmed up, GPS data updates and transfer alignment to weapon have started.
- DEGRD (green background) – Weapon is powered up and warmed up, GPS data updates and transfer alignment to weapon have started.
- MINREL (yellow background) – Weapon is not in IBIT, critical hardware has passed tests, minimum transfer alignment exists, minimum targeting exists with lat/lon/alt, warm up is complete and weapon is safe to release.
- AUR (green background) – Last weapon BIT passed, Transfer Alignment Quality (TXA) is 1 or 2, minimum GPS requirements have good crypto, keys, ephemeris and time, and weapon is safe to release.

Applicable to UAI Stores:

- IDLE (green background) – Weapon is verified but not powered.
- JET (green background) – Store is in the process of being jettisoned.
- JETTED (green background) – Store jettisoned.
- HUNG (red background) – Store hung.
- IBIT (green background) – Initiated Built-In Test (IBIT) in process.
- STPOVR (green background) – Store selected for release on a carriage was Stepped Over (STPOVR) during the release process.
- INIT (green background) – Weapon IBIT was successful, waiting for weapon to warm up and preparing to transfer align.
- DEGRD (yellow background) – Weapon is powered up and warmed up, GPS data updates and transfer alignment to weapon have started.
- MINREL (yellow background) – Weapon is not in IBIT, critical hardware has passed tests, minimum transfer alignment exists, minimum targeting exists (targeted with lat/lon/alt), warm up is complete and weapon is safe to release.
- SIMREL (yellow background) – Simulated Release (SIMREL) test control is active.
- AUR (green background) – Last weapon BIT passed, Transfer Alignment Quality (TXA) is 1 or 2, minimum GPS requirements has good crypto, keys, ephemeris and time, and weapon is safe to release.

Applicable to AWM-103A during 1760 test:

- RUNNING (green background) – Currently running tests.
- FAILED (red background) – Failed to run tests due to fatal errors.
- READY (green background) – Test set verified, master arm is enabled, and system is ready to run tests.
- IDLE (green background) – Test set verified, master arm is not enabled.

Store Type (Figure 1-3) provides an indication of what store is loaded and is displayed against a colored background. The color of the background provides a visual of the release status of the weapon on the station.

Store Type background colors and release status meaning:

- Green – ready for release
- Yellow – degraded release
- Red – release failure
- Black – not ready for release
- Black with no text – released

Inventory configuration Store Types with definitions:

- GBU12 – GBU-12/51
- GB12T – GBU-12 Trainer
- GBU12 Legacy – GBU-12/51
- GBU12T Legacy – GBU-12/51 Trainer
- 49V2 – GBU-49 Lot 3A with AB sensor
- 49V3 – GBU-49 Lot 5 ECCG with AB sensor
- HF-P – Hellfire IIP
- JDAM? – Unverified GBU-38 or GBU-54
- HF-? – Unverified AGM-114
- M? – Unverified M36
- 49?– Unverified GBU-49
- UAI? – Unverified GBU-12, GBU-49/48, JDAM, or LSDB/SDB
- GBU38 – GBU-38
- GB38T – GBU-38 Trainer
- GBU54 – GBU-54
- GB49T – GBU-49 Trainer
- GBU48 – GBU-48
- Non-UAI GBU49 Legacy – GBU-49
- Non-UAI GBU49T Legacy – GBU-49 Trainer
- Non-UAI GB54T – GBU-54 trainer
- Non-UAI JDAM Legacy – GBU-38 or GBU-54
- Non-UAI JDAMT Legacy – GBU-38 or GBU-54 Trainer
- M36P – AGM-114 M36P Training missile
- M36R – AGM-114 M36R Training missile
- AWM – AN/AWM-101A or AN/AWM-103/A test set



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- UAI – Universal Armament Interface
- UAIT – Universal Armament Interface Trainer
- LSDB – Laser Small Diameter Bomb

For AGM-114 missiles, following inventory verification the Store Type is replaced automatically with a missile variant Store Type. The new Store Type identifies the missile type, warhead type, guidance and missile functions.

HF-P+

- M-4 – AGM-114M-4
- N-4 – AGM-114N-4
- P-4 – AGM-114P-4
- P-4A – AGM-114P-4A
- P-4B – AGM-114P-4B
- P+IN – AGM-114P+, Inert warhead

HF-R

- R – AGM-114R
- R-2 – AGM-114R-2
- R-9E – AGM-114R-9E
- R-9F – AGM-114R-9F
- R-9X – AGM-114R-9X
- Q-4 – ATM-114Q-4, Inert warhead
- Q-6 – AGM-114Q-6
- R-IN – AGM-114-IN, Inert warhead
- 2410.0.0 R9S1 – AGM-114R-9S1
- 2411.0.0 R9H – AGM-114R-9H
- 2411.0.0 R-9S – AGM-114R-9S

1.3.1.2 Work Area. The Work Area is the main area in which inputs are made and is located between the System Mode and Loadout Display Areas. The LNK indicator, STS indicator, Title Bar, and Control and Page Sections are located within the Work Area (Figure 1-4). The LNK indicator is located in the upper left corner of the Work Area above the Control Section. The indicator provides a visual reference of the SMS link state between the GSMS and the RPA. There are four status indications that can be presented that are listed below:

- Red – no link present
- Blinking yellow – GSMS and RPA synchronizing
- Green – link present
- Blinking green – the link is processing SMS commands sent to the RPA

During power up, the GSMS will not be responsive until a link is present and the LNK indicator turns solid green. If the LNK indicator is not solid green, the GSMS and RPA are not synchronized. A blinking green LNK indicator indicates that GSMS command packets have not fully synched between the GSMS and RPA. A degraded LNK indicator state not resulting in lost link operations can delay commands between the GSMS and RPA. This delay could result in release of bombs after release cancellation when in CCRP mode. Because of this, the LNK indicator should be solid green before commanding bomb release.

The STS indicator is located to the right of the LNK indicator in the upper left corner of the Work Area above the Control Section (Figure 1-4). The STS indicator provides the state of the SMS health. There are five indications that can be presented:

- Red – warnings are present
- Blinking red – new warnings are present
- Yellow – cautions are present
- Blinking yellow – new cautions are present
- Green – no warnings or cautions are present

When a new caution or warning is displayed on the Warnings screen, the STS indicator blinks in the color of the most severe indication present until the user navigates to the Warnings screen. After the user navigates to the Warnings screen, the STS indicator becomes the solid color of the most severe indication present.

The Control Section is located to the left side of the Work section just below the SMS Link State Indicator and contains Control Buttons (Figure 1-4). The initial set of Control Buttons are determined by what System Mode is currently selected and can further be changed depending on what Control Button is selected. Certain System Mode and Control Button selections will provide a guarded button, represented with right angled top corners and rounded bottom corners. A guarded button must be pushed twice within a limited time (less than 5 seconds) for that selection to take effect.

The Title Bar is located at the top of the Work Area just above the Page Section (Figure 1-4). The Title Bar provides an indication of the System Mode and the current activity initiated by the Control Buttons or Page Section buttons for that mode, such as “Inventory: Station 2.”

The Page Section is located to the right of the Control Section, just below the Title Bar, and provides the main source of information display and manipulation (Figure 1-4). It displays various formats, depending upon the System Mode and Control Button options selected.



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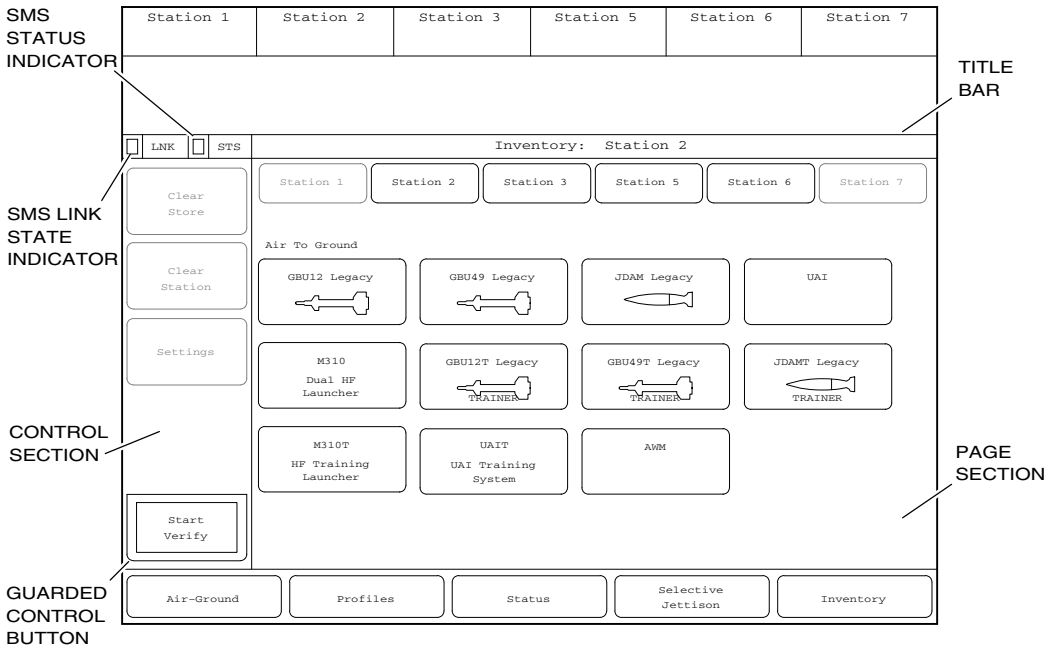


**Figure 1-4. Work Area.**

1.3.1.3 System Mode Area. The System Mode area extends across the bottom of the Weapons Display (Figure 1-2). This area is always present and contains five mode selection buttons labeled as follows, from left-to-right:

- Air-Ground
- Profiles
- Status
- Selective Jettison
- Inventory

Any of these buttons may be selected at any time. If a mode is exited and later reentered, the previously selected state will generally prevail. Exceptions:

- Entering Selective Jettison mode cancels an Air-Ground selection for release.
- Entering Air-Ground mode cancels selection for Selective Jettison.
- Entering Inventory mode and selecting Modify cancels selection for either release or selective jettison.

1.3.2 Inventory Mode.

**NOTE**

A GBU-51 must be inventoried as a GBU-12. The GSMS will reflect a configured GBU-12, but this does not impede the bomb's capabilities.

The Inventory Mode allows the pilot to provide a software mapping of munitions to stations and verify the validity of this mapping. This activity is usually performed prior to flight, but can be done in flight. When selected, the Control Section will be populated with up to four Control Buttons labeled from top-to-bottom:

- Clear Store – clears the current store selection without returning the station to the unloaded state when applicable.
- Clear Station – returns the selected station to the unloaded state.
- Settings – allows access to the inventory settings screen applicable to the specific weapon. For example, GBU-38/54 will have the Settings button available to access the sensor settings options.
- Start Verify (guarded) – commands the aircraft and GCS to compare physical configuration with software configuration and verify the two match. The Start Verify button will be replaced with the Modify button after a inventory verification has been completed.
- Modify (guarded) – allows the modification of the station configuration after stores verification.

The top of the Page Section contains a set of six Station Selection Buttons numbered 1–3 for the left wing stations, and 5–7 for the right wing stations. The Store Type options available are located below the Station Selection Buttons, and are determined by what stores the software is capable of utilizing (Figure 1-5 through 2411.0.0 Figure 1-8). The available Store Types are as follows:

- UAI GB12 – GBU-12/51
- UAI GB12T – GBU-12 Trainer



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- Non-UAI GBU12 Legacy – GBU-12/51
- Non-UAI GBU12T Legacy – GBU-12/51 Trainer
- UAI GB49T – GBU-49 Trainer
- UAI GB49 – GBU-49
- Non-UAI GBU49 Legacy – GBU-49
- Non-UAI GBU49T Legacy – GBU-49 Trainer
- AWM – AN/AWM-103/A test set
- UAI JDAM – GBU-38 or GBU-54
- UAI JDAMT – GBU-38 or GBU-54 Trainer
- Non-UAI JDAM Legacy – GBU-38 or GBU-54
- Non-UAI JDAMT Legacy – GBU-38 or GBU-54 Trainer
- M310 – M310 missile launcher
- M310T – M310 Training launcher
- 2411.0.0 M299 – M299 missile launcher
- 2411.0.0 M299T – M299 training launcher
- UAI UAI – Universal Armament Interface
- UAI UAIT – Universal Armament Interface Trainer
- 2411.0.0 AGM – any Air to Ground Missile type

Scrolling arrows will be present when more stores are available than will fit on the screen.

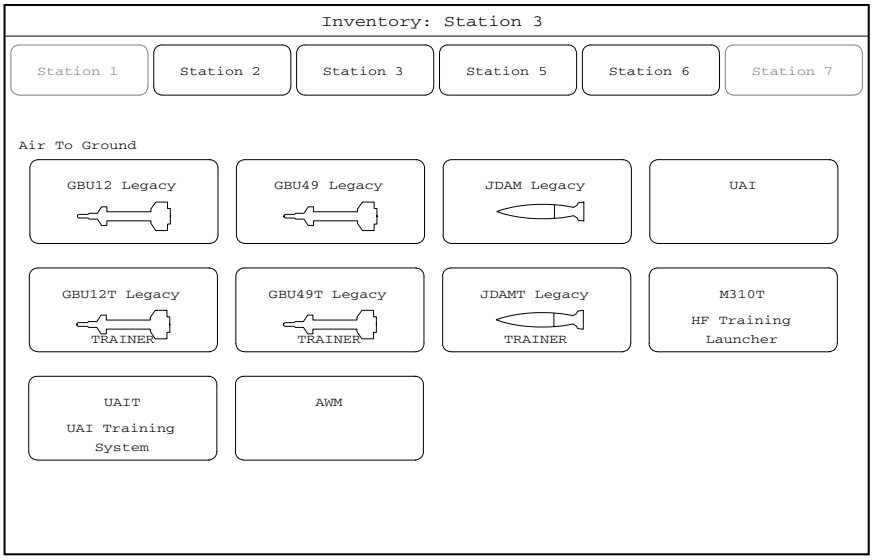


**Figure 1-5. Block 1 Inventory Selection Screen with SSIU.**



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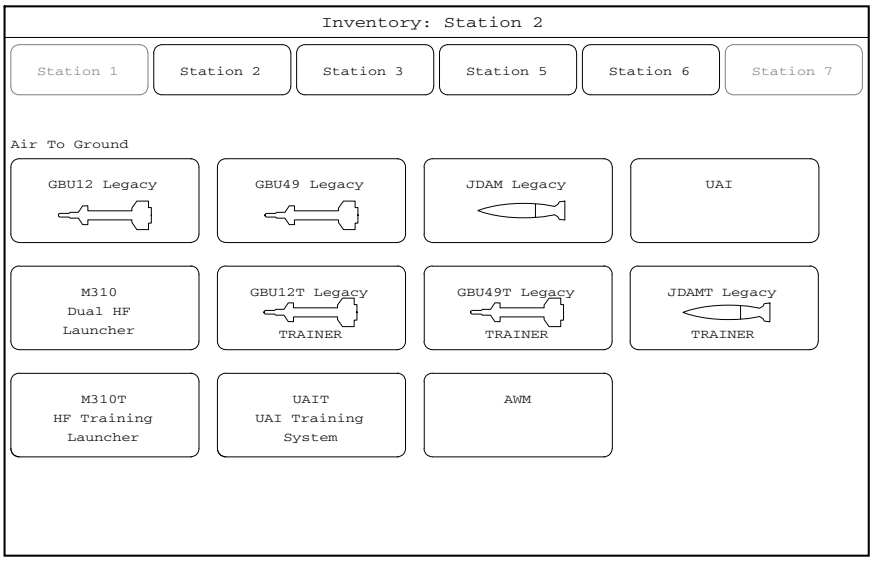


**Figure 1-6. Block 5 Inventory Selection Screen with SSIU.**

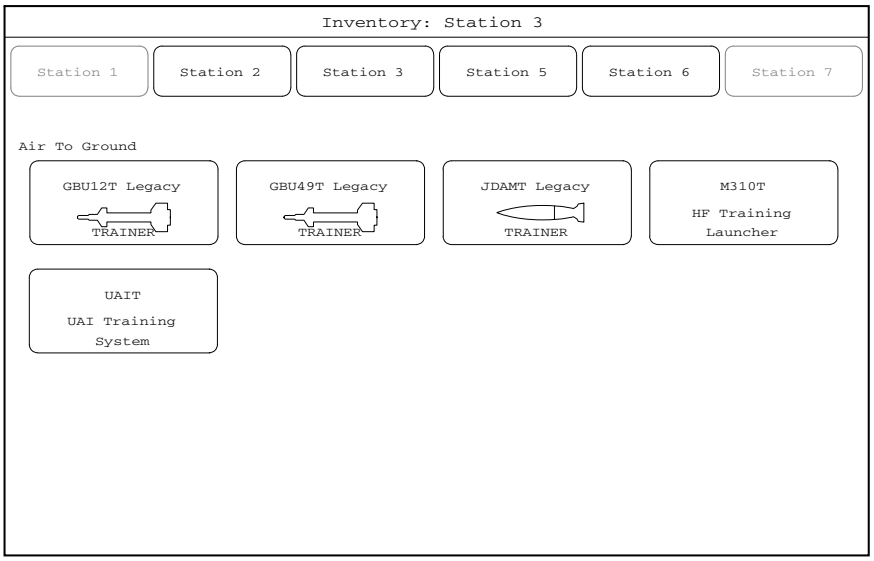


**Figure 1-7. UAIT Inventory Option.**



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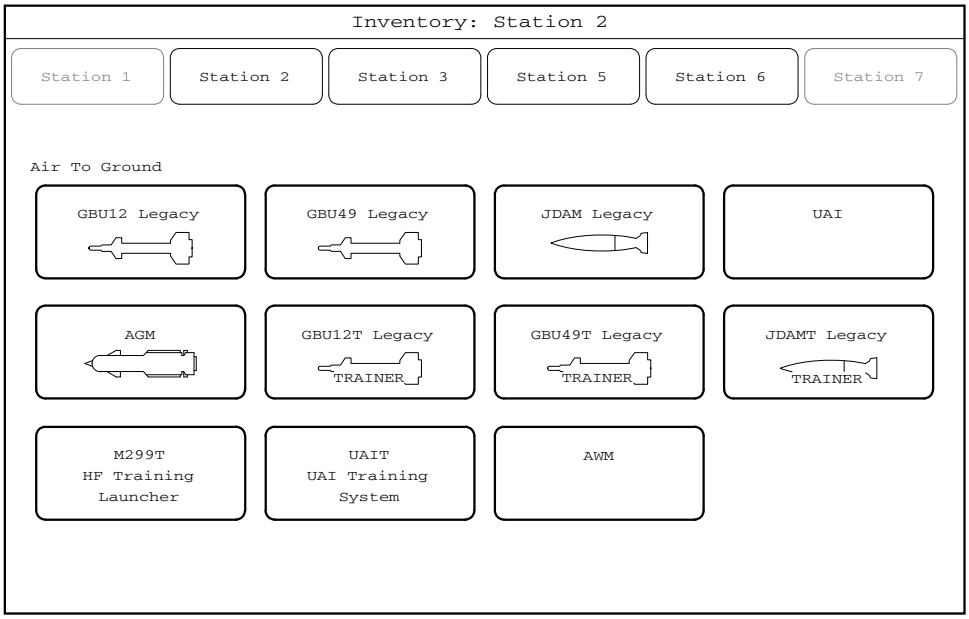


**Figure 1-8.** 2411.0.0 **Inventory Selection Screen.**

1.3.2.1 Inventory Configuration.

**CAUTION**

When inventorying a loadout, wait at least 5 seconds between selections of “Modify” and “Start Verify” to prevent ACSM Faults for UAI stores and carriages.

When performing an inventory, select only one button at a time and allow the weapon selection process to finish prior to selecting the next weapon (Figure 1-5 through 2411.0.0 Figure 1-8). Upon station selection, the corresponding Station Box in the Loadout Display will be outlined by animated blue dashes and will be highlighted along its horizontal axis by a lighter shading of the black background. Once the station is chosen, the user can specify the store for that station by touching one of the Store Type Buttons in the Page Section.

The inventory page includes a UAI weapon type inventory selection option for Stations 2, 3, 5, and 6. On Block 5 aircraft, Stations 2, 3, 5, and 6 all have the same selection options. On Block 1 aircraft, the M310 Dual HF Launcher is not available for Stations 3 and 5. The Legacy weapon types have been appended with the word “Legacy” so the user can differentiate them from the UAI version of the same store. To verify UAI weapon types, the user assigns the UAI inventory selection item to the station(s) that the UAI stores or carriage system is/are connected to. Once the user initiates the verification process, the system will auto-detect all UAI stores and/or carriage systems connected to the UAI assigned stations. There are no load out pictures for UAI Stores (Figure 1-9 and Figure 1-10). The inventory page also includes a UAIT weapon type inventory selection option for Stations 2, 3, 5, and 6 (Figure 1-7). When the UAIT inventory option is selected, the UAIT carriage store selection screen will display (Figure 1-17). When a weapon type that has multiple configuration options is selected, a configuration page will display (Figure 1-11).

2411.0.0 The inventory page allows for an auto-verification process of any Air to Ground Missile type. The AGM icon will be used to verify any Hellfire missile, M36 training missile, or AWM-103 rail tester. The M299T HF Training launcher icon will be used to verify up to four Hellfire training missiles on Stations 2 and 6 (2411.0.0 Figure 1-8). Prior to verification, the load out area will show "AGM ?" for all AGM icons that get assigned to stations (2411.0.0 Figure 1-12). After verification completes the load out area will display the detected missiles verified at each station. The launcher type will also be auto detected (2411.0.0 Figure 1-13).



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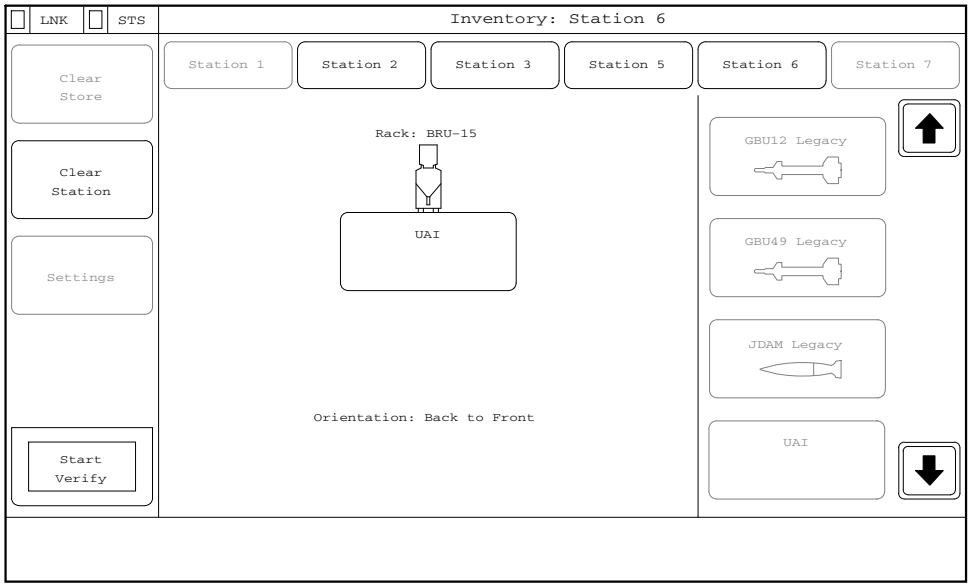


**Figure 1-9. UAI Selected (Prior to Starting Verification).**

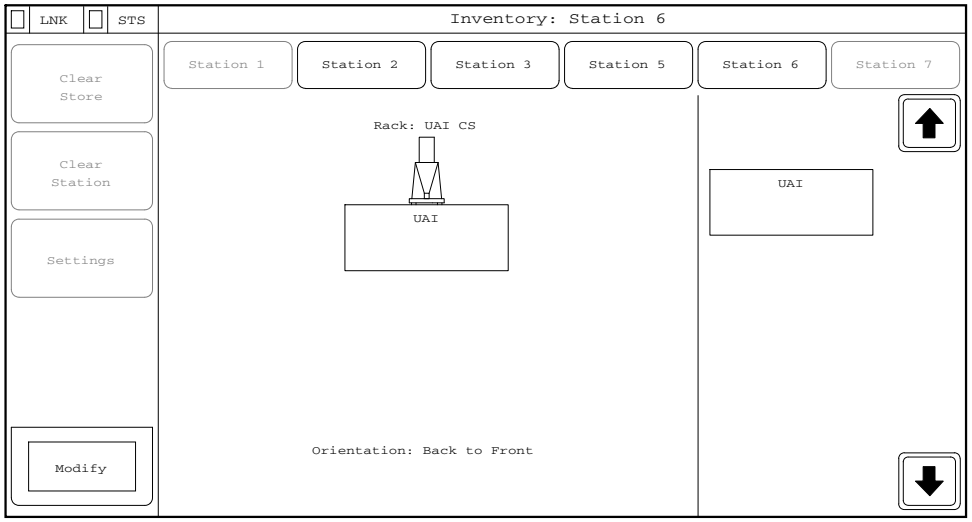


**Figure 1-10. UAI Verified Stores.**



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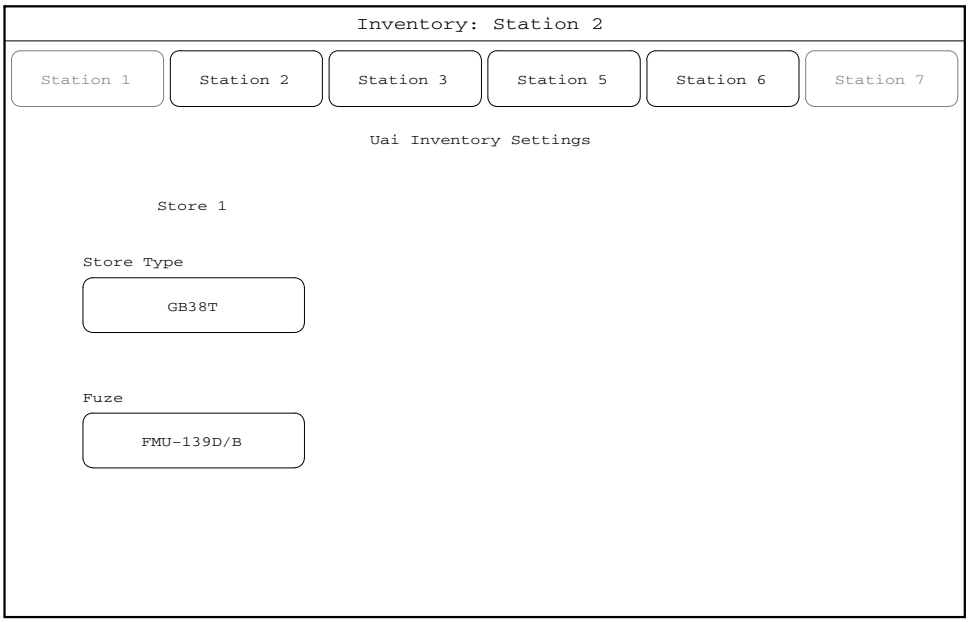


**Figure 1-11. UAIT Inventory Settings.**

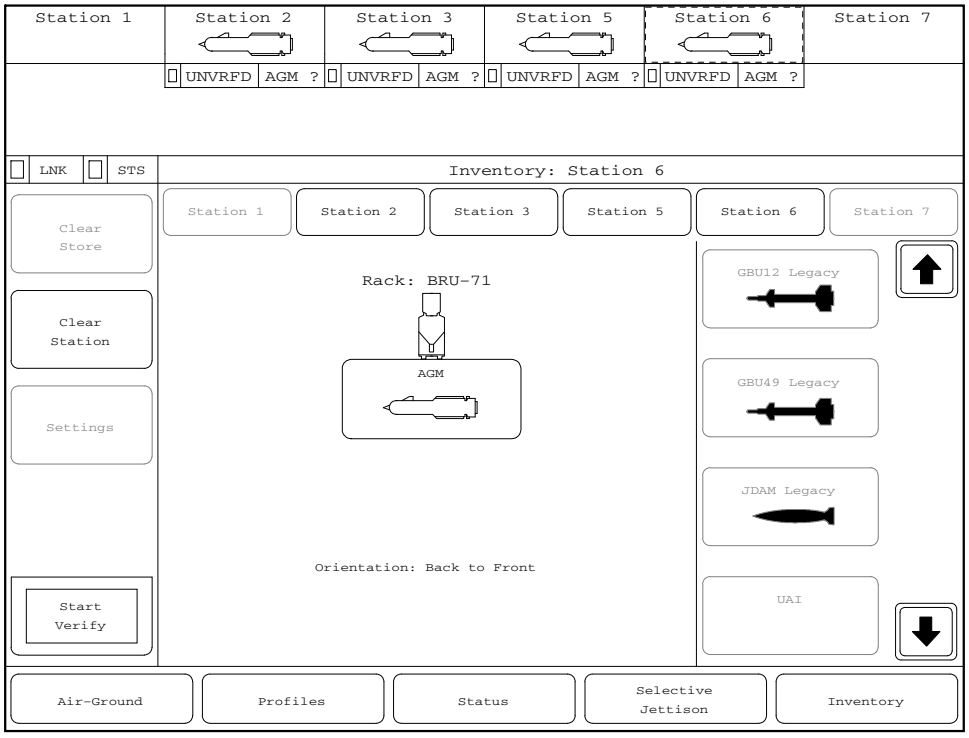


**Figure 1-12. 2411.0.0 AGM Unverified Stations.**



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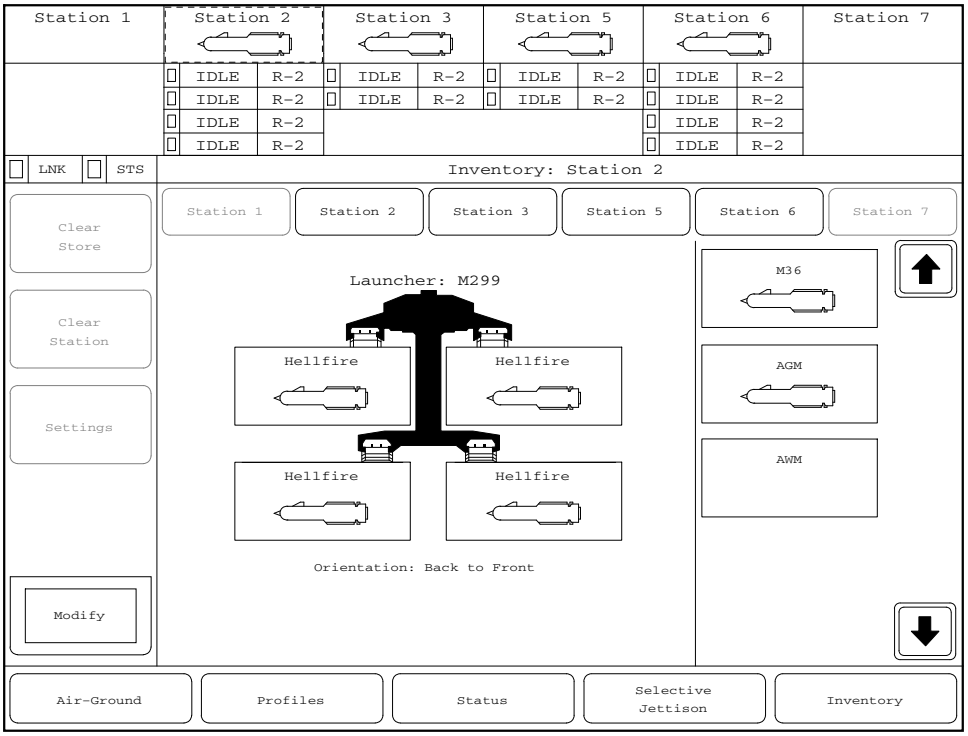


**Figure 1-13.** 2411.0.0 **AGM Verified On All Stations.**

When the Store Type button is selected, a menu will display with the available store configurations for that weapon type (Figure 1-14).

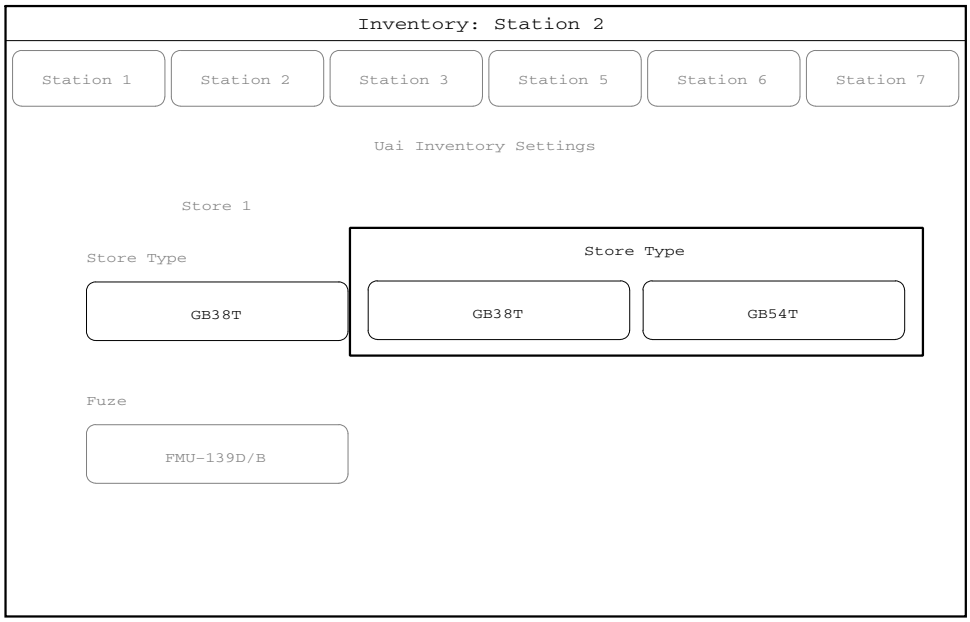


**Figure 1-14. Store Type Menu.**



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When the Fuze button is selected, a menu will display with the available fuze configurations for that weapon type. When the selection for the Store Code ID menu changes, the options for the Fuze will automatically be updated to the applicable fuzes for that store and defaulted to the first fuze option (Figure 1-15).

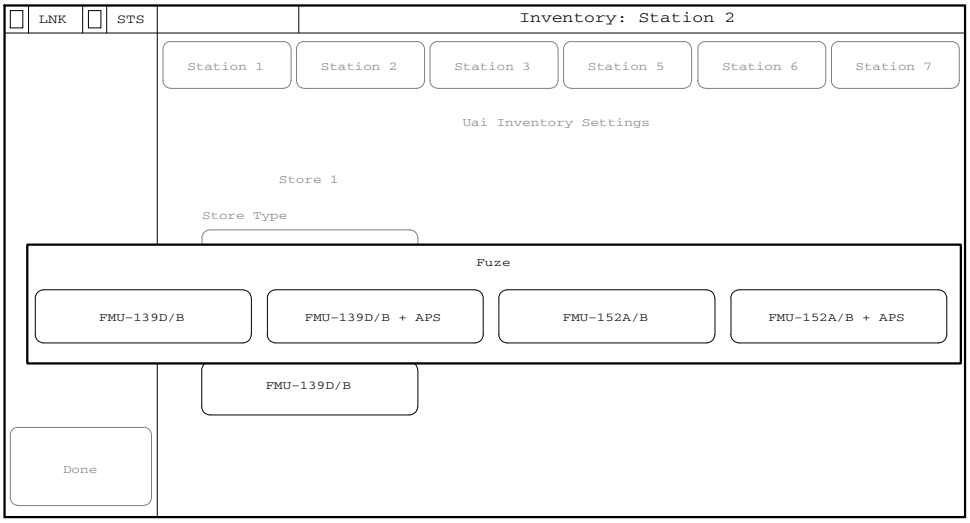


**Figure 1-15. Fuze Menu.**

If there is only one option for the Store Type or Fuze, the menu will not be selectable. Also note that a fuze configuration name has not been described for the Fuze and as such, it will display the fuze number identifier (Figure 1-16).

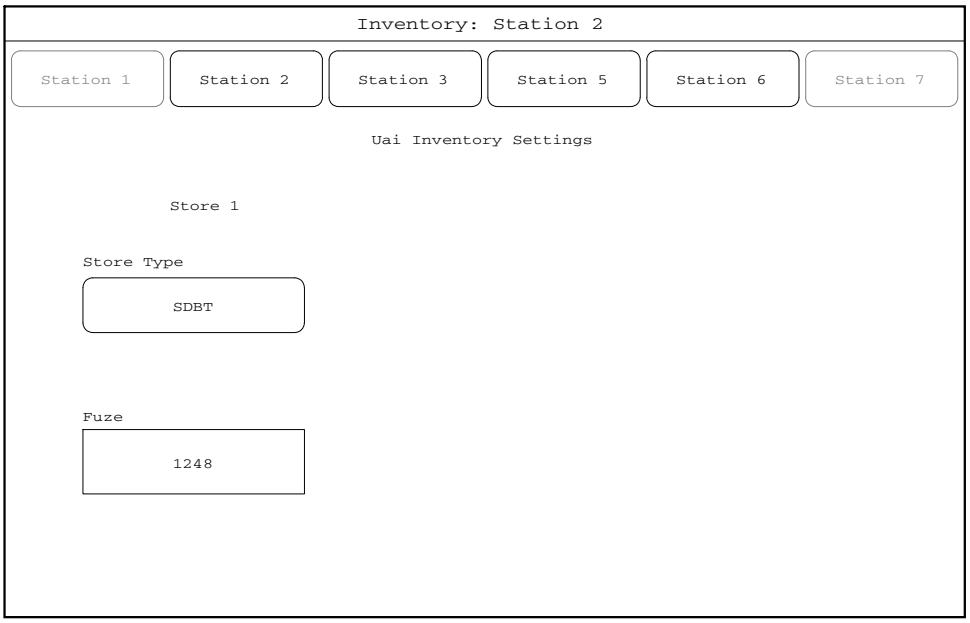


**Figure 1-16. Single Fuze ID Configuration.**

The UAIT carriage store selection page will display the weapon type of each store selected. After the loadout has been verified, the Carriage Store Select will look as follows with the option to view the configured stores with the Settings button (Figure 1-17 and Figure 1-18).



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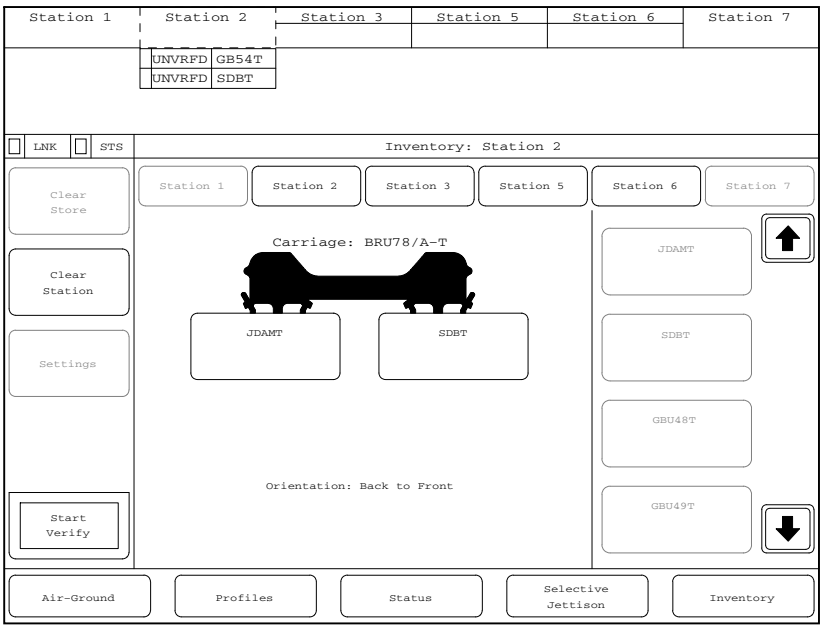


**Figure 1-17. UAIT Carriage Store Select.**

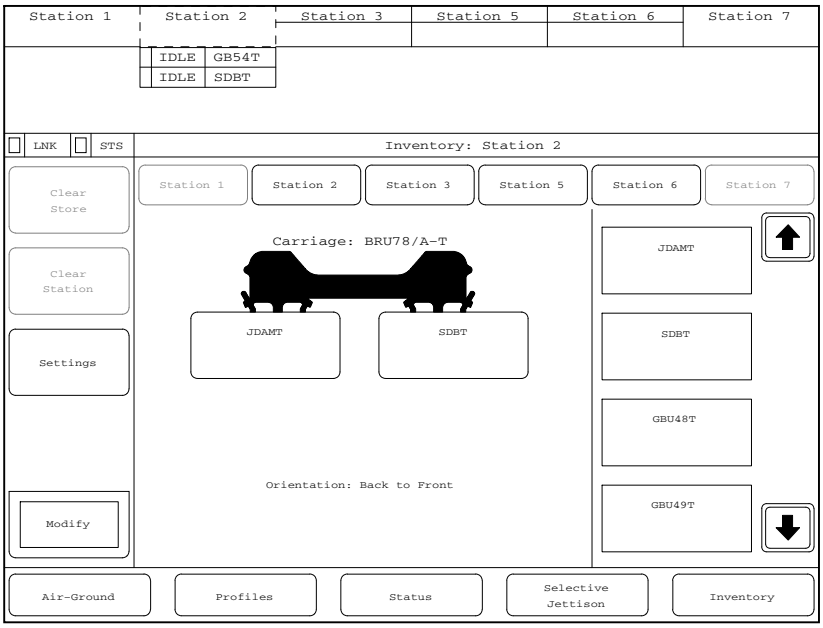


**Figure 1-18. Verified Carriage Store Select.**



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**NOTE**

- If test equipment and tactical munitions are included in the same load configuration, all will be powered off.
- During system power up, software will detect what bomb rack is present on each station.
- Cross-cued target auto mode should be disabled prior to verifying weapons and not be enabled until verification has been completed. Not disabling auto mode may result in an "SCC Software Fault Detected" caution on the Warnings screen.

1.3.2.1.1 AGM-114 Stores.

**CAUTION**

Block 5 If engine is running, the minimum engine speed RPM for weapon store(s) operations shall be greater than 96%. Do not attempt to power on, verify powered bomb-racks and/or missile launchers, or verify store(s) when aircraft is operating or transiting below 96%. If conducting store(s), powered bomb-rack and/or missile launcher operations and required to transition below 96% engine RPM, power off all store(s) and /or powered bomb-racks/missile launchers until 96% or greater has been achieved. Failure to do so may result in a loss of AC power at any stations with an inventoried store and an ACSM fault.

When the M310 button is selected from the Store Type options, an M310 icon with two selection buttons on the upper rails will appear in the center of the Page Section (2410.0.0 Figure 1-19).

At the right side of the Page Section, a set of AGM-114 icon buttons used to select the specific missile loaded on each substation will be displayed. 2411.0.0 When the M299 button is selected from the Store Type options, an M299 icon with two selection buttons on the upper rails and two selection buttons on the lower rails will appear in the center of the Page Section ( 2411.0.0 Figure 1-20). At the right side of the Page Section, a set of AGM-114 icon buttons used to select the specific missile loaded on each substation will be displayed. Those options and the respective representation are as follows:

- Hellfire – all variants
- AWM – AN/AWM-101A or AN/AWM-103/A test set
- M36 – M36E6 or M36E11 training missile

Block 1 Hellfire missiles can only be configured on Stations 2 and 6.

Block 5 Stations 2 and 6 can support up to four missiles per station if using M299 launchers. Stations 3 and 5 can support two missiles per station. The Hellfire type, fuze, PRF code and ripple order are displayed.

To allocate an AGM-114 store to a rail, touch the button representing the desired rail to be configured and then select the desired AGM-114 icon button. The Station Box will display a AGM-114 icon, and the required number of Store Information boxes will appear below the Station Box. Each Store Information Box corresponding to a configured rail will show a dark Store Power indicator, Store Status will indicate UNVRFD, and Store Type will indicate the selected missile. Mixed loads are authorized for use and are considered a combination of different tactical AGM-114 variants or training missiles and missile test equipment. However, tactical missiles cannot be configured with training and test equipment installed on the aircraft.

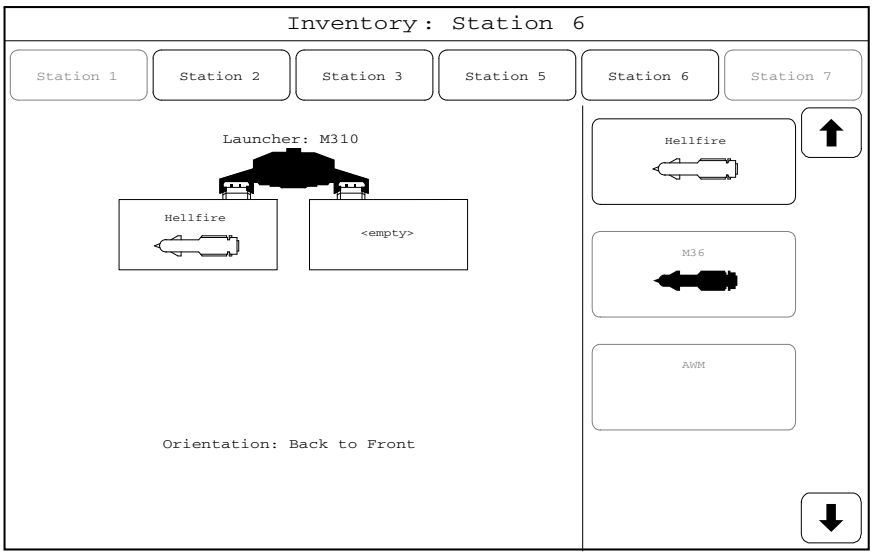


**Figure 1-19. 2410.0.0 AGM-114 Inventory Configuration.**



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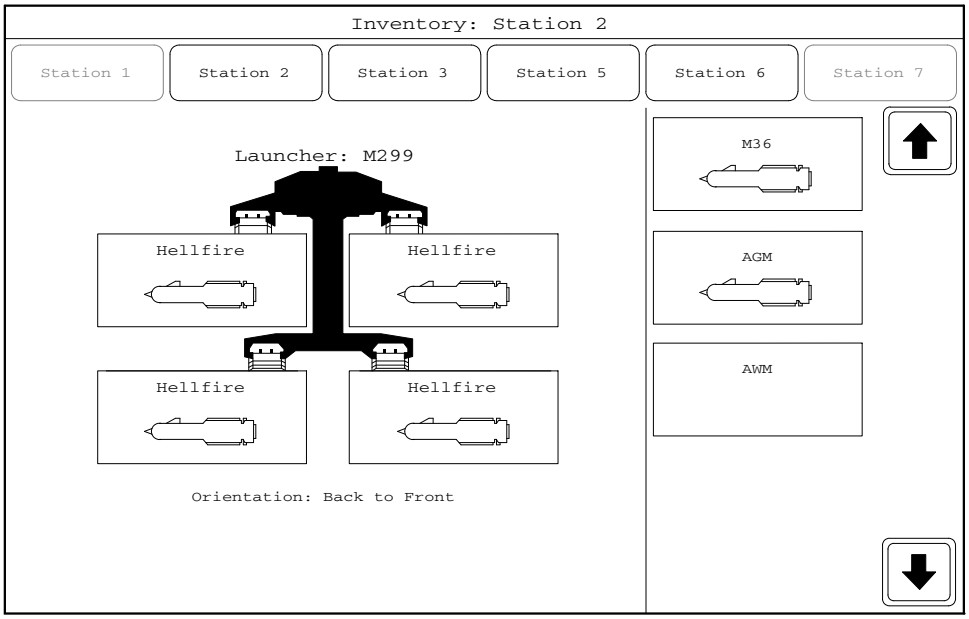


**Figure 1-20.** 2411.0.0 **AGM-114 Inventory Configuration.**

Block 1 During inventory verification when an M310 launcher is installed, the verification may fail due to a Payload Power Distribution Module (PPDM) fault. If store verification fails on stores loaded on the M310 launcher, the following indications will be present:

- INVLD will be displayed on a red background in the stores status box.
- The Inverter Power Control Fail error will be True on page 2 of the Hellfire stores status (Status/Detailed Status/Stores/Station N Errors).
- VIT 53 PPDM Status will show a value close to zero amps for HF-Inverter Power PPDM1 SW6 and PPDM1 SW8 rather than four amps.

Block 1 If inventory verification fails because of the above described PPDM fault, perform the following actions within five minutes of fault:

1. Display the Inventory screen.
2. Select Modify.
3. Restart inventory verification ensuring the Verify In-Process graphics are displayed.

**NOTE**

Block 1 If more than five minutes has elapsed since the PPDM fault has occurred, inventory verification may have to be done twice.

1.3.2.1.2 GBU Stores.

**CAUTION**

[ Block 5 UAI ] If engine is running, the minimum engine speed RPM for weapon store(s) operations shall be greater than 96%. Do not attempt to power on, verify powered bomb-racks and/or missile launchers, or verify store(s) when aircraft is operating or transiting below 96%. If conducting store(s), powered bomb-rack and/or missile launcher operations and required to transition below 96% engine RPM, power off all store(s) and/or powered bomb-racks/missile launchers until 96% or greater has been achieved. Failure to do so may result in a loss of AC power at any stations with an inventoried store and an ACSM fault.

To load a GBU store, select the desired Station Selection Button and then select the desired GBU store icon. Once selection has occurred, the Page Section will display a BRU-15/A or BRU-71 /A with the selected GBU Store Type. An icon for the selected GBU will also appear in the selected Station Box in the Loadout Display. A Store Information box will also appear below the Station box containing a dark Store Power indicator, Store Status will indicate UNVRFD, and Store Type will indicate the selected GBU.

The status layout for GBU weapons displays the status, fuze, PRF code and name. A station can support two GBUs if a carriage system is present. The PRF code will be blank if the GBU type does not support PRF code.



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1.3.2.2 Inventory Verification. After all munitions have been configured, the system will require an inventory comparison between the physical RPA configuration and the input GSMS configuration.

During the GBU-38, GBU-49, and GBU-54 verification process, the specific GBU-38, GBU-49, and GBU-54 store type is determined. The specific GBU-38, GBU-49, and GBU-54 store type verification determines weapons capabilities.

During the UAI stores and LSDB/SDB verification process, the specific UAI stores and LSDB/SDB store type is determined and store type verification determines weapons capabilities.

To verify stores configuration, the user will be required to select the Start Verify button to begin the process (Figure 1-21). Once verification has started "Verify In-Process" will be displayed in the Page Section (Figure 1-22).

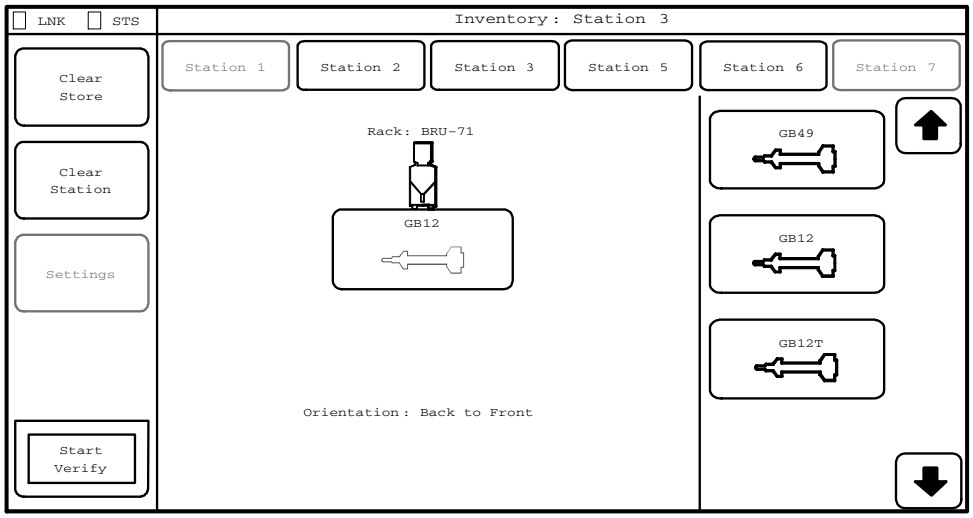


**Figure 1-21. Unverified Inventory Configuration – Typical.**

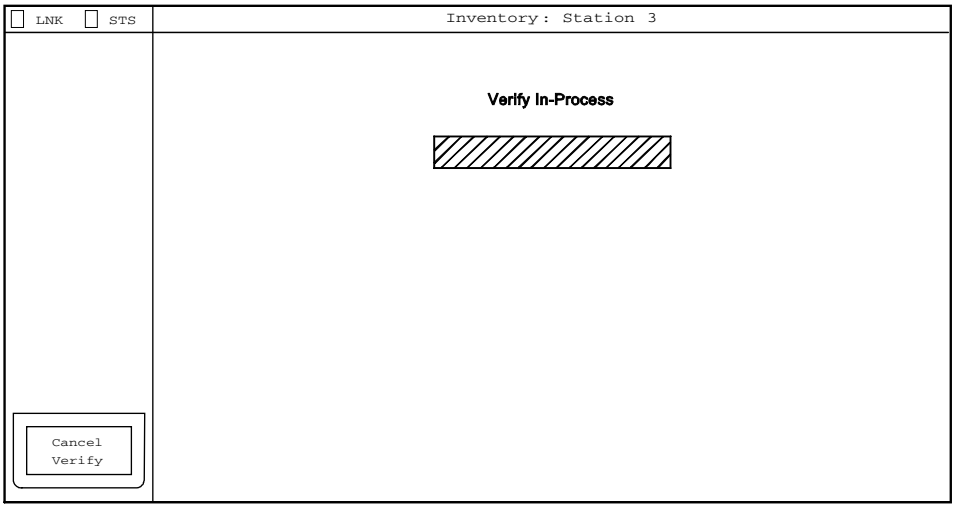


**Figure 1-22. Inventory Verification in Process.**

When inventory configuration verification concludes (Figure 1-23), all UNVRFD Store Status indications will change to the current state of the store. The Start Verify control button will become the Modify control button and the Clear Store and Clear Station control buttons will be disabled. Any stores that fail verification will display INVLD and will not be employable. To make changes to the inventory configuration, the Modify control button must be selected. The Clear Store and Clear Station control buttons will again be enabled.For Hellfire P+ (R guidance) and R missiles, following the completion of verification, HF-P store types will be changed automatically to display the correct missile type.



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**NOTE**

- If AGM-114 missiles are loaded, verification could take several minutes.
- If the SMS is powered off, inventory configuration must be verified when the SMS is powered on again.

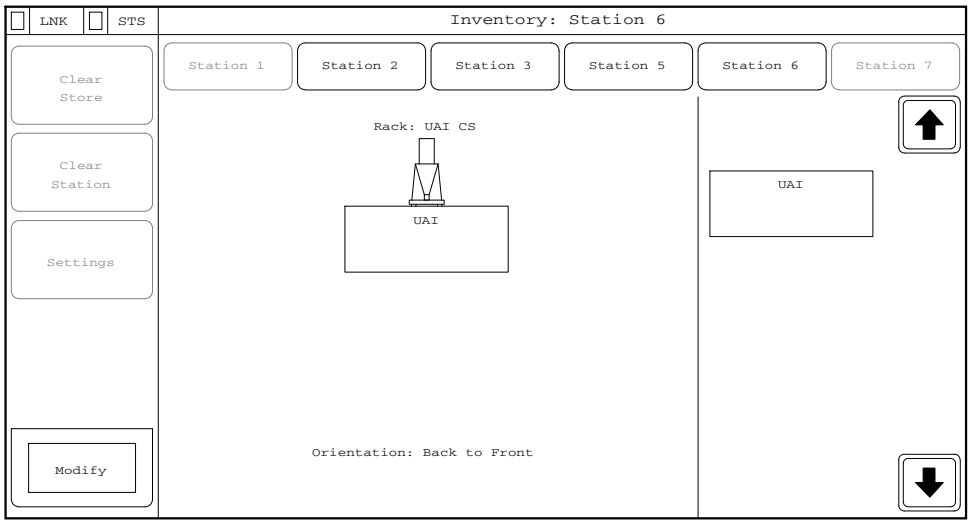


**Figure 1-23. Verified Inventory Configuration.**

1.3.3 Air to Ground Mode. The Air to Ground Mode allows the configuration of an individual weapon to provide a desired outcome. The Air to Ground mode is entered when the Air-Ground button in the System Mode Area of the GSMS is selected. In Air to Ground mode, the GSMS touchscreen Control Section contains five Control Buttons:

- Select Store
- Store Settings
- Select Target
- Release Settings
- Launch Status

1.3.3.1 Select Store.

**CAUTION**

Do not power on Hellfire missiles unnecessarily. Leaving a missile powered on for extended durations may decrease reliability.

The Air-Ground mode Select Store screen (Figure 1-24) appears when the Select Store control button is selected in Air to Ground mode. The Page Section of the Select Store screen contains as many Store Type rows as there are types of stores inventoried. Each row represents all stores of the same type and is identified with the Store Type icon and label. Depending on which stores have been loaded, the following may be displayed:

- A row for GBU-12 selection
- A row for GBU-12 trainer selection
- A row for GBU-49 selection
- A row for GBU-49 trainer selection
- A row for Hellfire selection
- A row for Hellfire trainer selection
- A row for JDAM selection
- A row for JDAM trainer selection
- A row for M36 selection
- A row for LSDB/SDB selection
- A row for GBU-12 UAI selection
- A row for GBU-48 UAI selection
- A row for GBU-49 UAI selection
- A row for JDAM UAI selection

Each Store Type row contains selector buttons labeled from left-to-right:

- Select
- Single or Ripple fire
- Power Off (guarded, disabled for GBU-12/51 and GBU-12T)



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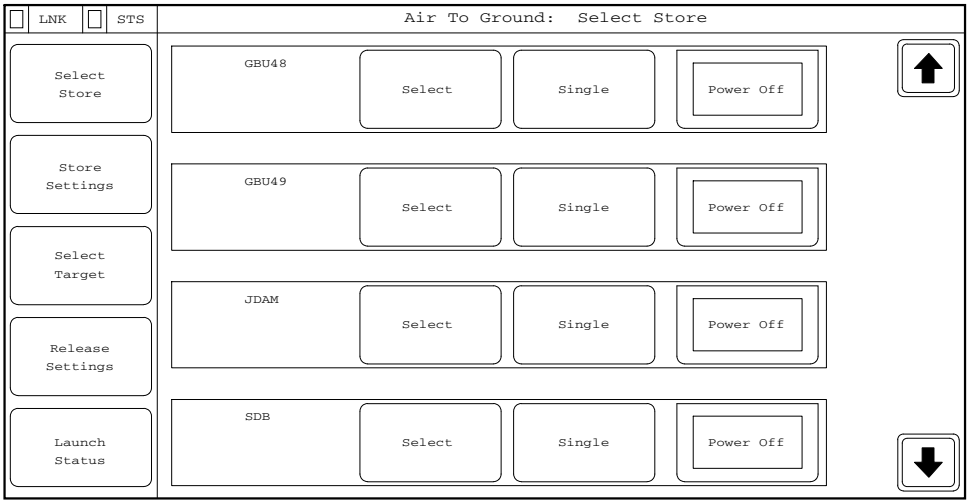


**Figure 1-24. Select Store.**

Scrolling arrows will be present when more stores are available than what will fit on the screen.

When the Select Store screen is entered no stores are selected for use and the Select button must be used to select the desired store. When a Store Type row Select button is selected, a store of that type is selected for release and the frame of the corresponding Store Information Box border becomes animated with yellow and green dashes. The specific weapon selection process is described further in the Release Order paragraph. The Select button is also used to deselect an already selected store when the store is not desired for use.

- If the Store Type selected is a powered store, all stores of that type are powered on.
- Current powered stores are the AGM-114, GBU-38, GBU-49, GBU-54, and UAI stores.
- When a store is powered on, it will run a BIT. After BIT, missiles will present a status and bombs will warm up prior to presenting a status.
- Selecting a Store Type while another is selected will deselect the previous Store Type and select the desired one, but will not power down the previous Store Type.
- Power Off must be used to power down weapon(s) that have been powered up and/or previously selected.
- A store determined by the SMS system to be inoperative cannot be selected.

During the power-up of a GBU-49 or JDAM, the fuze configuration is determined. If an FMU-152 fuze is not configured, the JPF Not Present caution is shown on the Warnings page. GBU-49 Store Settings will not have the Airburst, AB Height, Arm Delay, or Function Delay settings available and the respective buttons will be grayed out on Fuze Settings display screen. JDAM Store Settings will have the Function Delay, Airburst, and Arm Delay settings available and the respective buttons.

1.3.3.1.1 Release Configuration. Once a Store Type has been selected one store of that type is the default number of weapons to be used and is verified with the Single button located next to the Select button. Users have the ability to release more than one of that Store Type during the same delivery event. To create a multi-weapon or ripple release, the Single button must be selected. Once selected the Release Configuration page will be displayed in the Page Section in order to input values for a ripple release. The Release Configuration page provides all inputs necessary to configure a multiple weapon release event. The first column of options are the Ripple Settings which include the number of weapons and interval options. The second column provides the ability to set the release order of the ripple train. The Release Configuration Page (2410.0.0 Figure 1-25 or 2411.0.0 Figure 1-26) allows the selection of up to eight available stores for release.



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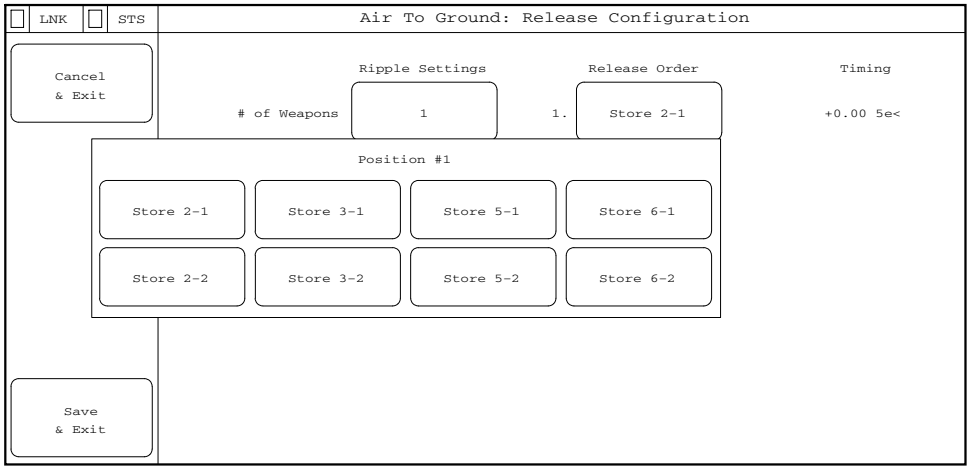


**Figure 1-25.** 2410.0.0 **Release Configuration Page.**

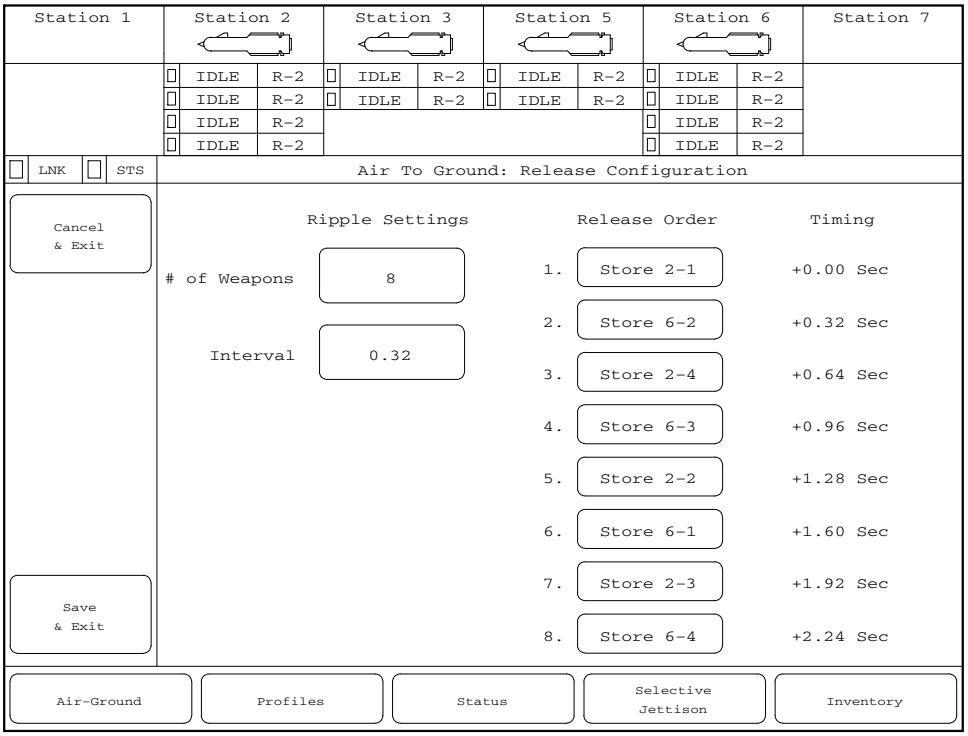


**Figure 1-26.** 2411.0.0 **Release Configuration Page.**



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1.3.3.1.1.1 Ripple Settings.

**CAUTION**

- Setting ripple release intervals greater than .32 seconds may have an adverse effect on missile accuracy due to the plume from the first missile.
- If the Ripple Release Interval is set too long, there is a risk of the target no longer being in the Weapon Engagement Zone (WEZ) for the Hellfire Missile and of overflying the Launch Acceptability Region (LAR) for GBU following the first release.
- Setting ripple release intervals less than .64 seconds on the same wing is not recommended due to close proximity to an adjacent missile launcher.
- 2411.0.0 When M299 launchers are mounted and loaded with four missiles, a missile launched from an upper rail may damage the seeker of the missile below it. Pilots should launch lower missiles before upper missile when practical.

**NOTE**

2411.0.0 During a ripple release, all stores in the super ripple must release from the same missile launcher at equal intervals. If the user selection violates this condition, GSMS will alter the ripple interval to create a valid ripple release. The user selected release order will not be modified.

The Ripple Settings column allows the user to change the number of weapons to be released and the timing interval of their release. The # of Weapons option has a selectable button that displays the current number of weapons selected for release. Pressing the button will bring up "Enter # of Weapons" keypad (Figure 1-27) which allows the selection of the number of munitions to be released in the ripple train. The numeric buttons available for selection will be determined by how many of that specific weapon is loaded on the aircraft. The 0 and – buttons are not be available for selection. The OK and backspace arrow button are only available after a numeric selection has been made.

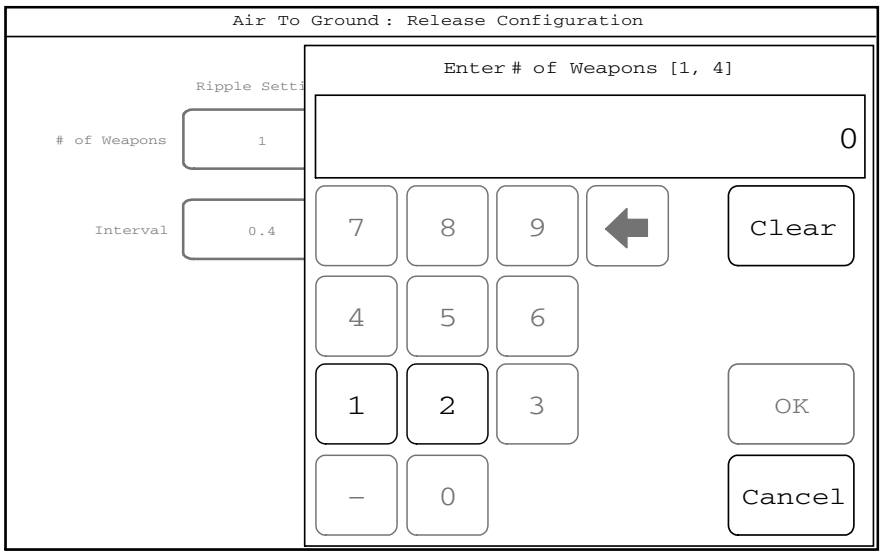


**Figure 1-27. Number of Weapons – Typical.**

The Interval option has a selectable button that displays the current timing interval. Selecting the button allows the user to change the amount of time between each individual release during a ripple release. This setting is activated when the number of weapons exceeds one. The default value for any selected weapon type is shown in Table 1-3. Pressing the button will bring up the "Enter Interval" keypad (Figure 1-28) which allows the setting of release intervals. Pressing the button will bring up the Enter Interval keypad (Figure 1-28) which allows the setting of the ripple interval as listed in Table 1-3). The minimum to the maximum interval values for each munition type are displayed in Table 1-3.

2411.0.0 When the maximum number of eight missiles, are selected for a single ripple release, the default orders is shown in Table 1-2.



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**Table 1-2. 2411.0.0 AGM Default Station-Store Order.**

| Store Order | Station-Store |
|---|---|
| 1 | 2-4 |
| 2 | 6-3 |
| 3 | 2-3 |
| 4 | 6-4 |
| 5 | 2-1 |
| 6 | 6-2 |
| 7 | 2-2 |
| 8 | 6-1 |
| 9 | 3-1 |
| 10 | 5-2 |
| 11 | 3-2 |
| 12 | 5-1 |

2411.0.0 To create a valid ripple release, all stores in super ripple must release from the same missile launcher at equal intervals. If this condition is not met, the software will alter ripple interval timing and create a valid ripple release using the following two requirements:

1. If there is an existing super ripple interval between two stores that is less than the interval between the second store in the super ripple and a third store on that same missile launcher, the software will increase the interval between the second store and the third store to two seconds (2411.0.0 Figure 1-29). This two second interval starts a new launch command which can either be a single fire command or a new super ripple release with the third and fourth stores on the missile launcher (if applicable).
2. If there is an existing super ripple interval between two stores that is greater than the interval between the second store in the super ripple and a third store on that same missile launcher, the software will increase the interval between the second store and the third store to equal the super ripple timing between the first two stores (2411.0.0 Figure 1-30).

These requirements will allow to select the store for release in any desired release order. All altered timings will be displayed in yellow to alert the user of the timing updates.



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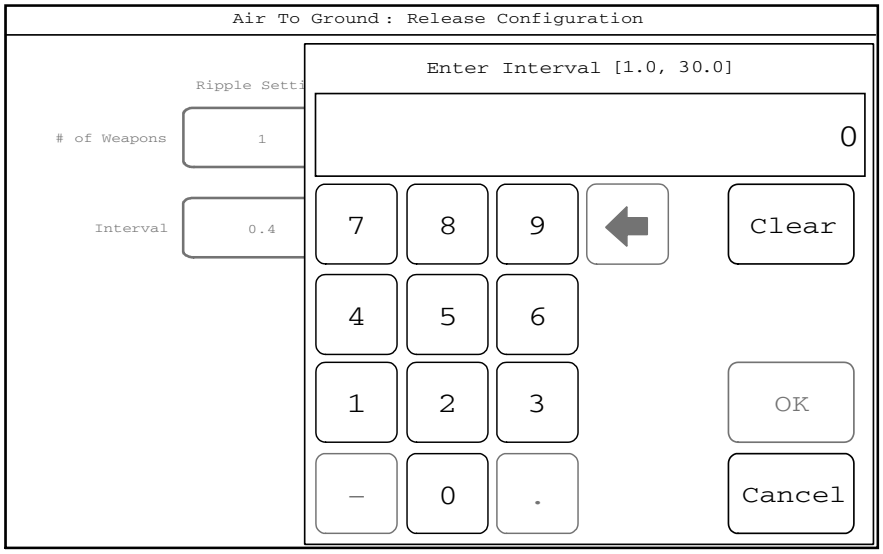


**Figure 1-28. Ripple Interval – Typical.**

**Table 1-3. Ripple Interval Parameters.**

| Intervals are in seconds | GBU on Pylon | GBU on DCS | With Rapid Fire Capable (RFC) Launcher | |
|---|---|---|---|---|
| | | | AGM-114, Same Wing | AGM-114, Opposite Wings |
| Default: | 0.3 | 0.5 | .64 | .32 |
| Minimum: | 0.3 | 0.5 | .64 | .32 |
| Range | 0.3 – 30 | 0.5 – 30 | .64, .96, 1.28, 1.6, 2 – 8 | .32, .64, .96, 1.28, 1.6, 2 – 8 |
| Maximum: | 30 | 30 | 8 | 8 |



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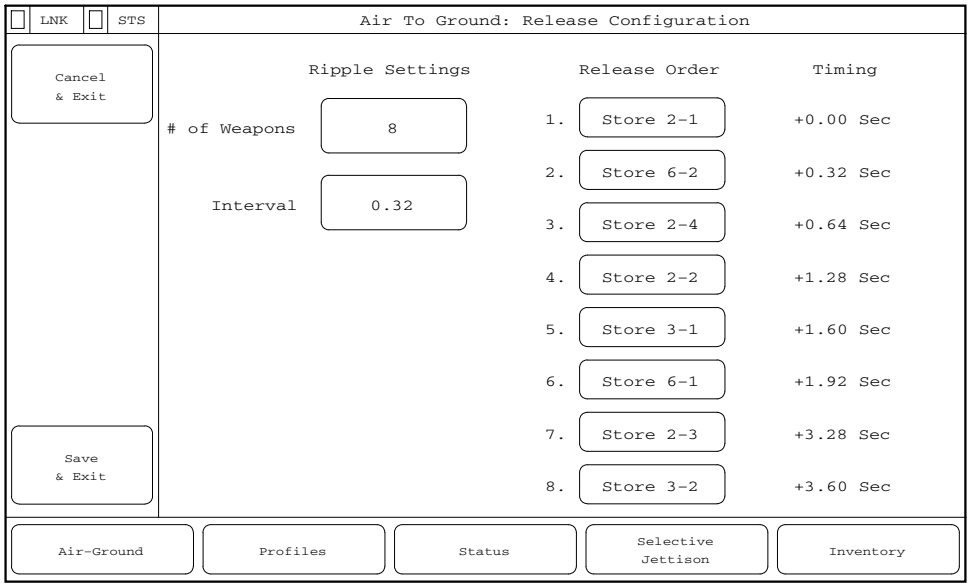


**Figure 1-29.** 2411.0.0 **AGM-114 Two Second Interval Enforcement.**

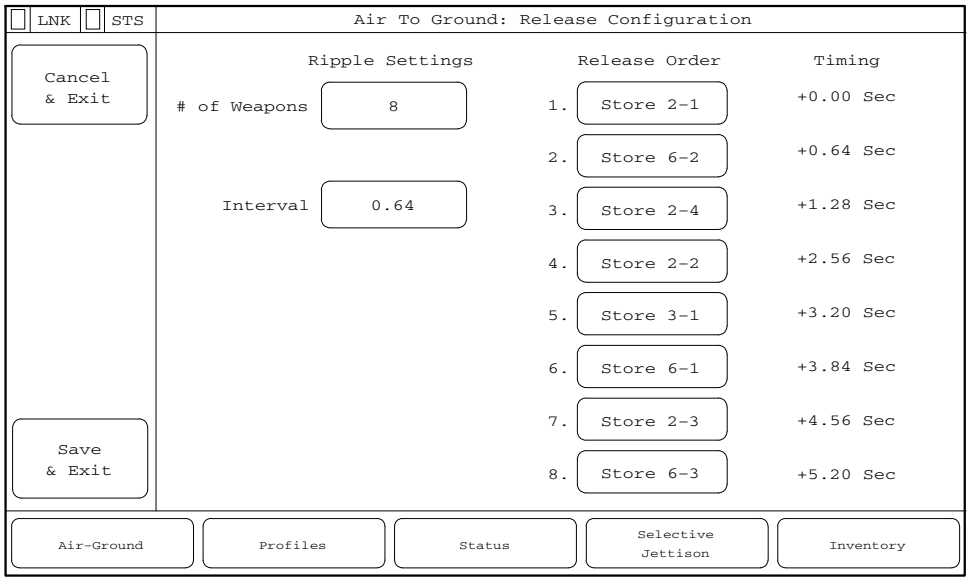


**Figure 1-30.** 2411.0.0 **AGM-114 Interval Matching Enforcement.**



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1.3.3.1.1.2 Release Order.

**NOTE**

When changing the release order on the Release Configuration page, the release position option screen will display “Position #1”, regardless of what button is being selected. Once the screen closes, the release order selected by the user will be displayed on the Release Order button on the Release Configuration page.

The Release Order is displayed to the right of the Page Area and provides a series of selectable buttons to manipulate the order in which specific weapons of the same type are to be released. Associated with the Release Order is the Timing listing displayed to the right of the buttons. The settings selected for the number of weapons and the interval directly impact the Release Order and Timing. Timing displays in seconds and tenths of seconds when that weapon will be released relative to the time the first weapon is released by the aircraft.

The top button in the Release Order column will have “1.” located to its left and corresponds to the first munition to be released. The next button, if present, will have “2.” located to its left and corresponds to the second munition in the ripple train, and so forth. The face of each Release Order button will initially display a default (software assigned) store number, showing the munition assigned to that release position. The operator can override these assignments.

Hellfire default assignments are left-to-right, outer-to-inner, so a ripple release of four AGM-114 missiles have a default release order of 2-1, 6-2, 2-2, 6-1 (2410.0.0 Figure 1-31). 2411.0.0 When using M299 launchers and launching a maximum ripple of eight AGM missiles, the default release orders is shown in 2411.0.0 Table 1-2. GBU default assignments are left-to-right and outer-to-inner, so a ripple release of four GBUs have a default release order of Stations 2, 6, 3, 5 (Figure 1-33).

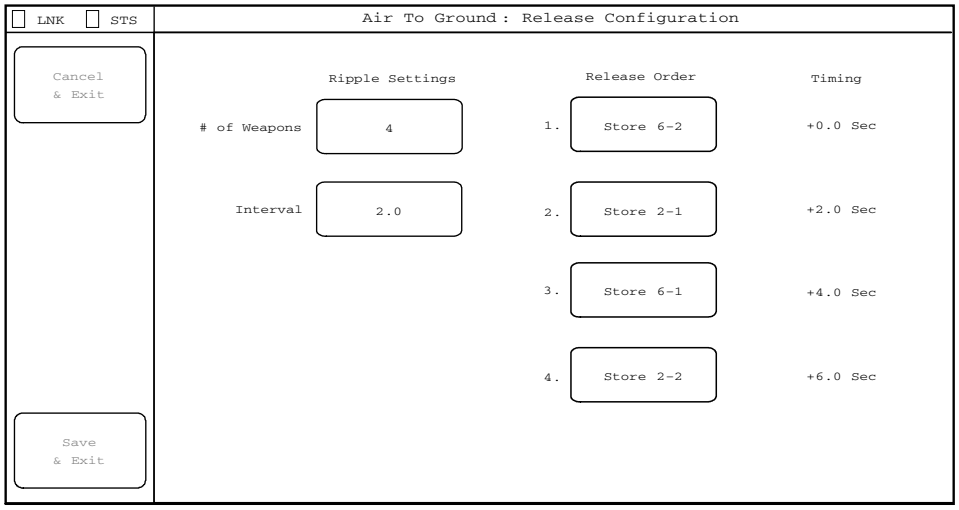


**Figure 1-31. 2410.0.0 Release Order for AGM-114.**



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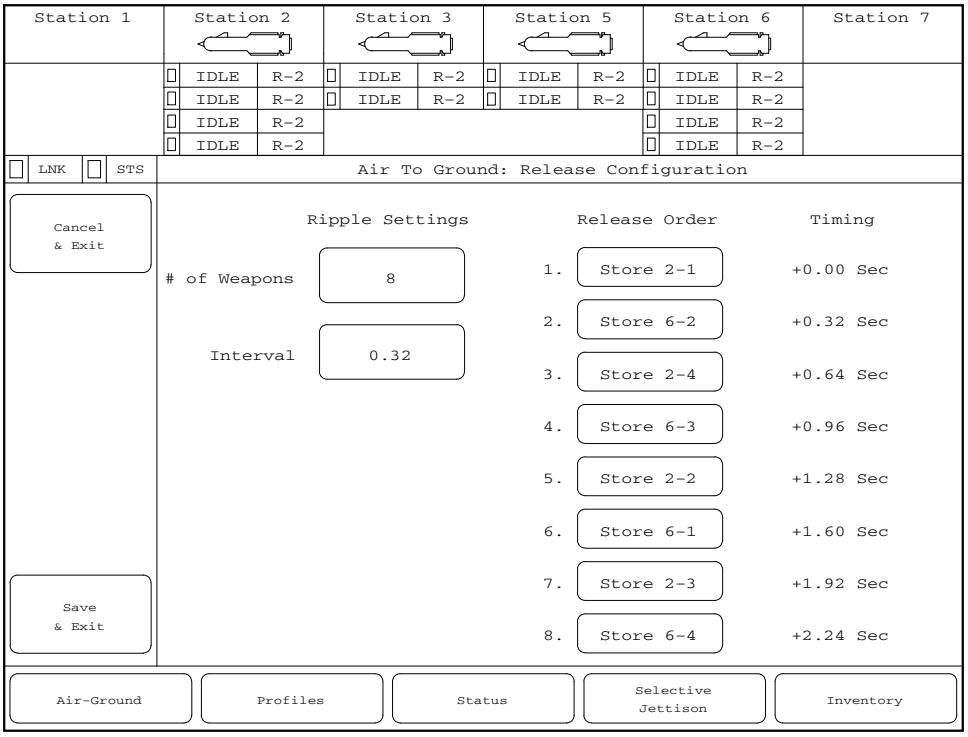


**Figure 1-32.** 2411.0.0 **Release Order for AGM-114.**

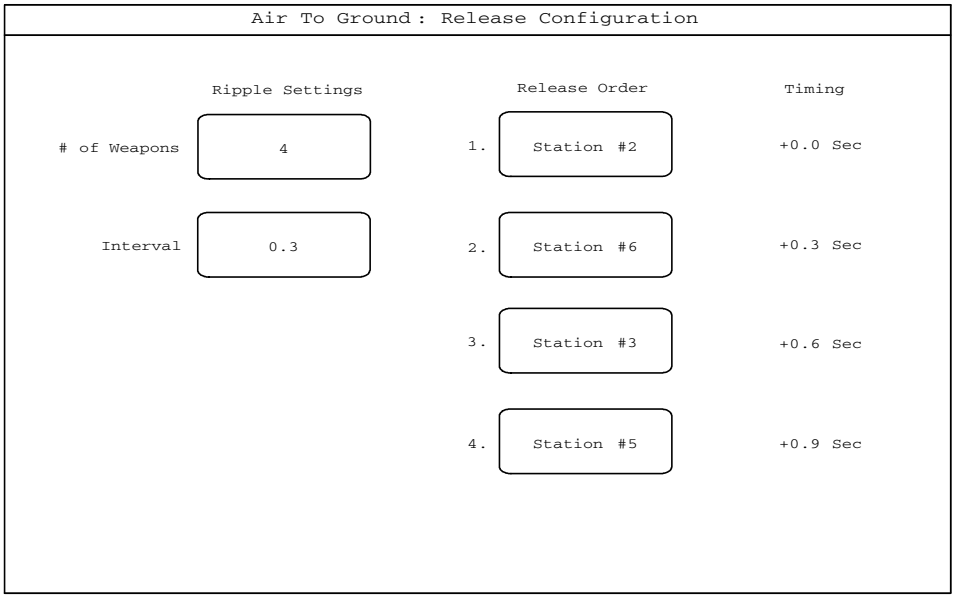


**Figure 1-33. Release Order for GBU-12/51 – Typical.**



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If a different release order is desired, selecting any button on the Release Order listing will bring up a release position option screen (2410.0.0 Figure 1-34 through Figure 1-36). This screen allows the pilot to select from any station carrying the selected weapon type that is available for release. Selecting any of the buttons on the release position option screen will cause the selected station or store to become that release position in the release order. If the selected station or store is already in the release order, the release order will change according to the latest selection. When release order has been changed to launch from the same wing the ripple order will be changed from .32 to .64 seconds although the Interval button will continue to read .32.

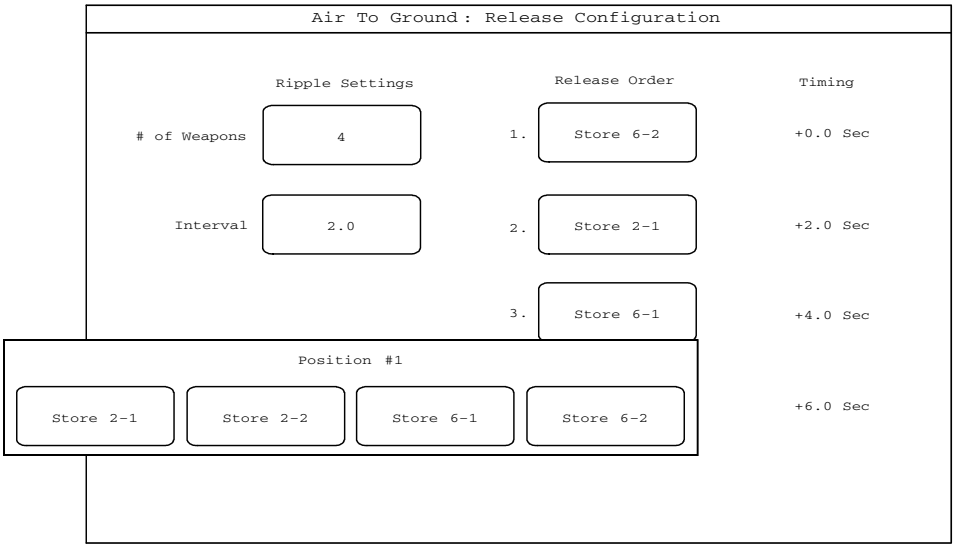


**Figure 1-34. 2410.0.0 Release Position Option Screen for AGM-114.**

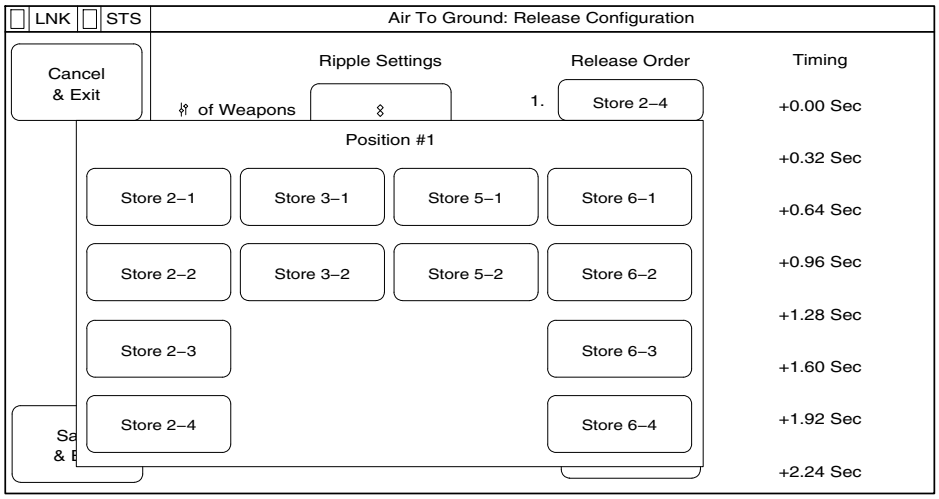


**Figure 1-35. 2411.0.0 Release Position Option Screen for AGM-114.**



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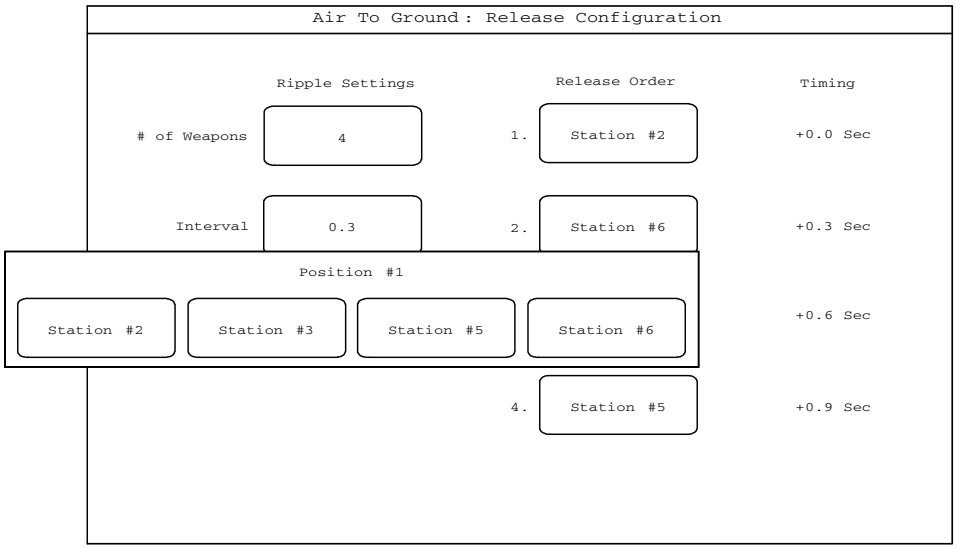


**Figure 1-36. Release Position Option Screen for GBU-12/51 – Typical.**

Once ripple settings have been made, the operator has the option of making these effective via the Save & Exit control button or abandoning any settings made via the Cancel & Exit button. After use of either of these buttons, the GSMS will return to the Select Store page. After a ripple train has been specified the Single/Ripple button for that munition type in the Select Store screen will display "Ripple: N" (N is number of munitions) and below that "Int: " followed by the interval for the ripple train (Figure 1-37).

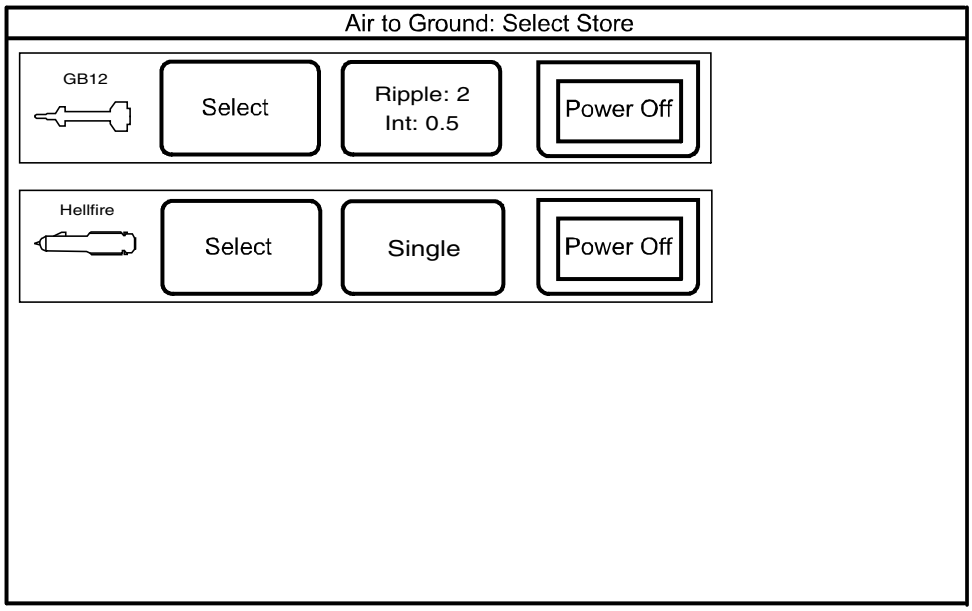


**Figure 1-37. Select Store with Ripple Settings – Typical.**

1.3.3.2 Store Settings.

**NOTE**

A positive value in the Offset Down field will reduce the target elevation.

The Air-Ground mode Store Settings screen (Figure 1-38) appears when the Store Settings Control button is selected in Air to Ground Mode. The Store Settings screen allows the setting of parameters for one specific store with the option to be applied to all stores of that type or all stores selected for a ripple release.



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Store settings can be configured at any time after inventory verification, without either prior target or store selection. Store settings are uplinked to the Stores Control Computer (SCC) and sent to the munitions once they have been powered up. Station: N (N is station number) is displayed in the top right of the Page Section. This indicates which station is currently selected and its selected store attributes displayed. The selected store will also have a flashing border surrounding the Store Information Box. Below the station number identification is a pair of horizontal arrows described below:

- Right Arrow – permitting station navigation from left to right.
- Left Arrow – permitting station navigation from right to left.

For each UAI store type, a Configuration Data File determines what store settings are applicable. The setting can be a list option or a range value.

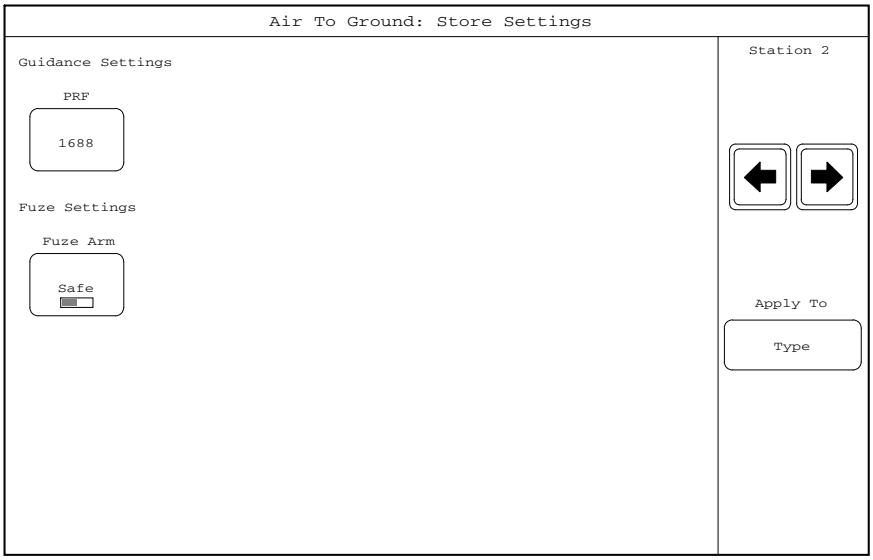


**Figure 1-38. Store Settings for GBU-12/51.**

When a station has the capability to carry more than one store, a second pair of vertical arrows will be displayed below the pair of horizontal arrows in the Page Section (Figure 1-43 through 2410.0.0 Figure 1-45). This pair of vertical arrows provides the ability to navigate from one store to the next on the same station. "Store: N" (N is store number) identifies which store is selected and is located above the vertical arrows. The store identification is also represented by a flashing border surrounding the Store Information Box. Below the store navigation arrows, located in the lower right side of the Page Area, are up to two push buttons under the Apply To title. The buttons are Type or Ripple Train and become selectable when a change is made to the store settings. These buttons allow the change to be applied to other weapons based on the information below:

- Type – apply to all loaded stores of the same type as the currently selected store.
- Ripple Train – apply to all stores in the ripple train to which the currently selected store is assigned.

Attempting to perform a store settings Apply to Type or Ripple Train action on a GBU that can be configured with the FMU-139 /152 may cause the Apply To Type/Ripple Train dialog window to be displayed for the affected station(s). Table 1-4 provides a list of possible indications on the Apply To Type/Ripple Train dialog window. The selected and displayed store that has received store setting modifications is considered the source store. All other stores that are attempting to be modified are considered the destination store.

The AGM-114 Store Settings screen does not have the Apply To buttons located in the lower right side of the Page Area. They are now located in the respective rows of and directly associated with the Launch Mode and PRF Code settings. The buttons are selectable at all times except for Ripple Train which is only available when a ripple has already been set up, this allows changes to be applied to similar missile types based on the information described above.

The Store Target Coordinate has been removed from the JDAM and GBU-49 Store Settings pages (Figure 1-39 through Figure 1-42). The Target Settings group box now contains Offset North, Offset East, and Offset Down (Figure 1-48 through Figure 1-50). The offsets are entered in units of meters.



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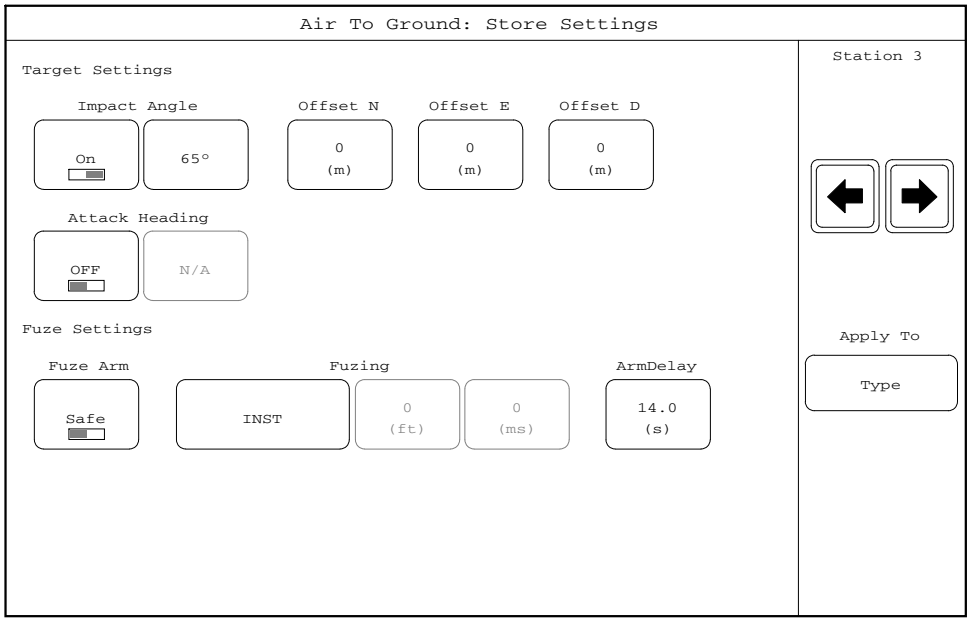


**Figure 1-39. Store Settings for GBU-38.**

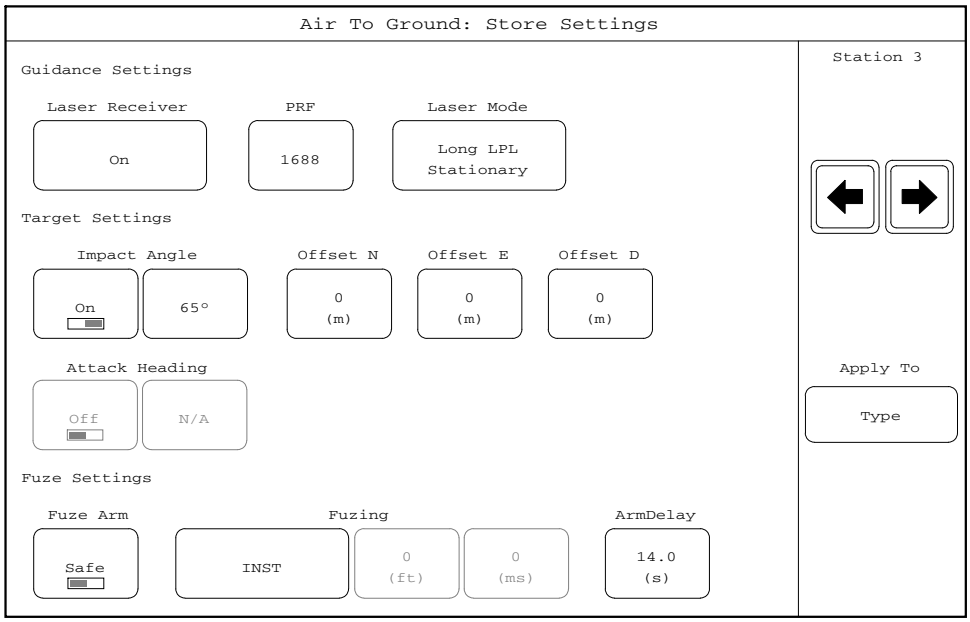


**Figure 1-40. Store Settings for GBU-54.**



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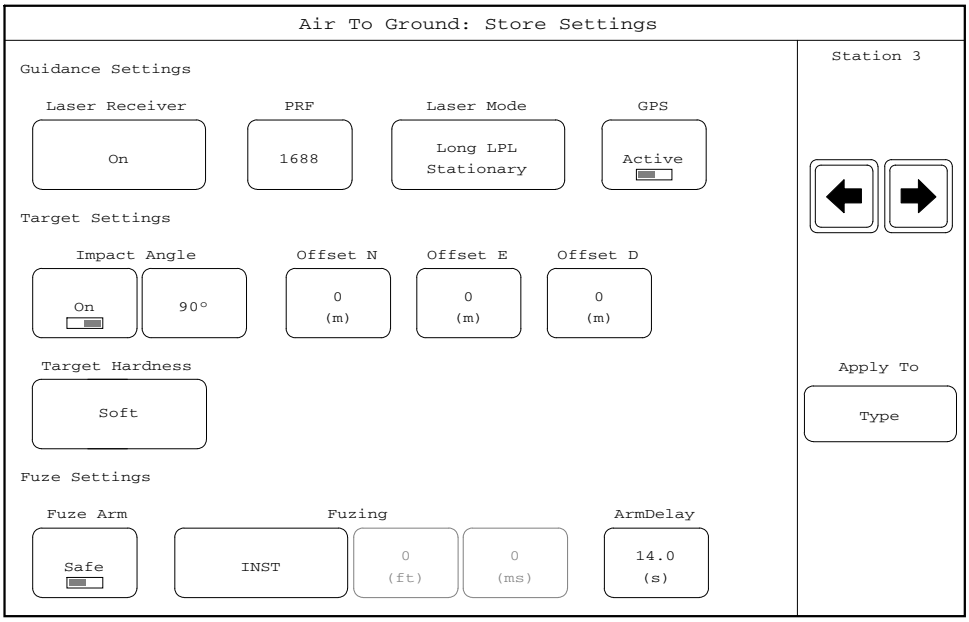


**Figure 1-41. Store Settings for GBU-48.**

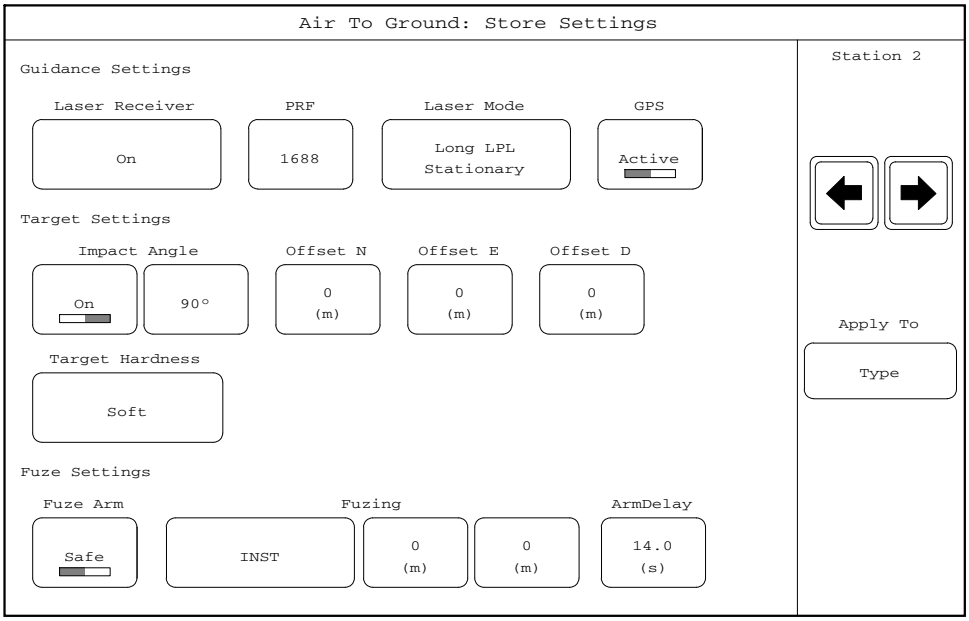


**Figure 1-42. Store Settings for GBU-49.**



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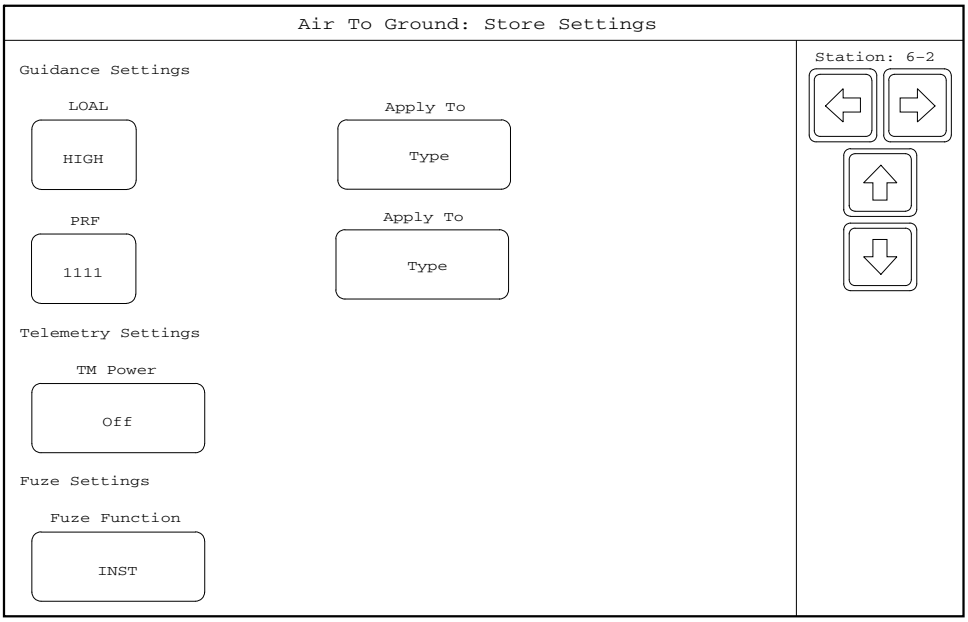


**Figure 1-43. Store Settings for AGM-114.**

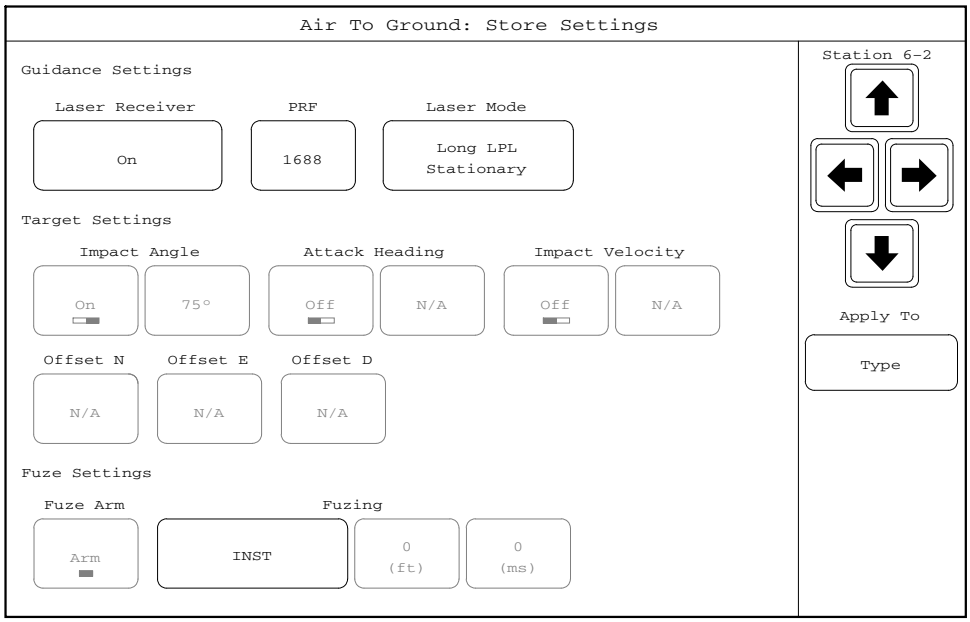


**Figure 1-44.** 2410.0.0 **Store Settings for LSDB.**



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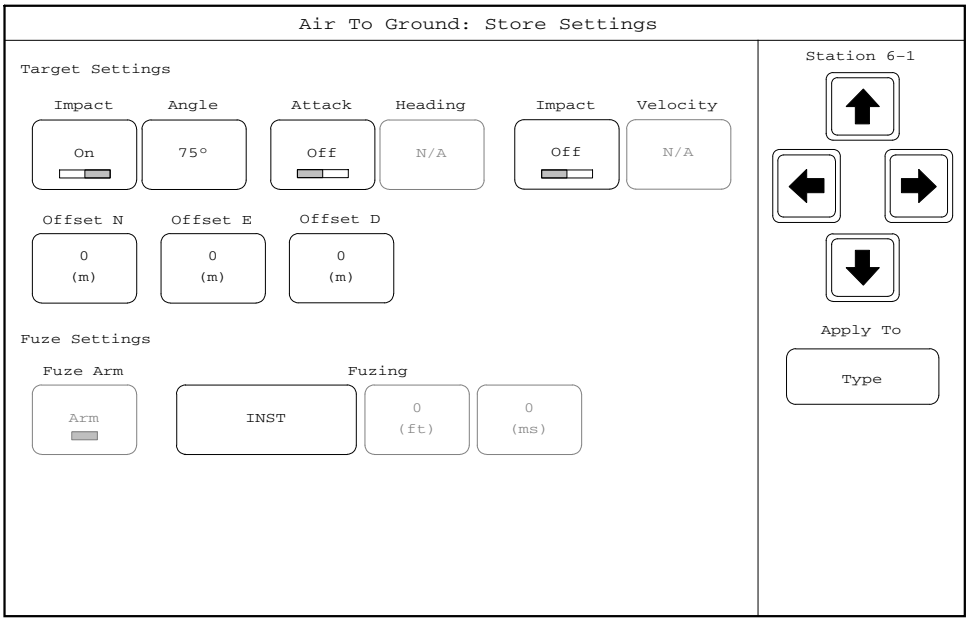


**Figure 1-45.** 2410.0.0 **Store Settings for SDB.**

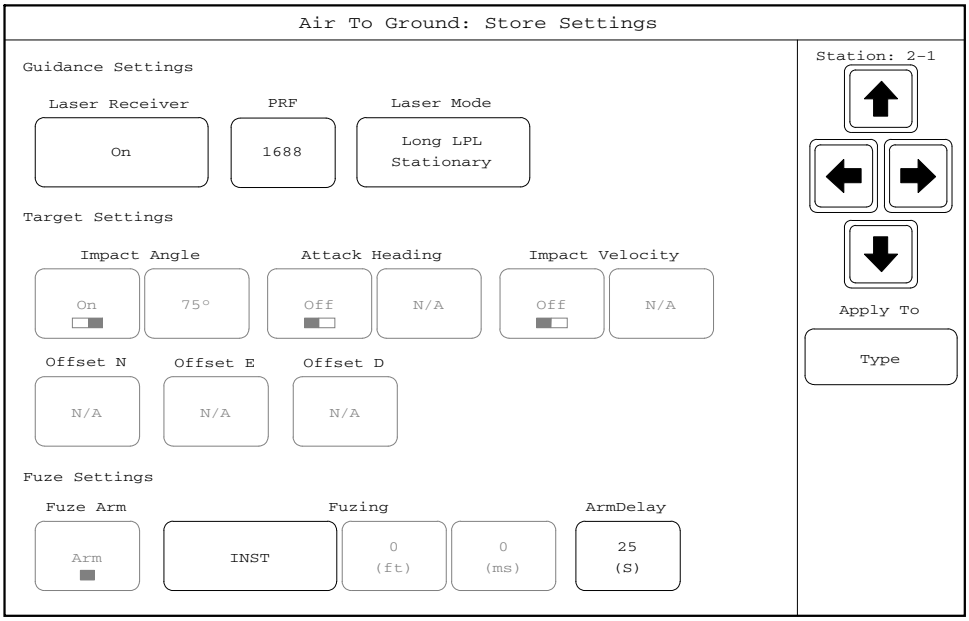


**Figure 1-46.** 2411.0.0 **Store Settings for LSDB.**



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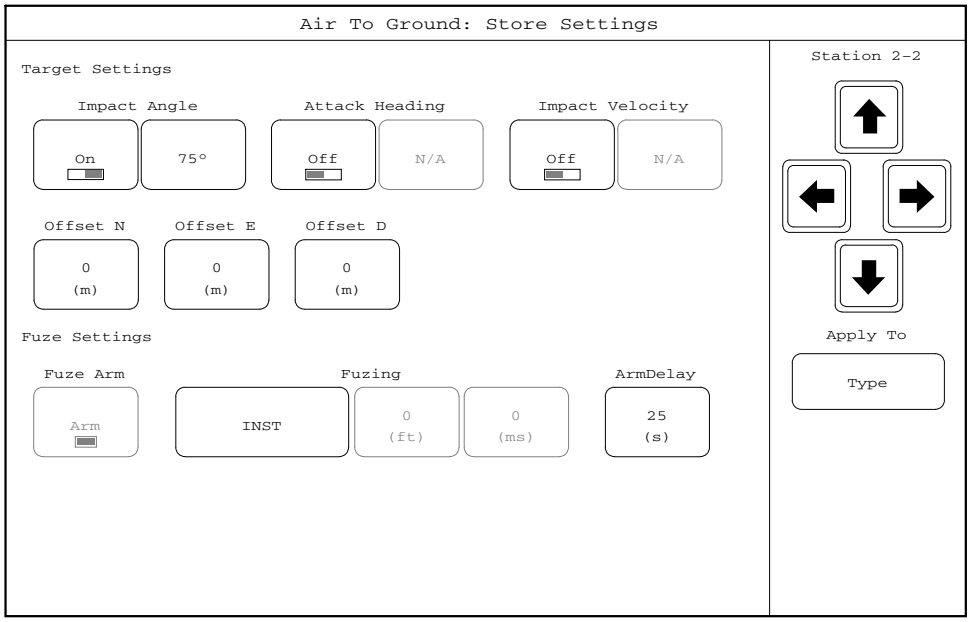


**Figure 1-47.** 2411.0.0 **Store Settings for SDB.**

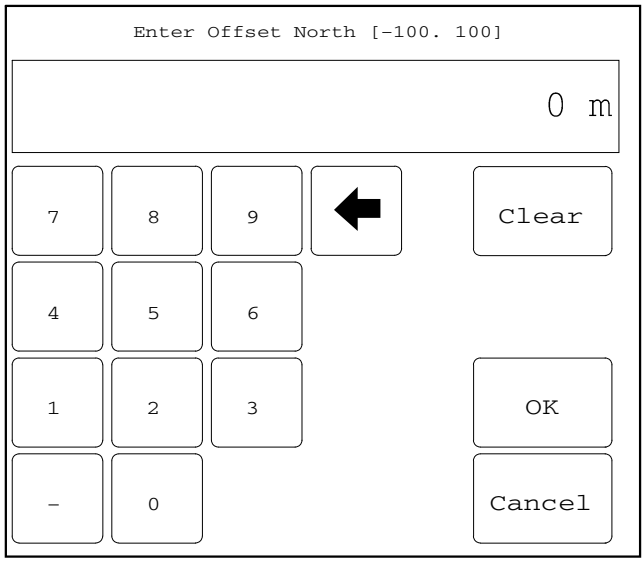


**Figure 1-48. Offset North Input Dialog.**



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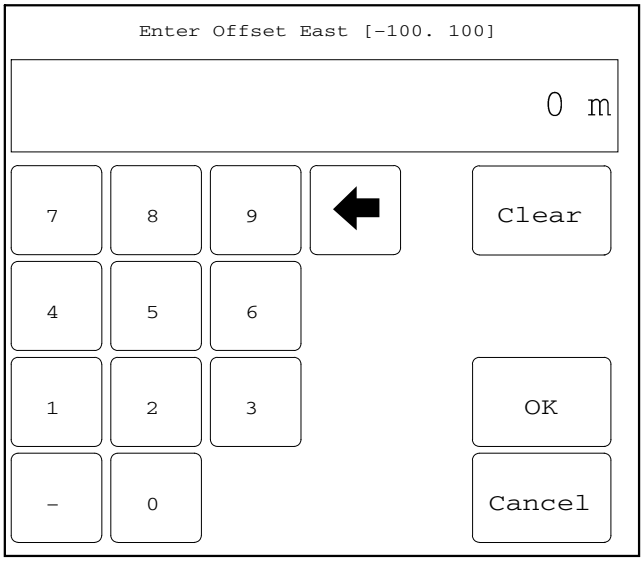


**Figure 1-49. Offset East Input Dialog.**

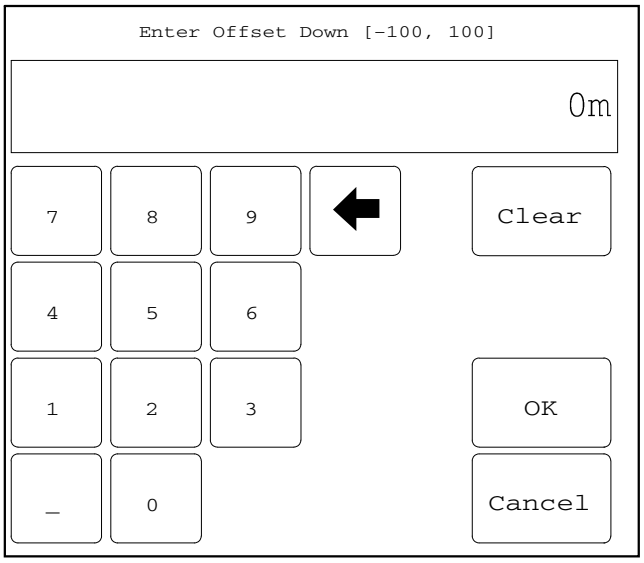


**Figure 1-50. Offset Down Input Dialog.**

**Table 1-4. FMU-139/FMU-152 Apply To Type/Ripple Train Situations.**

| Indication Displayed | Source Store | Destination Store | Explanation |
|---|---|---|---|
| None | No FMU-139/FMU-152 detected | No FMU-139/FMU-152 detected | None |


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**Table 1-4. FMU-139/FMU-152 Apply To Type/Ripple Train Situations – Continued.**

| Indication Displayed | Source Store | Destination Store | Explanation |
|---|---|---|---|
| JPF settings not changed on STA N | No FMU-139/FMU-152 detected | FMU-139/FMU-152 detected, desired store settings selected | The destination GBU-49 GBU-38/54 will keep any FMU-152 settings that have been sent to the fuze. |
| JPF settings not changed on STA N | No FMU-139/FMU-152 detected | FMU-139/FMU-152 detected, default store settings selected | The destination GBU-49 GBU-38/54 will keep any FMU-152 settings. |
| JPF Settings cannot be applied to STA N | FMU-139/FMU-152 detected, desired store settings applied | No FMU-139/FMU-152 detected | Destination GBU-49 GBU-38/54 will have delay settings buttons grayed out. |
| None | FMU-139/FMU-152 detected desired store settings applied | FMU-139/FMU-152 detected, desired store settings applied | The destination GBU-49 GBU-38/54 store settings will be changed to the source GBU-49 GBU-38/54 store settings. |
| None | FMU-139/FMU-152 detected, desired store settings applied | FMU-139/FMU-152 detected, default store settings selected | The destination GBU-49 GBU-38/54 will have any default delay settings replaced with desired store setting delays from source GBU-38/54 GBU-49. |
| JPF Settings cannot be applied to STA N | FMU-139/FMU-152 detected, default store settings selected | No FMU-139/FMU-152 detected | Destination GBU-49 will have delay settings buttons grayed out. |
| None | FMU-139/FMU-152 detected, default store settings selected | FMU-139/FMU-152 detected, desired store settings selected | The destination GBU-49 GBU-38/54 will have default store settings applied. |
| None | FMU-139/FMU-152 detected, default store settings selected | FMU-139/FMU-152 detected, default store settings selected | Both GBU-49's GBU-38/54 will keep default store settings. |

When a store is part of a ripple train, "Seq: N/N" appears below the Apply To button(s). The first number displayed is the position number of the currently selected weapon in the ripple train. The second number displayed is the total amount of stores being utilized in the ripple train.

The main portion of the Page Section contains a series of store setting buttons. These settings will directly impact the store while others are for informational purposes. Below are lists of store settings available by weapon type.

The AGM-114 has the following store settings available:

Guidance Settings

- LOAL
- PRF

Telemetry Settings

- TM Power

Fuze Settings

- Fuze Function

The GBU-12/51 store settings are as follows:

Guidance Settings

- Laser Receiver
- PRF

Fuze Settings

- Fuze Arm

The GBU-38 store settings are as follows:

Target Settings

- Impact Angle Enable
- Impact Angle
- Attack Heading Enable
- Attack Heading
- Offset North



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- Offset East
- Offset Down

Fuze Settings for FMU-139/152

- Fuze Arm
- Fuzing Type
- Fuzing Height
- Fuzing Delay
- Arm Delay

The GBU-54 store settings are as follows:

Guidance Settings

- Laser Receiver
- PRF
- Laser Mode

Target Settings

- Impact Angle Enable
- Impact Angle
- Attack Heading Enable
- Attack Heading
- Offset North
- Offset East
- Offset Down

Fuze Settings for FMU-139/152

- Fuze Arm
- Fuzing Type
- Fuzing Height
- Fuzing Delay
- Arm Delay

The JDAM store settings are as follows:

GBU-54 Guidance Settings

- PRF
- Last Pulse
- Guidance

Target Settings

- Impact Angle
- GBU-38 Impact Azimuth
- Offset North
- Offset East
- Offset Down

Fuze Settings

- Fuze Arm
- Airburst
- Function
- Arm

UAI The GBU-48 store settings are as follows:

Guidance Settings

- Laser Receiver
- PRF
- Last Pulse
- GPS
- Laser Mode

Target Settings

- Impact Angle Enable
- Impact Angle
- Target Hardness
- Offset North
- Offset East
- Offset Down

Fuze Settings

- Fuze Arm
- Fuzing Type
- Fuzing Height
- Fuzing Delay
- Arm

The GBU-49 store settings are as follows:

Guidance Settings

- Laser Receiver
- PRF
- Last Pulse
- Guidance
- Laser Mode
- GPS

Target Settings

- Impact Angle Enable
- Impact Angle
- Hardness
- Offset North
- Offset East
- Offset Down

Fuze Settings

- Fuze Arm
- Fuzing Type
- Fuzing Height
- Fuzing Delay
- Non-UAI Airburst
- Non-UAI Function
- Non-UAI Arm
- Arm Delay

The SDB store settings are as follows:

Target Settings

- Impact Angle Enable
- Impact Angle
- Attack Heading Enable
- Attack Heading
- Impact Velocity Enable
- Offset – North
- Offset – East
- Offset – Down


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Fuze Settings

- Fuze Arm
- Fuzing Type
- Height
- Delay
- 2411.0.0 Arm Delay

The LSDB store settings are as follows:

Guidance Settings

- Laser Receiver
- PRF
- Laser Mode

Target Settings

- Impact Angle Enable
- Impact Angle
- Attack Heading Enable
- Attack Heading
- Impact Velocity Enable
- Offset – North
- Offset – East
- Offset – Down

Fuze Settings

- Fuze Arm
- Fuzing Type
- Height
- Delay
- 2411.0.0 Arm Delay

Many settings apply to multiple weapons while some settings are specific to one weapon. The following paragraphs describe each store settings available. Settings that apply to multiple weapons will be covered first followed by weapon specific settings.

1.3.3.2.1 PRF Code. The AGM-114, and GBU-12/51 require a Laser Rangefinder Designator (LRD) to guide them to a target. The GBU-49 and GBU-54 does not require an LRD to guide it to a target, but has the option to utilize laser guidance.

A four digit Pulse Repetition Frequency (PRF) code is used to determine which laser spot, designated by an LRD, the weapon guides to. The PRF Code setting provides the ability to set or change the PRF code the weapon will track. Selecting the PRF Code button will bring up the PRF Code Keypad (Figure 1-51).

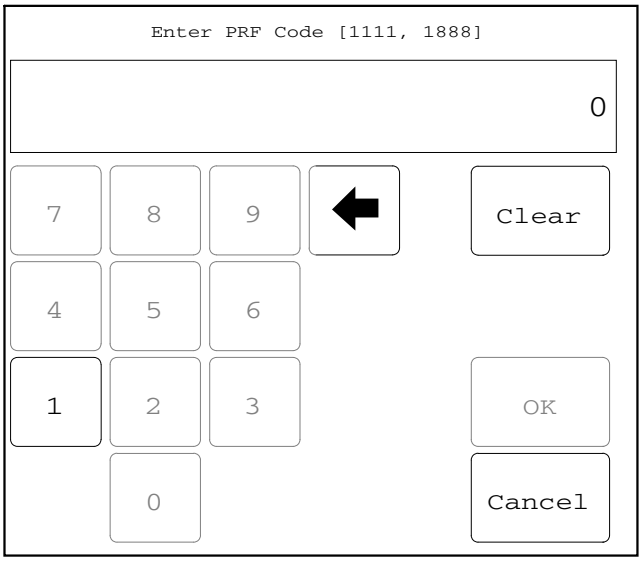


**Figure 1-51. PRF Code Keypad – Typical.**

Once the keypad is displayed the desired four digit PRF code is entered into the keypad and accepted by selecting OK. The entered PRF code will then be displayed on the PRF Code button.

The PRF code entered into the AGM-114 Launch Settings will be transmitted to the missile launcher and then to the missile as the PRF code to track. The PRF code input to the missile must match the PRF code being used by the LRD lasing the target location.

The PRF code entered into the GBU-12/51 PRF Setting does not change the PRF code set by the load crew. It provides a way to pass on the information in event of a hand-off to another GCS or as a reference to what is coded into the GBU-12/51. The PRF code entered must be the same PRF code that is coded by the load crew into the bomb's Computer Control Group (CCG) Code Select Switches.

The PRF code entered into the GBU-49 Laser Settings will be sent to the bomb as the code to track. Load crews will set a PRF code into the weapon Enhanced Computer Control Group



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(ECCG) Code Select Switches and is considered the default code for the weapon. If the weapon is required to track a different PRF code, the entered code will be transmitted to the weapon as the required code to track. The allowable PRF code range for the GBU-49 is between 1511 and 1788. The allowable PRF code range for the AGM-114 and GBU-54 is between 1111 and 1788.

1.3.3.2.2 Weapon Fuzing. The GBU-12, GBU-38, GBU-49, GBU-48, GBU-51, and GBU-54, have fuzing installed in order for them to function in the desired manner. The fuze configuration consist of a tail fuze or a combination of a nose and tail fuze. The fuzes are equipped with lanyards that are attached to the bomb rack arming units that hold the lanyards in place during release. When the release command is given, the bomb rack arming units that have been selected will retain the lanyards, which will activate the respective fuzes or when the Jettison command is given, the bomb rack arming unit will release all lanyards, thus not activating the fuzes. The fuze setting configured in the GSMS will determine which arming units will retain their fuze lanyards during release.

Selecting the Fuze button under Fuze Settings on the Store Settings window will bring up the Fuze Arming window, which provides the fuze arming options. The bomb rack type installed on the station will determine which Fuze Arming window will be displayed (Figure 1-52 through Figure 1-54).

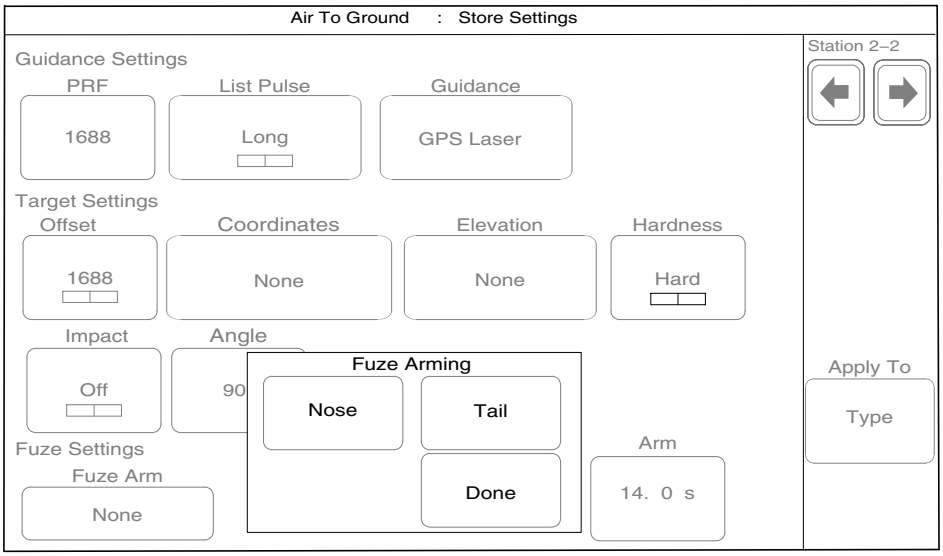


**Figure 1-52. Fuze Arming for BRU-15/A – Typical.**

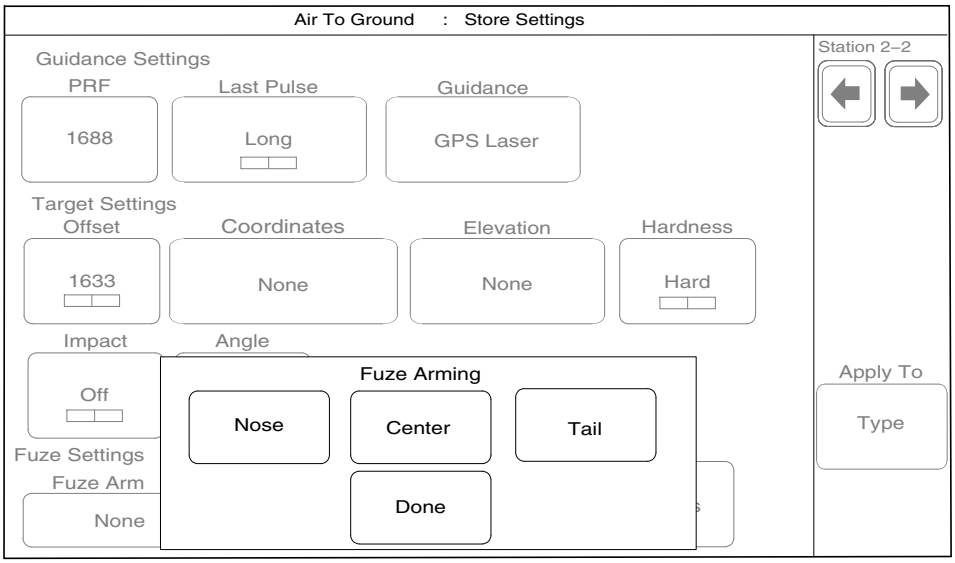


**Figure 1-53. Fuze Arming for BRU-71/A – Typical.**



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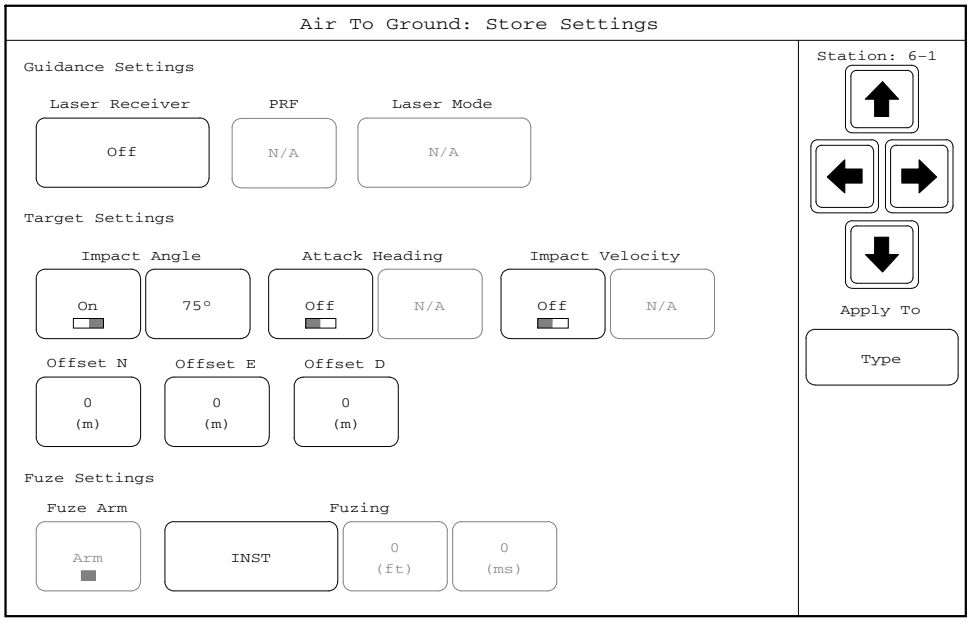


**Figure 1-54. Fuze Arming for BRU-78/A – Typical.**

The Fuze Arming window options and resulting functions are listed below:

BRU-15/A

- Nose – Fuze lanyard attached to the forward arming unit will be retained.
- Tail – Fuze lanyard attached to the aft arming unit will be retained.
- Done – Saves fuzing configuration and closes window.

BRU-71/A

- Nose – Fuze lanyard attached to the forward arming unit will be retained.
- Center – Fuze lanyard attached to the center arming unit will be retained.
- Tail – Fuze lanyard attached to the aft arming unit will be retained.
- Done – Saves fuzing configuration and closes window.

BRU-78/A

- Fuze Arm – Arm/Disarm
- Done – Saves fuzing configuration and closes window.

One or more fuzing option may be selected which is indicated by the button turning blue upon selection. Once the desired fuzing option is selected, the Done button must be selected to close the window. The fuzing selection will be displayed in the Fuze Arm button.

If no fuze is selected for either the BRU-15/A or BRU-71/A, there will be no yellow cautions displayed. For weapons loaded on a BRU-15/A, if the tail fuze is not armed but another fuze is, the yellow caution "Tail Fuze Not Armed" will be displayed. For weapons loaded on a BRU-71/A, if the center fuze is not armed but another is, the yellow caution "Center Fuze Not Armed" will be displayed.

All tail fuze lanyards are attached to the BRU-15/A tail arming unit or BRU-71/A center arming unit.The GBU-12, GBU-49, and GBU-51 have different fuzing options available. Ensure the proper fuze arming settings are used when these weapons are employed.

1.3.3.2.3 AGM-114 Launch Mode, TM Power and Fuze Function.

**NOTE**

Hellfire missiles receive targeting data from the MTS.

The Launch Mode and TM Power settings are set or changed by selecting the respective button on the AGM-114 Store Settings Screen (Figure 1-43).

Selecting the Launch Mode button will display the Launch Mode selection window. Selecting the LOAL button will display the Lock On After Launch window. The three available methods are listed and described below (Figure 1-55):

LOAL

- The Direct setting results in the most direct attack trajectory.



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- The High setting results in a more vertical attack trajectory than Direct and is the default setting.
- The Low setting results in a more horizontal attack trajectory then Direct.

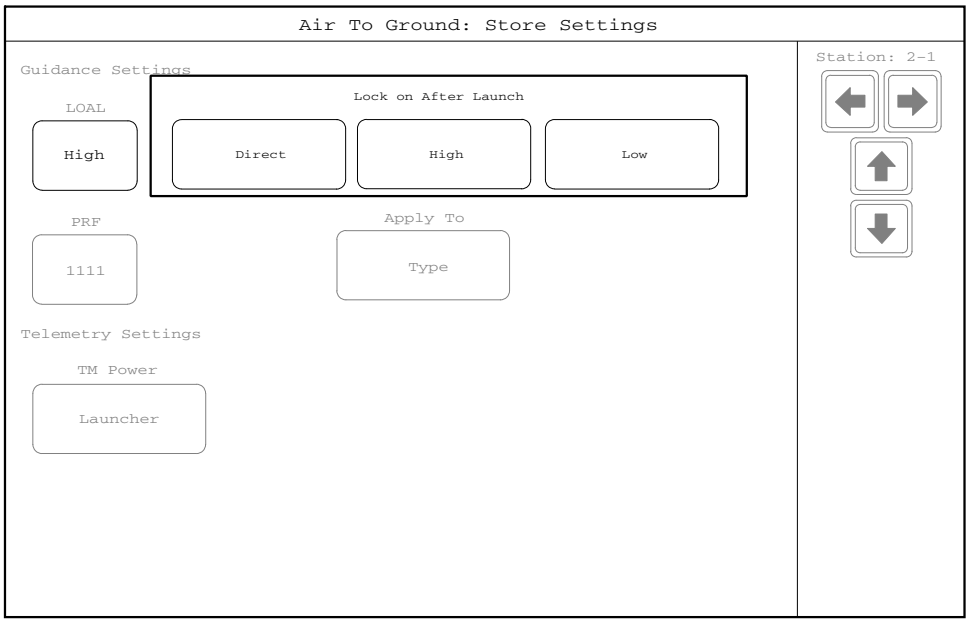


**Figure 1-55. Launch Modes for AGM-114.**

If the missile is equipped with a warhead replacement Telemetry (TM) kit, the TM Power button (Figure 1-56) allows the pilot to choose the source of power for the telemetry kit. The options and definitions are as follows:

TM Power

- Off – no power source
- Launcher – missile TM power is provided through the launcher
- Battery – missile TM power is provided by the missile battery

Launcher power should be used prior to missile release and the battery should not be turned on until the last available moment prior to release.



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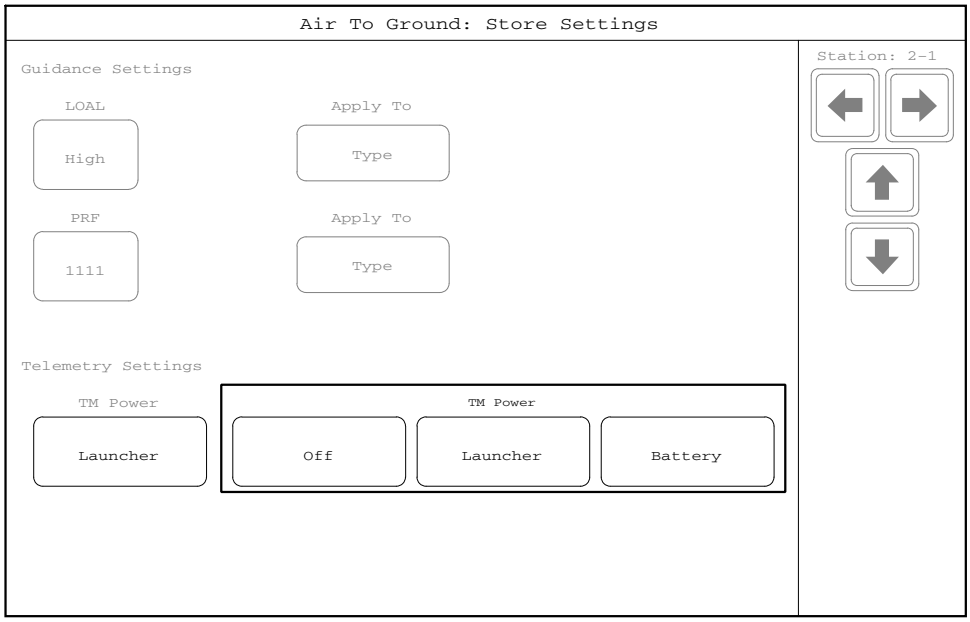


**Figure 1-56. TM Power for AGM-114.**

Fuze Function will only be displayed on the store settings screen when an AGM-114 R missile is loaded. Selecting the Fuze Function button will bring up the Delay window (Figure 1-57). The Delay window provides sixteen selections; PRF, INST, AB, ten delays and three Spare for use later with new capability development. The primary method for selecting the fuze function will be performed utilizing the Delay window. Instantaneous (INST) is the default setting for all missiles with programmable fuzing capability and will cause the missile to detonate on impact. When Airburst is selected the missile delay will be INST with Airburst capability enabled. Enabling Airburst allows the weapon to detonate above the target instead of detonating on impact. The setting for the Airburst is approximately 1.3 (± 1) meters above the target. For AGM-114 without airburst capabilities the airburst button will not be displayed but is present for all AGM-114 missiles with airburst capabilities. Fuze Function settings set using the Delay window will be maintained through missile power cycles.



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**Figure 1-57. Fuze Function Settings AGM-114 – Typical.**

1.3.3.2.3.1 Fuze Function PRF. A secondary option for selecting fuze functions is done by pressing the PRF button on the Fuze Function window which enables the PRF fuzing capability. Fuze functions can then be input using the PRF Code button and PRF Code keyboard on the Store Settings screen. These codes are specific to this employment, refer to Table 1-5.

**NOTE**

- Fuze Function PRF codes are specific to Hellfire R missiles. Other PRF codes may cause missile BIT failures. Refer to Table 1-5 for correct PRF codes.
- If the PRF button is selected on the Delay window and no Fuze Function PRF Code is set, the missile will default to INST.
- When using Fuze Function PRF codes, Hellfire R missiles will revert back to the instantaneous (default) fuze setting any time the missile is powered on from an off state.

**Table 1-5. Hellfire Selectable Fuze Functions.**

| Mode | GSMS Display | Fuze Function PRF Code | Delay (ms) |
|---|---|---|---|
| PRF Override | PRF | N/A | Allows Fuze Function via PRF code, “backdoor” functionality |
| Delay K, Default | INST | 1755 | Instantaneous |
| HOB enable | AB | 1744 | Airburst |
| Delay 1 | 3.5 ms | 1766 | 3.5 ms |
| Delay 2 | 6.9 ms | 1622 | 6.9 ms |
| Delay 3 | 10 ms | 1777 | 10 ms |
| Delay 4 | 20 ms | 1788 | 20 ms |
| Delay 5 | 30 ms | 1633 | 30 ms |
| Delay 6 | 40 ms | 1644 | 40 ms |



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**Table 1-5. Hellfire Selectable Fuze Functions – Continued.**

| Mode | GSMS Display | Fuze Function PRF Code | Delay (ms) |
|---|---|---|---|
| Delay 7 | 60 ms | 1655 | 60 ms |
| Delay 8 | 80 ms | 1666 | 80 ms |
| Delay 9 | 120 ms | 1677 | 120 ms |
| Delay 10 | 150 ms | 1688 | 150 ms |
| Spare 1 * | Spare 1 | TBD | TBD |
| Spare 2 * | Spare 2 | TBD | TBD |
| Spare 3 * | Spare 3 | TBD | TBD |

* Reserved for future development.

1.3.3.2.4 GBU-49/48 Store Settings.

**CAUTION**

2411.0.0 GBU-48 Lot 5 LAR does not support attacking a moving target.

**NOTE**

- When inputting settings, if an incorrect value or conflicting command is entered, the Setting Disabled window will appear with a list of conditions causing the conflict. Selecting the OK pushbutton will allow the user to navigate back to the previous screen and make the appropriate corrections.
- MSL-84 uses the WGS 84/10 (EGM84) geoid as its reference, HAE-84 uses the WGS-84 ellipsoid as its reference, and MSL-96 uses the EGM-96 geoid as its reference. CCTGP targets have an automatic elevation reference selection of “MSL-96”. When DTED is being utilized, the elevation is MSL-96. Manual and store targets have an elevation reference option of “MSL-84”, “HAE-84”, or “MSL-96”.
- When target hardness is set to Hard, the GBU will alter its final trajectory for maximum penetration. A small amount of accuracy will be lost.

The Store Settings window provides the ability to adjust guidance, target, and fuze settings on the weapon (Figure 1-58 and Figure 1-59). The Guidance Settings section allows the selection of which guidance system the weapon will utilize. Target Settings give the ability to input specific target information, and Fuze Settings allow the selection of different functions when the weapon reaches a desired height or impacts a target. The Store Settings overview screen consists of three sections which are as follows:

- Guidance Settings
- Target Settings
- Fuze Settings



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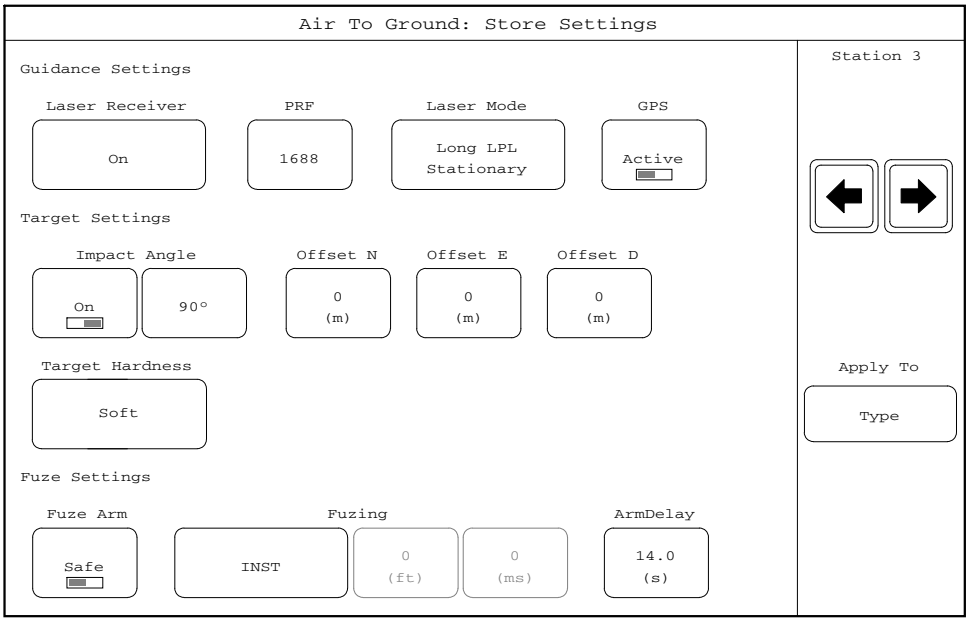


**Figure 1-58. Store Settings for GBU-48.**

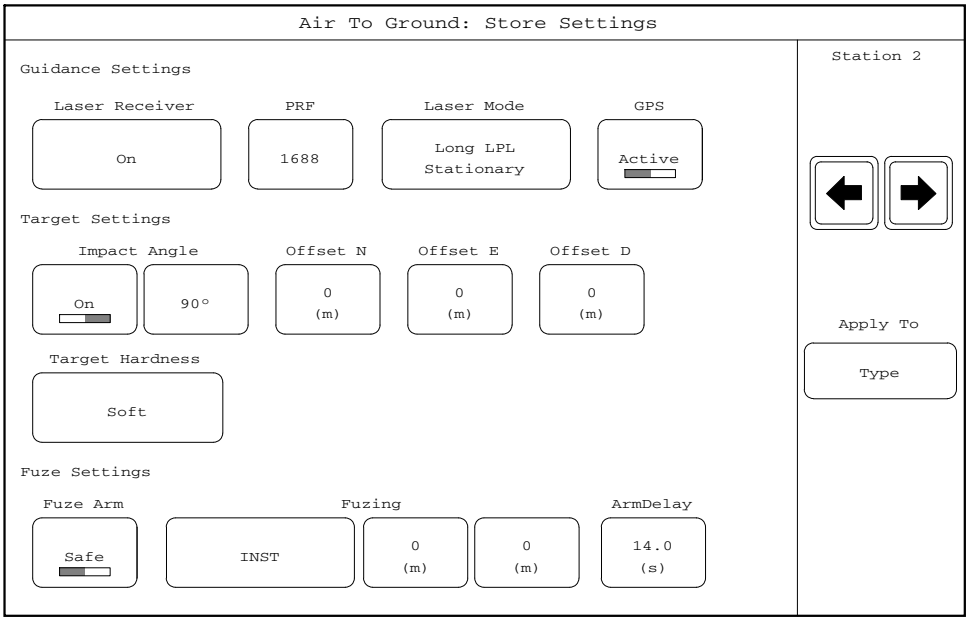


**Figure 1-59. Store Settings for GBU-49.**

The top section in the Store Settings window is the Guidance Settings section. This section displays currently selected guidance settings and provides the option to change guidance settings. The first button in the Guidance Settings section is Laser Receiver. This button allows the user to turn on or off the laser receiver. When set to “On”, the PRF and Laser Mode buttons will be enabled and the store will use both GPS and laser guidance when released from the aircraft. GPS guidance will be the primary source of guidance until the laser seeker detects laser energy with the set programmed PRF code. Once the laser energy is detected, GPS will be the secondary source of guidance. When no laser is in use for targeting, the weapon cannot track


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the programmed PRF code, or laser energy is lost, GPS will be used for guidance until the laser energy is reacquired or weapon impacts. GPS will be the primary source of guidance if the laser receiver button is set to “Off” and disable the PRF and Laser Mode buttons.

Next to the Laser Receiver button is the PRF button. This button displays the currently selected PRF code. When pressed, it will bring up the PRF Code Keypad (Figure 1-51) which provides the ability to set or change the PRF code the selected weapon will track. Load crews will set a PRF code into the weapon ECCG Code Select Switches and is considered the default code for the weapon. If the weapon is required to track a different PRF code, the entered code will be transmitted to the weapon as the required code to track. The PRF code range used is between 1511 and 1788 with a default code of 1688.

The third button in the Guidance Settings section is Laser Mode. This will display a “Long” or “Short” Last Pulse Logic setting and either “Stationary” or “Moving” for target setting. The Last Pulse Logic setting commands the weapon to track the closest or furthest point of laser energy reflection. The Last Pulse Logic can be set to “Short” or “Long”. When selecting the target setting, stationary setting should be used against stationary targets and moving setting should be used against moving targets. The default setting for Laser Mode is “Long LPL Stationary”.

Next to the Laser Mode button is the GPS button. This button allows the GPS setting for the weapon to be active or inhibited when released from the aircraft. Active is the default setting.

The middle section of the Store Settings window is the Target Settings section. The Target Settings section displays currently selected target settings and provides the option to change current offset target coordinates, target hardness setting, and impact angle of the selected weapon. The three sections under Target Settings are as follows:

- Impact Angle: Controls impact angle options.
- Offset N, Offset E, Offset D: Controls offset inputs to target.
- Target Hardness: Selects target hardness setting.

The Impact Angle consists of two buttons. The first is an On/Off toggle button that provides the ability to turn impact angles on or off. The second button displays the current impact angle that has been entered. Pressing this button brings up Impact Angle Settings window (Figure 1-60), and allows input of desired impact angle. The impact angle will default to 90°. Impact angles of less than 20° are not valid. Impact angle will be disabled for GBU-49 Lot 5 weapons when target type is set to moving.

The Target Offset Settings section has three buttons: Offset North, Offset East and Offset Down. Pressing these buttons brings up the Target Offset Settings window and allows input of desired Offset Settings. Target Offset Default for North, East and Down will be 0 m. The user will be allowed to enter an Offset of between -16384 and +16383 m. If Laser Receiver is turned On, Offset Settings will be ignored and will not allow an input.

The Target Hardness button is used for determining the orientation of the bomb immediately prior to impact. Selecting this button toggles the hardness selection between Soft and Hard. Setting target hardness to soft will ensure the bomb hits with the highest possible accuracy. With hardness set to hard, the bomb will adjust the bomb orientation to align the bomb body axis with the weapon trajectory. This will provide better penetration through hard targets, but accuracy will be slightly decreased. The default hardness setting is “Soft”.

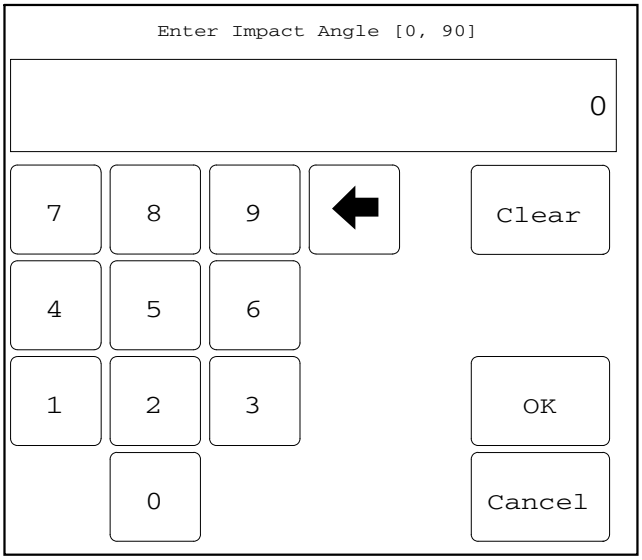


**Figure 1-60. Impact Angle Setting.**



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The bottom section in the Store Settings window is the Fuze Settings section (Figure 1-59). This section displays selected fuze settings that have been input to the selected weapon. The first button in the Fuze Settings section is the Fuze Arm button. This setting enables the user to ARM or SAFE the store. The next three buttons are Fuzing Type, Fuzing Height, and Fuzing Delay which provide the user to adjust the weapon impact time period of detonation on the target. Each button allows to select a specific setting or numerical value as follows:

- Fuzing Type: INST, DELAY, AIRBURST.
- Fuzing Height: -50 – 15 meters.
- Fuzing Delay: 0, 5, 15, 25, 35, 45, 60, 90, 180, 240 ms; 15, 30, 45 min; 1,4, 8, 12, 16, 20, 24 Hr.

When Fuzing Type is set to “INST”, Fuzing Height and Fuzing Delay are disabled. When Fuzing Type is set to “DELAY”, Fuzing Height is disabled. When Fuzing Type is set to “AIRBURST”, Fuzing Height is enabled and Fuzing Delay is disabled.

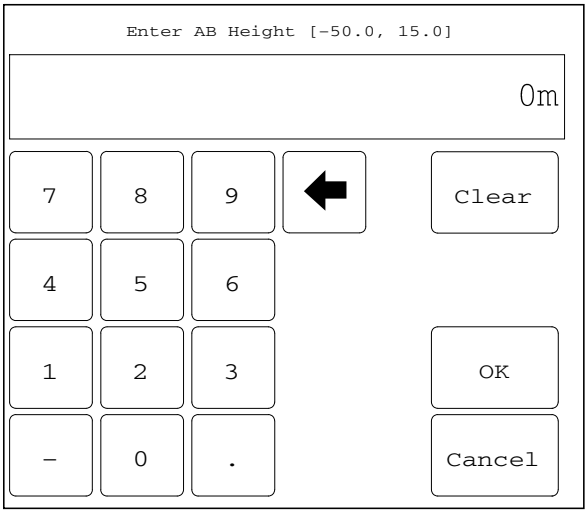


**Figure 1-61. Fuzing Height.**

The fifth button on the Fuze Settings section is the Arm Delay button. The button is used to delay the amount of time between fuze activation (FZU-63/B initiator spun up) and arming. Selecting the button will bring up the Enter Arm Delay keypad (Figure 1-62). This key pad provides 16 selections to choose from to set the arming delay on the FMU-152. The physical settings on the FMU-152 are considered the default settings, but these settings can be overridden using the Arm Delay button. When making selections with the Arm Delay button 14 seconds will be the default setting. Arm times less than 14 seconds could result in aircraft damage due to premature airburst. If an error occurs and the information set into the GSMS is not received or accepted by the FMU-152, the default setting will be the faceplate setting on the FMU-152 set by ground crews.



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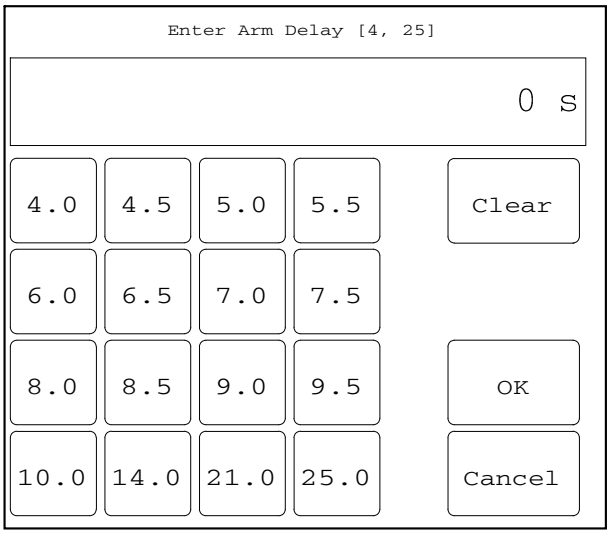


**Figure 1-62. Arm Delay Keypad.**

1.3.3.2.5 JDAM Store Settings.

**NOTE**

- With an elevation value input, switching target elevation source between MSL-84, HAE-84, or MSL-96 will not automatically convert the input value. When switching between elevation sources, verify elevation value and adjust accordingly. Failure to do so may result in incorrect target elevation.
- MSL-84 uses the WGS 84/10 (EGM84) geoid as its reference, HAE-84 uses the WGS-84 ellipsoid as its reference, and MSL-96 uses the EGM-96 geoid as its reference. CCTGP targets have an automatic elevation reference selection of "MSL-96". When DTED is being utilized, the elevation is MSL-96. Manual and store targets have an elevation reference option of "MSL-84", "HAE-84", or "MSL-96".

The JDAM Store Settings window provides the ability to adjust guidance, target, and fuze settings on the JDAM (Figure 1-63 and Figure 1-64). The Guidance Settings section allows the selection of which guidance system the GBU-54 will utilize. The GBU-38 does not provide the ability to select guidance for the weapon. Target Settings give the ability to input specific target information, and Fuze Settings allow the selection of different functions when the JDAM reaches a desired height or impacts a target. The JDAM Store Settings overview screen consists of three sections, which are as follows:

- GBU-54 Guidance Settings
- Target Settings
- Fuze Settings



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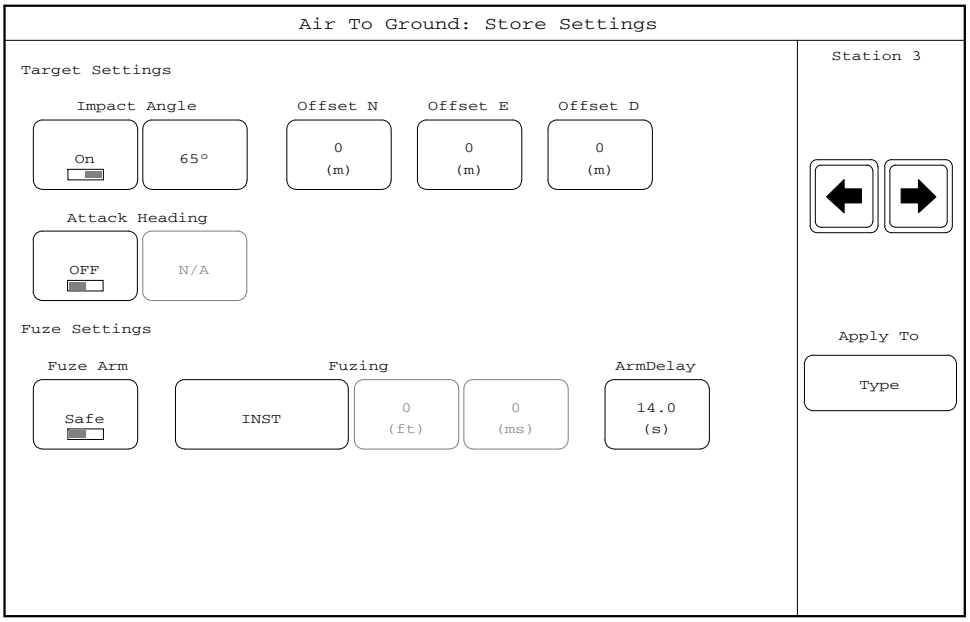


**Figure 1-63. Store Settings for GBU-38.**

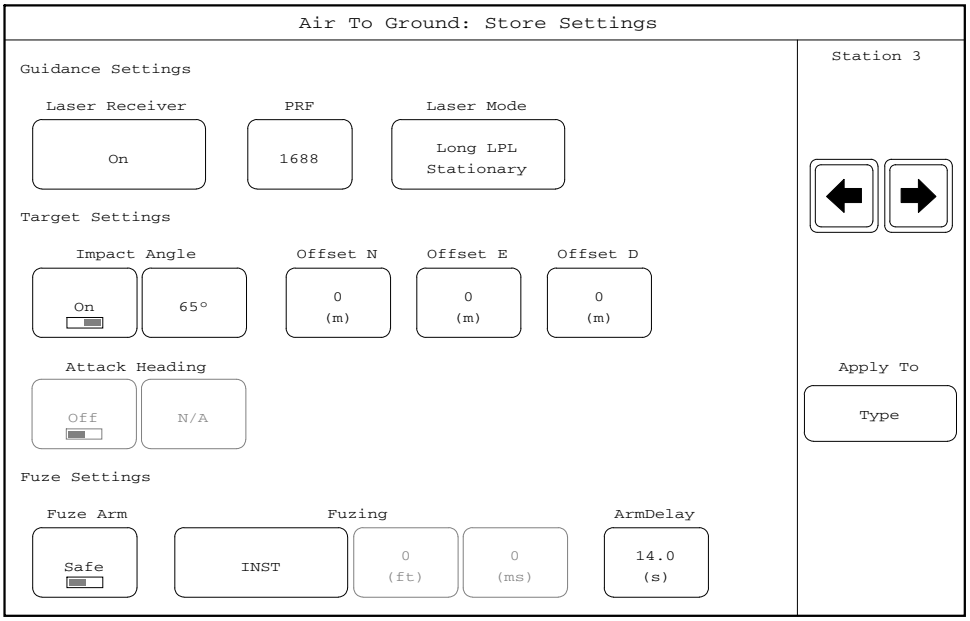


**Figure 1-64. Store Settings for GBU-54.**

GBU-54 The top section in the Store Settings window is the Guidance Settings section. This section displays currently selected guidance settings and provides the option to change guidance settings for GBU-54 (Figure 1-64).

The first button in the Guidance Settings section is Laser Receiver (Figure 1-64). This button allows the user to turn on or off the laser receiver. When set to “On”, the PRF and Laser Mode buttons will be enabled and the store will use both GPS and laser guidance when released from the aircraft. GPS guidance will be the primary source of guidance until the laser seeker detects


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laser energy with the set programmed PRF code. Once the laser energy is detected, GPS will be the secondary source of guidance. When no laser is in use for targeting, the weapon cannot track the programmed PRF code, or laser energy is lost, GPS will be used for guidance until the laser energy is reacquired or weapon impacts. GPS will be the primary source of guidance if the laser receiver button is set to "Off" and disable the PRF and Laser Mode buttons.

Next to the Laser Receiver button is the PRF button. This button displays the currently selected PRF code. When pressed, it will bring up the PRF Code Keypad (Figure 1-51) which provides the ability to set or change the PRF code the selected weapon will track. Load crews will set a PRF code into the weapon ECCG Code Select Switches and is considered the default code for the weapon. If the weapon is required to track a different PRF code, the entered code will be transmitted to the weapon as the required code to track. The PRF code range used is between 1511 and 1788 with a default code of 1688.

The third button in the Guidance Settings section is Laser Mode. This will display a "Long" or "Short" Last Pulse Logic setting and either "Stationary" or "Moving" for target setting. The Last Pulse Logic setting commands the weapon to track the closest or furthest point of laser energy reflection. The Last Pulse Logic can be set to "Short" or "Long". When selecting the target setting, stationary setting should be used against stationary targets and moving setting should be used against moving targets. The default setting for Laser Mode is "Long LPL Stationary". When Laser Mode is set to Long LPL Moving or Short LPL Moving, the Impact Angle range will adjust to be 65 – 90 degrees. When Laser Mode is set to Long LPL Stationary or Short LPL Stationary, the Impact Angle range will adjust to be 30 – 90 degrees.

The middle section of the JDAM Store Settings window is the Target Settings section. The Target Settings section provides options to adjust settings of the weapon to impact a target. The options are as follows:

- Impact Angle: Two buttons that control impact angle options.
- Offset N, Offset E, Offset D: Controls offset inputs to target.
- Attack Heading

Impact Angle consists of two buttons. The first is an On/Off toggle button that provides the ability to turn impact angles on or off. The second button displays the current impact angle that has been entered. Pressing this button brings up Impact Angle Settings window (Figure 1-65), and allows input of desired impact angle. When Release Cue Mode is set to LAR, Impact Angle will default to on. When target type is set to Stationary, the user will be allowed to enter an Impact Angle between 30° – 90°. When guidance is set to GPS, the user will be allowed to enter an Impact Angle of between 30°– 90°. When a target type is set to Moving, the user will be allowed to enter an Impact Angle between 65° – 90° for GBU-54 only.

The Target Offset Settings section has three buttons: Offset North, Offset East and Offset Down. Pressing these buttons brings up the Target Offset Settings window and allows input of desired Offset Settings. Target Offset Default for North, East and Down will be 0 m. The user will be allowed to enter an Offset of between -16384 and +16383 m (Figure 1-66). If Laser Receiver is turned On, Offset Settings will be ignored and will not allow an input.

The Attack Heading consists of two buttons. The first is an On/Off toggle button that provides the ability to turn Attack Heading on or off. The second button displays the current Attack Heading that has been entered. If Laser Receiver is Off, pressing this button (Figure 1-67), will allow input from 1-360 degrees range for Attack Heading. Attack Heading will default to Off, N/A. If Laser Receiver is On, Attack Heading is ignored and cannot be input.

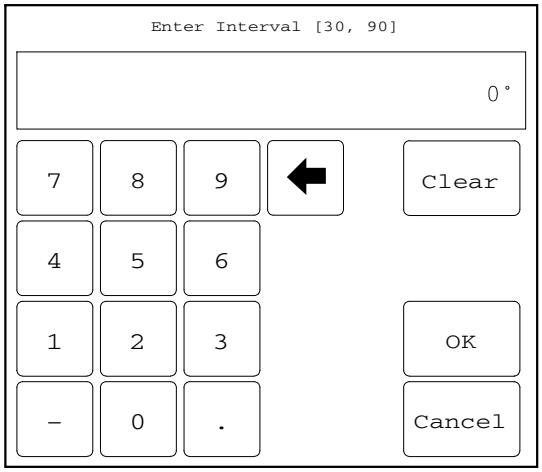


**Figure 1-65. Impact Angle Setting for JDAM – Typical.**



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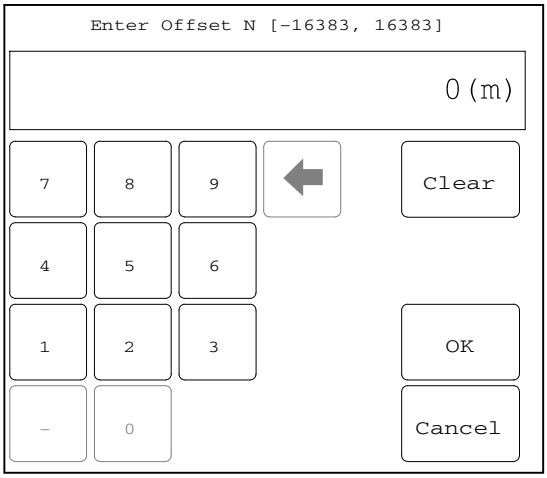


**Figure 1-66. Target Offset Settings for JDAM.**

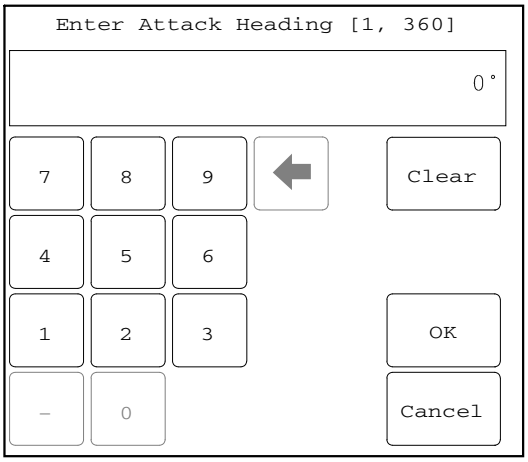


**Figure 1-67. Attack Heading.**

The bottom section in the Store Settings window is the Fuze Settings section (Figure 1-63 and Figure 1-64). This section displays selected fuze settings that have been input to the selected weapon. The first button in the Fuze Settings section is the Fuze Arm button. This setting enables the user to ARM or SAFE the store. The next four buttons are Fuzing Type, Fuzing Height, Fuzing Delay, and Arm Delay which provide the user to adjust the weapon impact time period of detonation on the target. Each button allows to select a specific setting or numerical value as shown in Table 1-6 and Table 1-7.

**Table 1-6. GBU-38 Fuze Settings.**

| Setting | Range, List Options | Default Setting | Range, List Options | Default Setting |
|---|---|---|---|---|
| | FMU-152 | | FMU-139 | |
| Fuze Arm | Safe, Arm | Safe | Safe, Arm | Safe |
| Fuzing Type | INST, DELAY, AIRBURST | INST | INST, DELAY, AIRBURST | INST |
| Fuzing Height | 0, 20 ft. | 0 ft. | 0, 20 ft. | 0 ft. |


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**Table 1-6. GBU-38 Fuze Settings – Continued.**

| Setting | Range, List Options | Default Setting | Range, List Options | Default Setting |
|---|---|---|---|---|
| Fuzing Delay | 0, 5, 15, 25, 35, 45, 60, 90,180, 240 ms; 15, 30, 45 min.; 1, 4, 8, 12, 16, 20, 24 Hr | 0 ms. | 0, 5, 10, 15, 20, 25, 30, 35, 45, 60, 90,180, 240 ms | 0 ms. |
| Arm Delay | 4.0, 4.5, 5.0, 5.5, 6.0, 6.5, 7.0, 7.5, 8.0, 8.5, 9.0, 9.5, 10.0, 14.0, 21.0, 25.0 | 14.0 | 4.0, 4.5, 5.0, 5.5, 6.0, 6.5, 7.0, 7.5, 8.0, 8.5, 9.0, 9.5, 10.0, 14.0, 21.0, 25.0, 30 | 14.0 |

**Table 1-7. GBU-54 Fuze Settings.**

| Setting | Range, List Options | Default Setting | Range, List Options | Default Setting | Range, List Options | Default Setting | Range, List Options | Default Setting |
|---|---|---|---|---|---|---|---|---|
| | FMU-152 (with APS) | | FMU-152 (No APS) | | FMU-139 (with APS) | | FMU-139 (No APS) | |
| Fuze Arm | Safe, Arm | Safe | Safe, Arm | Safe | Safe, Arm | Safe | Safe, Arm | Safe |
| Fuzing Type | INST, DELAY, AIRBURST | INST | INST, DELAY | INST | INST, DELAY, AIRBURST | INST | INST, DELAY | INST |
| Fuzing Height | 0, 10, 20, 30 ft | 0 ft. | 0 ft. | 0 ft. | 0, 10, 20, 30 ft | 0 ft. | 0 ft. | 0 ft. |
| Fuzing Delay | 0, 5, 15, 25, 35, 45, 60, 90,180, 240 ms. 15, 30, 45 min. 1, 4, 8, 12, 16, 20, 24 Hr. | 0 ms. | 0, 5, 15, 25, 35, 45, 60, 90,180, 240 ms. 15, 30, 45 min. 1, 4, 8, 12, 16, 20, 24 Hr. | 0 ms. | 0, 5, 15, 25, 35, 45, 60, 90,180, 240 ms. | 0 ms. | 0, 5, 15, 25, 35, 45, 60, 90,180, 240 ms. | 0 ms. |
| Arm Delay | 4.0, 4.5, 5.0, 5.5, 6.0, 6.5, 7.0, 7.5, 8.0, 8.5, 9.0, 9.5, 10.0, 14.0, 21.0, 25.0 | 14.0 | 4.0, 4.5, 5.0, 5.5, 6.0, 6.5, 7.0, 7.5, 8.0, 8.5, 9.0, 9.5, 10.0, 14.0, 21.0, 25.0, 30 | 14.0 | 4.0, 4.5, 5.0, 5.5, 6.0, 6.5, 7.0, 7.5, 8.0, 8.5, 9.0, 9.5, 10.0, 14.0, 21.0, 25.0, 30.0 | 14.0 | 4.0, 4.5, 5.0, 5.5, 6.0, 6.5, 7.0, 7.5, 8.0, 8.5, 9.0, 9.5, 10.0, 14.0, 21.0, 25.0, 30.0 | 14.0 |

GBU-38 When Fuzing Type is set to “INST”, Fuzing Height and Fuzing Delay are disabled. When Fuzing Type is set to “DELAY”, Fuzing Height is disabled. When Fuzing Type is set to “AIRBURST”, Fuzing Height is forced to 20 ft. and Fuzing Delay is enabled. When Fuzing Type is set to “AIRBURST” for FMU-139 or FMU-152, the Fuzing Delay values are limited to 45 ms or less.

GBU-54 When Fuzing Type is set to “INST”, Fuzing Height and Fuzing Delay are disabled. When set to “DELAY”, Fuzing Height is disabled. If “AIRBURST” is set, Fuzing Height and Fuzing Delay is enabled. For FMU-139/152 is set to “AIRBURST”, the Fuzing Delay values are limited to 45 ms or less. If no Adjustable Proximity Sensor (APS) is present, Fuzing Height is set to 0 ft. and is disabled.

The fifth button in the Fuze Settings section is the Arm Delay button. This button is used to delay the amount of time between fuze activation (FZU-48A/B or FZU-63/B initiator spun up) and arming. Selecting this button will bring up the Enter Arm Delay keypad (Figure 1-68). This key pad provides 16 selections for FMU-152 and 17 selections for FMU-139 to choose from to set



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the arming delay on the FMU-139/152. The physical settings on the FMU-152 or FMU-139 are considered the default settings, but these settings can be overridden using the Arm Delay button. When making selections with the Arm Delay button 14 seconds will be the default setting. Arm times less than 14 seconds may result in aircraft damage due to premature airburst. If an error occurs and the information set into the GSMS is not received or accepted by the FMU-139/152, the default setting will be the faceplate setting on the FMU-139/152 set by ground crews.

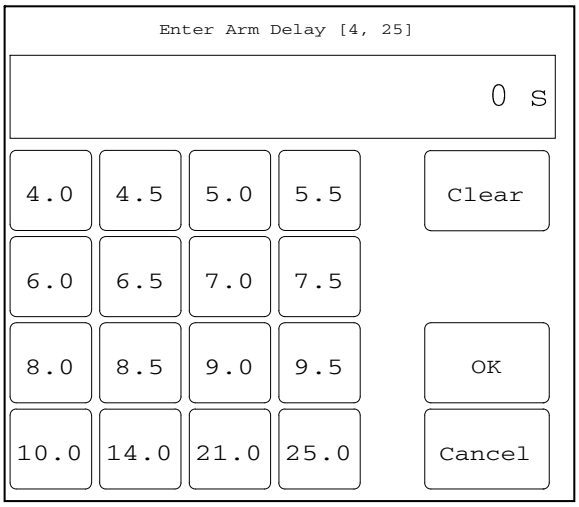


**Figure 1-68. Arm Delay Keypad for JDAM – Typical.**

1.3.3.2.6 LSDB/SDB Store Settings.

**CAUTION**

2411.0.0 Arm delay times shorter than 25 seconds may cause GBU-39/B and GBU-39B/B weapons with Block 15 software to dud.

**NOTE**

2411.0.0 GBU-39/B and GBU-39B/B LAR displays are calculated for release altitudes up to 35,000 feet HAT. Releases above 35,000 feet HAT will not reflect an increased LAR due to release altitude increase.

The Laser Small Diameter Bomb (LSDB) Store Settings window provides the ability to adjust guidance, target, and fuze settings on the LSDB (2410.0.0 Figure 1-69 or 2411.0.0 Figure 1-70). The Guidance Settings section allows the selection of which guidance system the LSDB will utilize. Target Settings give the ability to input specific target information, and Fuze Settings allow the selection of different functions. The LSDB Store Settings overview screen consists of three sections, which are as follows:

- Guidance Settings
- Target Settings
- Fuze Settings

Similar to the LSDB, the Small Diameter Bomb (SDB) Store Settings window provides the ability to adjust target and fuze settings but does not provide any guidance adjustments.



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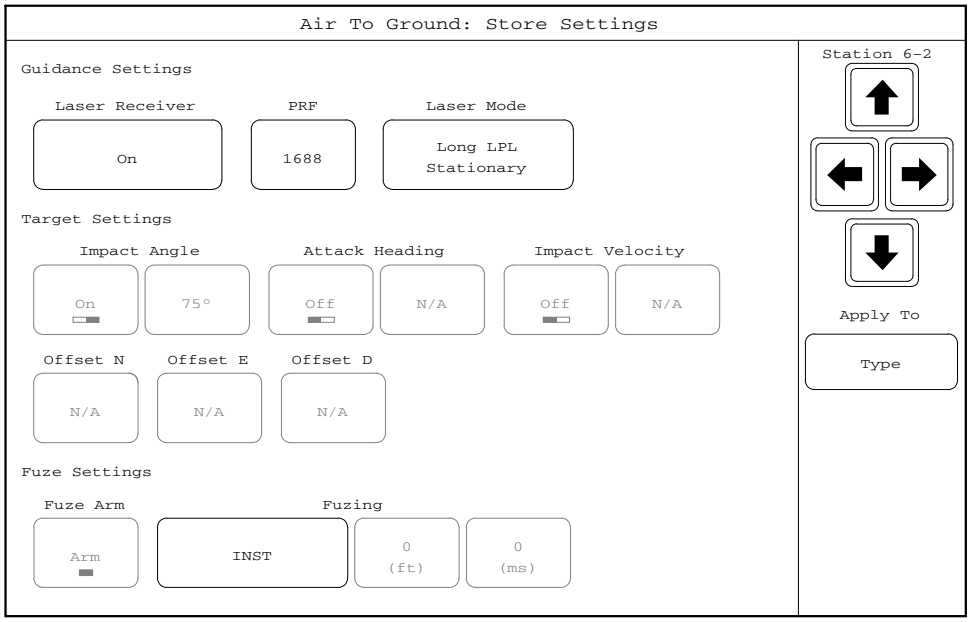


**Figure 1-69.** 2410.0.0 **Store Settings for LSDB.**

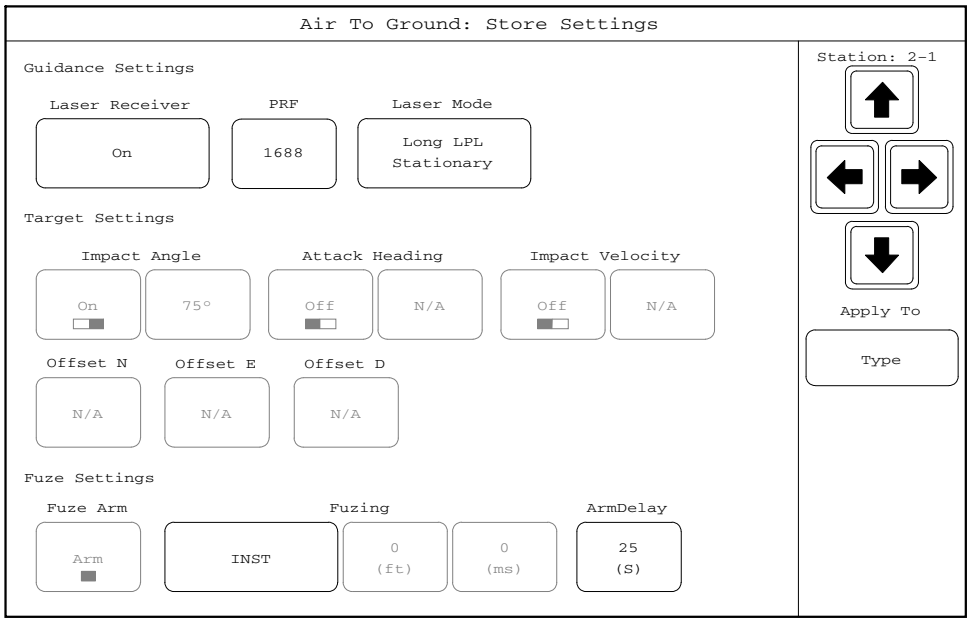


**Figure 1-70.** 2411.0.0 **Store Settings for LSDB.**

The top section in the Store Settings window is the Guidance Settings section. This section displays currently selected guidance settings and provides the option to change guidance settings of LSDB. The first button in the Guidance setting section is the Laser Receiver button which enables the pilot to turn on or off the laser receiver. When the laser receiver is on, the store uses both GPS and laser guidance, but when the laser receiver is off, only GPS guidance is used. When the weapon is programmed to have both guidance systems active, GPS guidance is used as the primary source for guidance until the laser seeker detects laser energy of the programmed PRF code. Once the laser energy is detected, GPS guidance is used as the secondary



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source for guidance. When no laser is in use for targeting, the weapon cannot find the programmed PRF code, or laser energy is lost, GPS guidance will be used until the laser energy is found or weapon impacts.

The second button is the PRF button. This button displays the currently selected PRF code. When pressed, it will bring up the PRF Code Keypad (Figure 1-51), which provides the ability to set or change the PRF code the selected weapon will track. Load crews will set a PRF code into the weapon if the weapon is required to track a different PRF code. The entered code will be transmitted to the weapon as the required code to track. The PRF code range used is between 1111 and 1788 with a default code of 1688, which does not permit unallowable PRF values.

Next to the PRF button is the Laser Mode button. This button displays the last pulse logic and stationary vs. moving target settings. Select stationary when attacking stationary targets and moving when attacking moving targets. The Last Pulse Logic setting commands the weapon to track the closest or the farthest point of laser energy reflection. When the Laser Receiver is set to on, impact angle, attack heading, impact velocity, and target offsets are disabled. When Laser Receiver is set to off, the PRF and Laser Modes are disabled. The default Laser Mode for LSDB is Long LPL Stationary.

The middle section of the Store Settings window is the Target Settings section. The Target Settings section displays currently selected target settings and provides the option to change current Impact Angle, Attack Heading, Impact Velocity and Target Offset options of the selected weapon.

The Impact Angle section has two buttons. The first is an On/Off toggle button that provides the ability to turn Impact Angles on or off. The second button displays the current Impact Angle that has been entered. Pressing this button brings up the Impact Angle Settings window (Figure 1-71) and allows input of desired impact angle. LSDB supports impact angles between 15-85 degrees with a default value of 75 degrees. If Laser Receiver is On, Impact Angle is ignored and cannot be input.

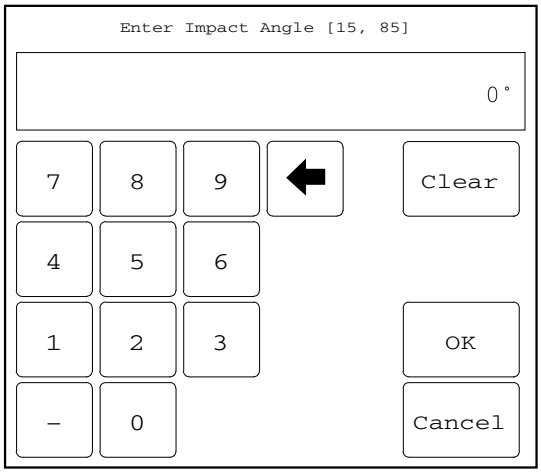


**Figure 1-71. Impact Angle Setting for LSDB.**

The Attack Heading section has two buttons. The first is an On/Off toggle button that provides the ability to turn Attack Heading on or off. The second button displays the current Attack Heading that has been entered. If Laser Receiver is Off, pressing this button (Figure 1-72), will allow input from 0-359 degrees range for Attack Heading. Attack Heading will default to Off, N/A. If Laser Receiver is On, Attack Heading is ignored and cannot be input.



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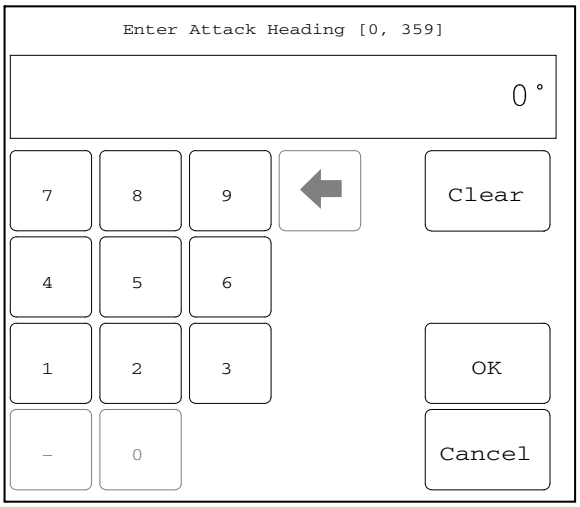


**Figure 1-72. Attack Heading for LSDB.**

The Impact Velocity section has two buttons. The first is an On/Off toggle button that provides the ability to turn Impact Velocity on or off. The second button displays the current Impact Velocity (Figure 1-73). Turning the impact velocity setting to ON will command the weapon to achieve at least 850 ft/s impact velocity. Turning the setting to Off will allow the weapon to impact the target at any impact velocity. The only impact velocity value supported by LSDB is 850 ft/s and cannot be changed by the pilot. Additionally, if Laser Receiver is On, Impact Velocity will be ignored and unable to be turned on.

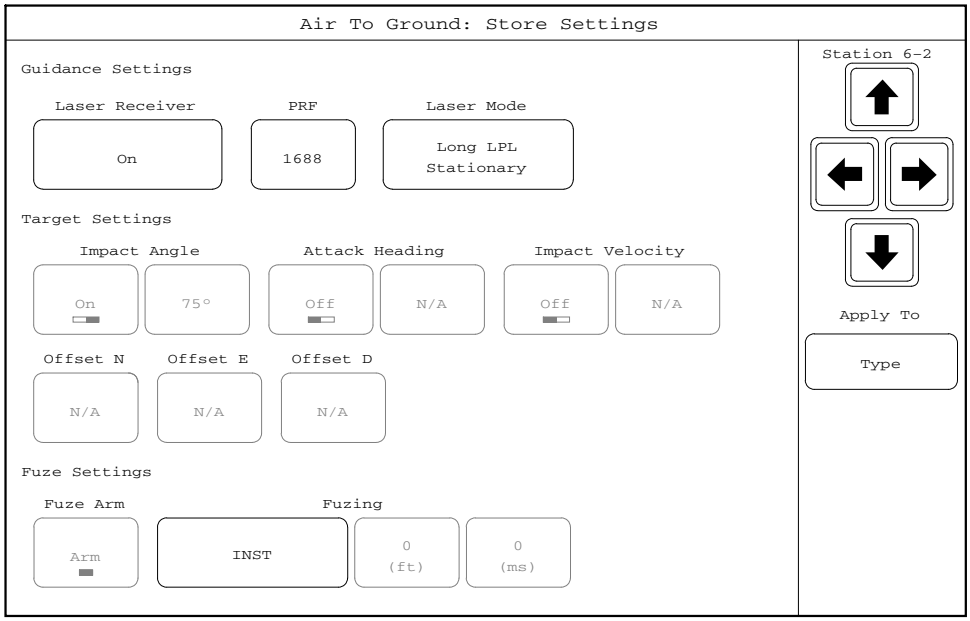


**Figure 1-73. Impact Velocity Setting for LSDB.**

The Target Offset Settings section has three buttons: Offset N, Offset E, and Offset D. Pressing these buttons brings up the Target Offset Settings window (Figure 1-74), and allows input of desired Offset Settings. Target Offset Default for N, E, and D will be 0 m. The user will be allowed to enter an Offset of between -16736 and +16735 m. If Laser Receiver is turned On, Offset Settings will be ignored and will not allow an input.



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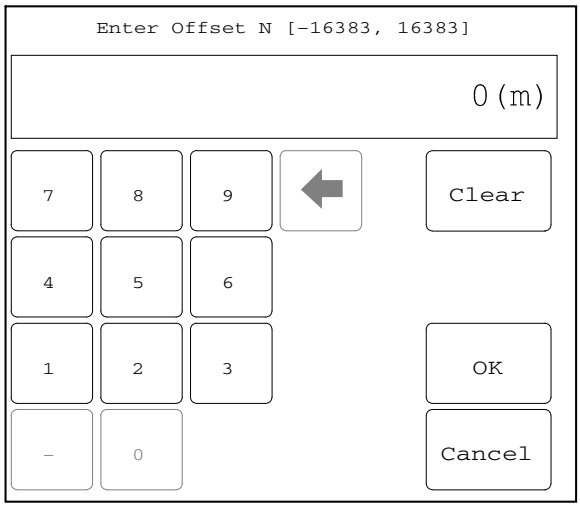


**Figure 1-74. Target Offset Settings for LSDB.**

The bottom section in the Store Settings window is the Fuze Settings section (Figure 1-75). This section displays selected fuze settings that have been input to the selected weapon with instantaneous remaining the default setting. The first button in the Fuze Settings section is the Fuze Arm button. This setting enables pilots to ARM or SAFE the store. Fuze Arm is always set to Arm and greyed out for LSDB because LSDB does not have fuze arming lanyards. The next button in the Fuze Settings section is the Function Mode button. The Function Mode button allows selection of the weapons specific fuzing modes. Selecting the Function Mode button will bring up the Fuze Function Modes (Figure 1-57). The three options are as follows:

- Instantaneous: Height 0 m and Delay 0 ms. Disabled height and delay buttons.
- Delay: Height 0 m. Delay setting options are 0, 5, 15, 25, 35, 45, and 60 ms. Default is 0 ms.
- Airburst: Height 7ft and 14ft.

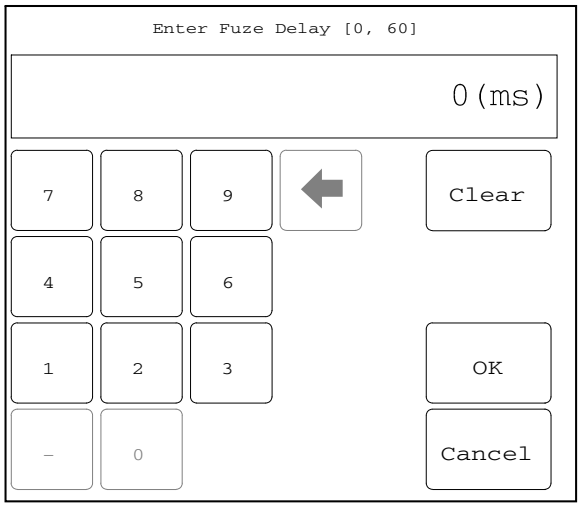


**Figure 1-75. Fuze Setting for LSDB.**

2411.0.0 The last button on the right of the Fuze Settings section is the Arm Delay button (2411.0.0 Figure 1-70). The Arm Delay button is used to delay the amount of time to arm the weapon once released from the aircraft. The available arm delay times are as follows for Block 21 SDB/LSDB weapons:

- 14 seconds


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- 21 seconds
- 25 seconds (default)

The GSMS defaults the arm delay time for all SDB/LSDB variants to 25 seconds. When SDB/LSDB weapons are inventoried, there are no indicators to know which block version of the weapon is inventoried. If a Pre-Block 21 SDB/LSDB weapon is inventoried, the weapon will only support a 14 second arm delay regardless of what is selected by the pilot on the Store Settings page. MQ-9 crews should coordinate with appropriate weapons loading crew to determine the block variant available prior to flight.

1.3.3.3 Select Target.

**NOTE**

- With an elevation value input, switching target elevation source between MSL-84, HAE-84, or MSL-96 will not automatically convert the input value. When switching between elevation sources, it is necessary to verify elevation value and adjust accordingly. Failure to do so may result in incorrect target elevation.
- MSL-84 uses the WGS 84/10 (EGM84) geoid as its reference, HAE-84 uses the WGS-84 ellipsoid as its reference, and MSL-96 uses the EGM-96 geoid as its reference. CCTGP targets have an automatic elevation reference selection of "MSL-96". When DTED is being utilized, the elevation is MSL-96. Manual and store targets have an elevation reference option of "MSL-84", "HAE-84", or "MSL-96".

The Select Target screen (Figure 1-76) appears when the Select Target control button is depressed while in the Air to Ground Mode. Select Target provides the pilot the ability to select a predetermined target from a list or create a new target based on known coordinates, MTS generated coordinates, or Synthetic Aperature Radar (SAR) generated coordinates.

The Page Section can contain up to two groups of targets. The top portion is the Cross-Cued Target. This target is determined by coordinates provided by the MTS.

The lower group of targets are the Saved Targets. These targets have been created and saved through the target creation process or through the SAR target creation process. To the right of the saved targets are a set of vertical arrows that permit scrolling through the saved target list. Below the target list is the Create Target Button.

Each target entry in either of the target lists contains the following data:

- Target Name
- Target Latitude
- Target Longitude
- Target Source
- Target Elevation
- Target Elevation Source: MSL-84, HAE-84, MSL-96

Selecting any of the targets causes an animated dashed outline to be present around the target. Target selection may be done at any time during the store release setup process (Figure 1-76).

2411.0.0 The Select Target page "Select" buttons have been increased in size to accommodate up to eight munitions getting displayed on the button after assignment to a target. The following two figures show the feedback once assigned to a store(s) ( 2411.0.0 Figure 1-77 and 2411.0.0 Figure 1-78).



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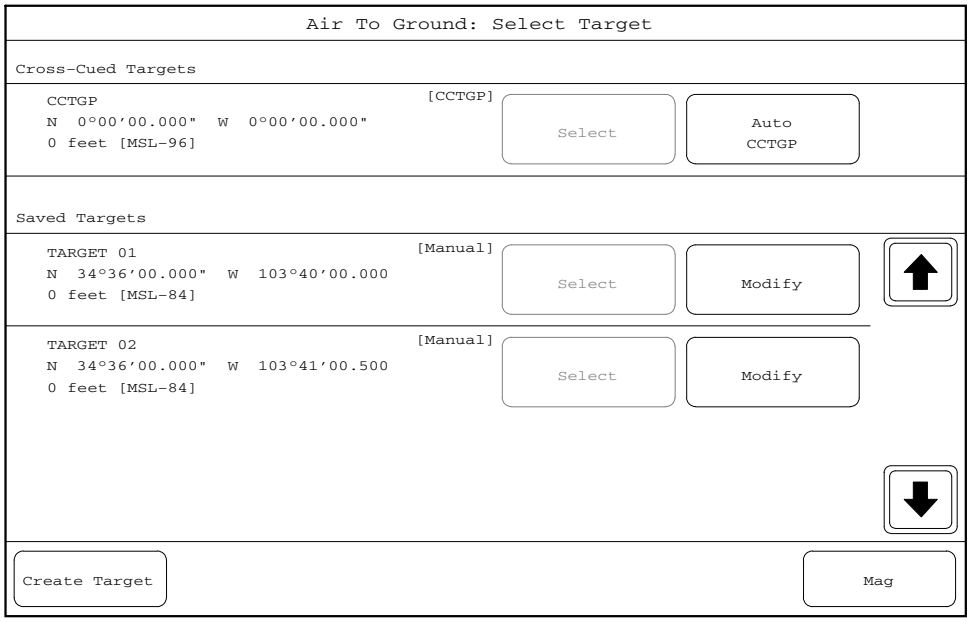


**Figure 1-76. Target Selection Screen.**

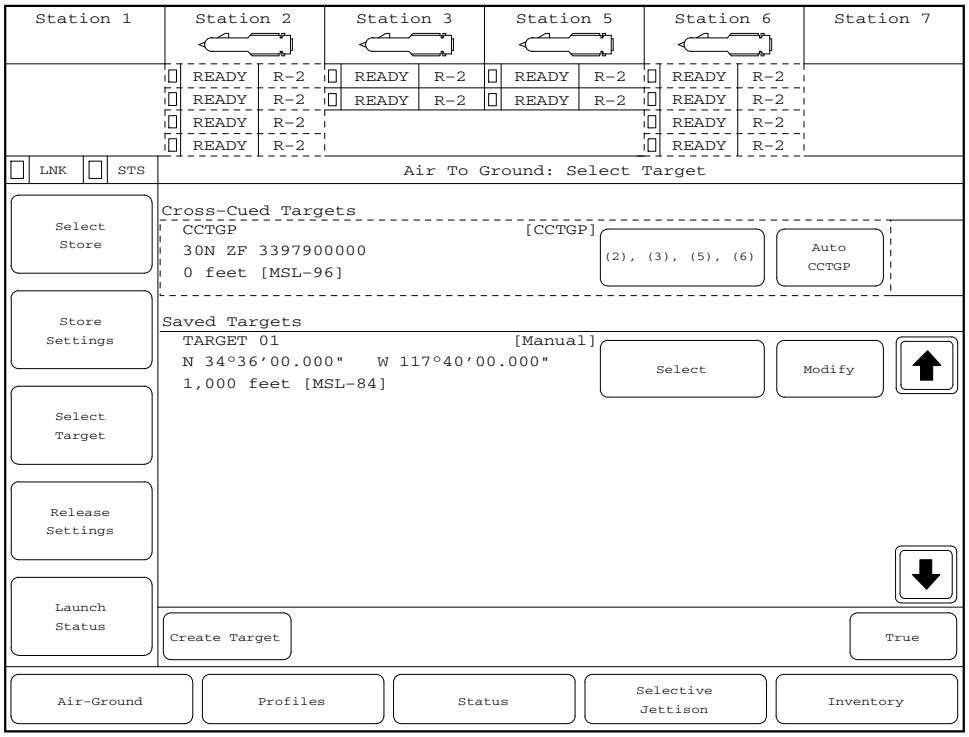


**Figure 1-77.** 2411.0.0 **Target Selection Screen - Single Target.**



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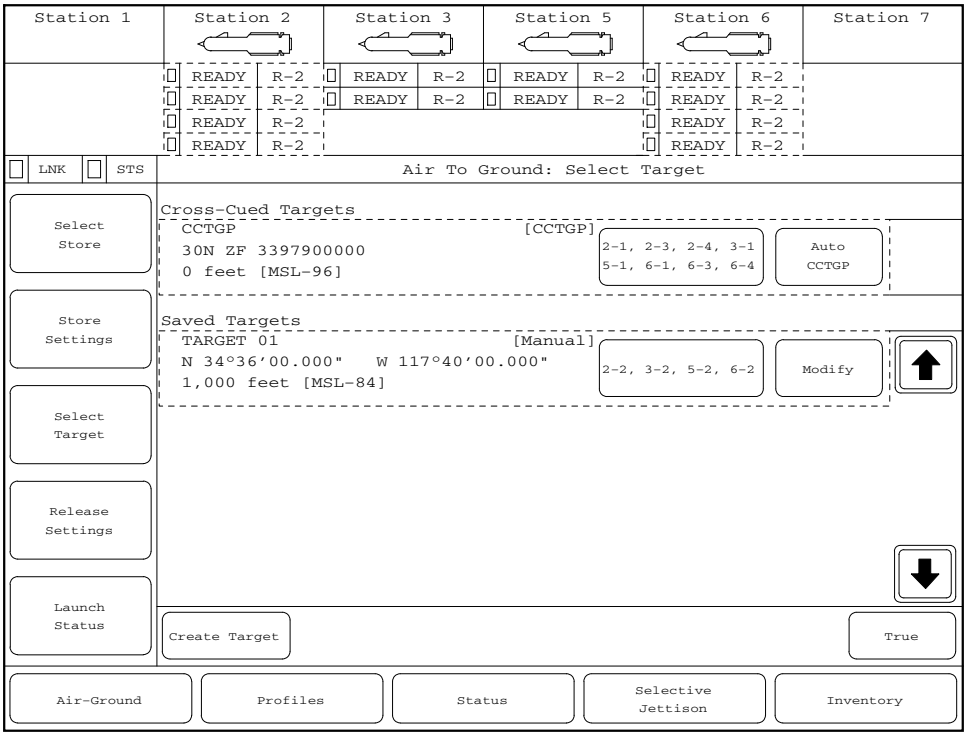


**Figure 1-78.** 2411.0.0 **Target Selection Screen - Multi-Target.**

The Cross-Cued Targets list only contains one target. The Cross Cued Target name will be identified as CCTGP and will have two target location updating options. Those options are Select Auto Mode and Select Manual Mode and can be turned on or off by selecting the appropriate button (Figure 1-76). The selection buttons are gray and will become blue when selected. Selecting the Select Auto Mode button will select the cross-cued target, and the target information will be updated with the MTS crosshair coordinates at a rate of 1 Hz. Pressing the Select Auto Mode button again will result in no targeting mode selected and no further target updates (the buttons are gray). Selecting the Select Manual Mode button will select the cross-cued target and update the target information once with the current MTS crosshair coordinates. If another update is required, press the Select Manual Mode button again to update the target coordinates. In either update case, the selection of the button causes a target update

Located on the lower right corner of the Select Target screen is a global reference heading toggle button (Figure 1-76). This toggle button allows the user to set global heading to “Mag” for magnetic north or “True” for true north (Figure 1-79 and Figure 1-80). The default reference type is set to “True”. The selection of magnetic north does not affect the numerical input value of the Hellfire Impact Heading, GBU-38 Impact Azimuth, or Run-In Course. The effect of changing the reference type will only be noticed on the HUD and Tracker when the magnetic offset preset value is not equal to zero. The GBU-38 Impact Azimuth will always be converted to true north prior to sending it to the weapon and CWL interface.



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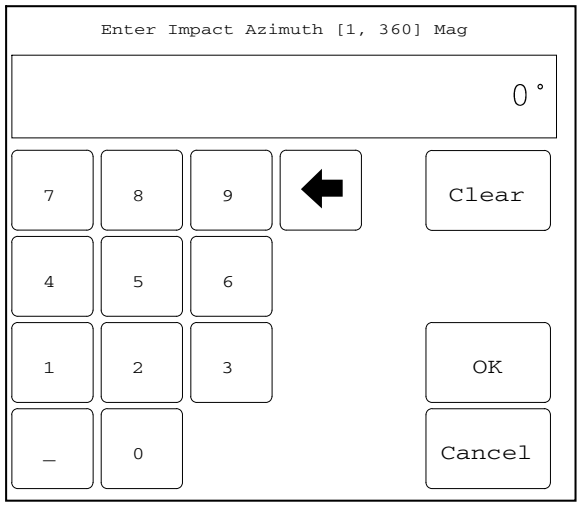


**Figure 1-79. Impact Azimuth Input – Global Heading Set to Mag.**

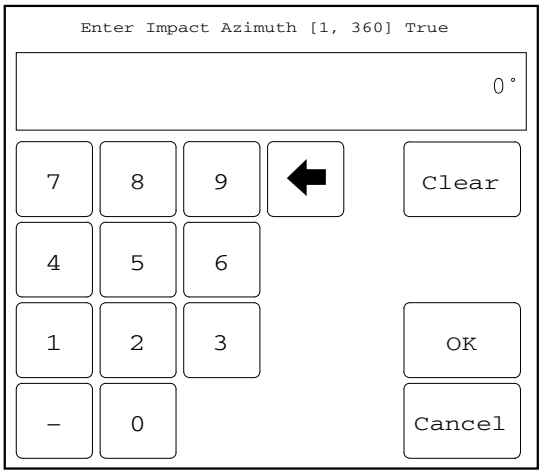


**Figure 1-80. Impact Azimuth Input – Global Heading Set to True.**

The Auto CCTGP and Manual CCTGP buttons have been combined into one toggle button, and the user can choose which mode to use by depressing the button to the appropriate mode. To the left of the Auto CCTGP and Manual CCTGP toggle button is a Select button (Figure 1-76). By pressing the Select button, a store menu window will appear allowing the user to assign individual stores to a target (Figure 1-81). After a store is selected for a target, the Select button will update to show all the stores assigned to that target. Pressing the Select All button will select all available stores to the target and close the store menu automatically. When both stores on station are selected for the same target, that station number will be displayed with a parenthesis around the station number. For all other cases, the individual store is indicated (Figure 1-82 or Figure 1-83).


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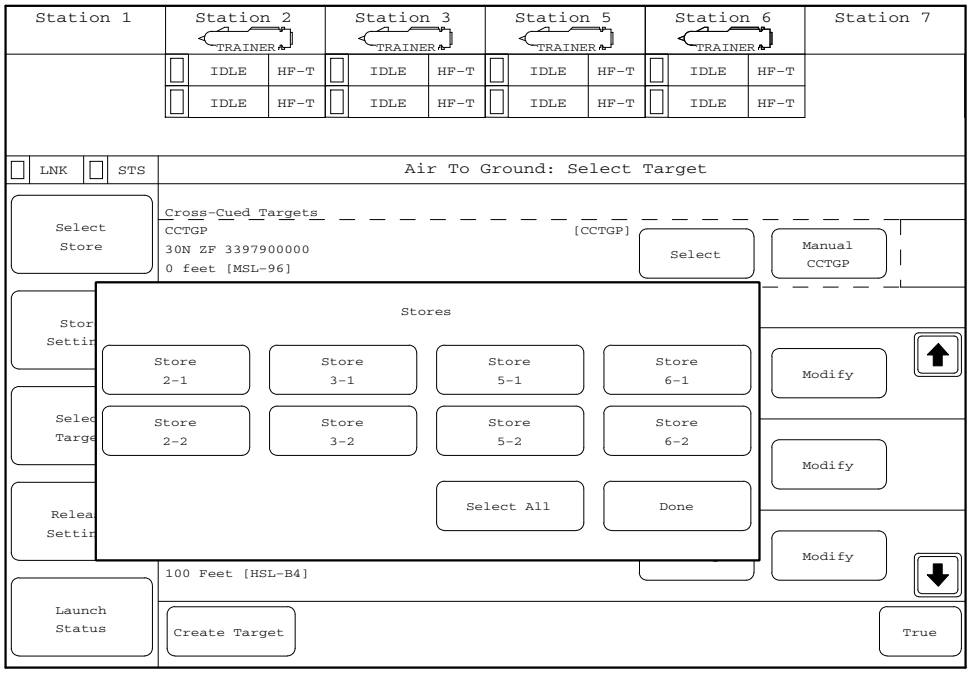


**Figure 1-81. Select Target Page Store Menu.**

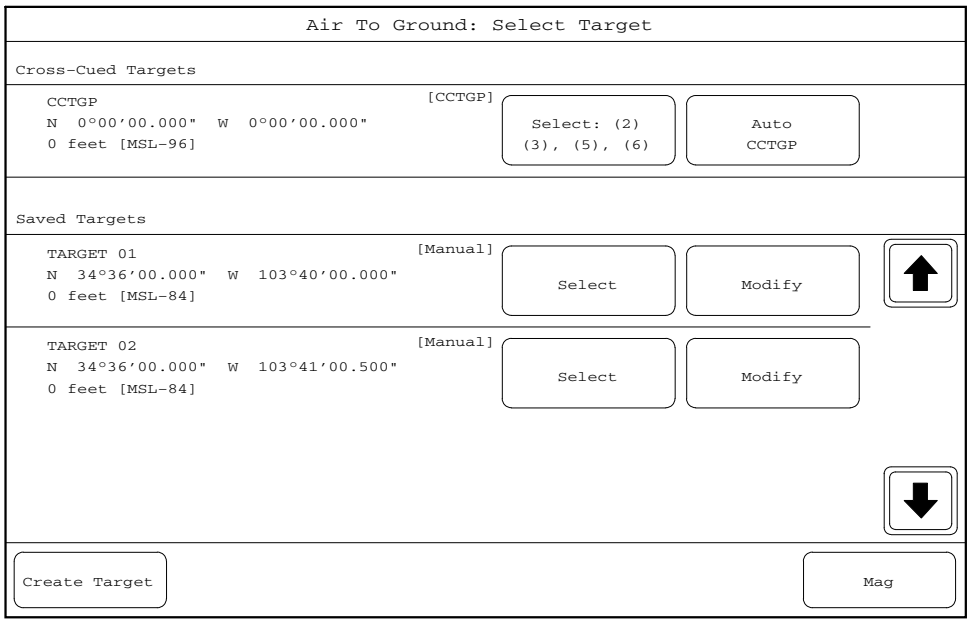


**Figure 1-82. Select Target Page with All Stores Selected.**



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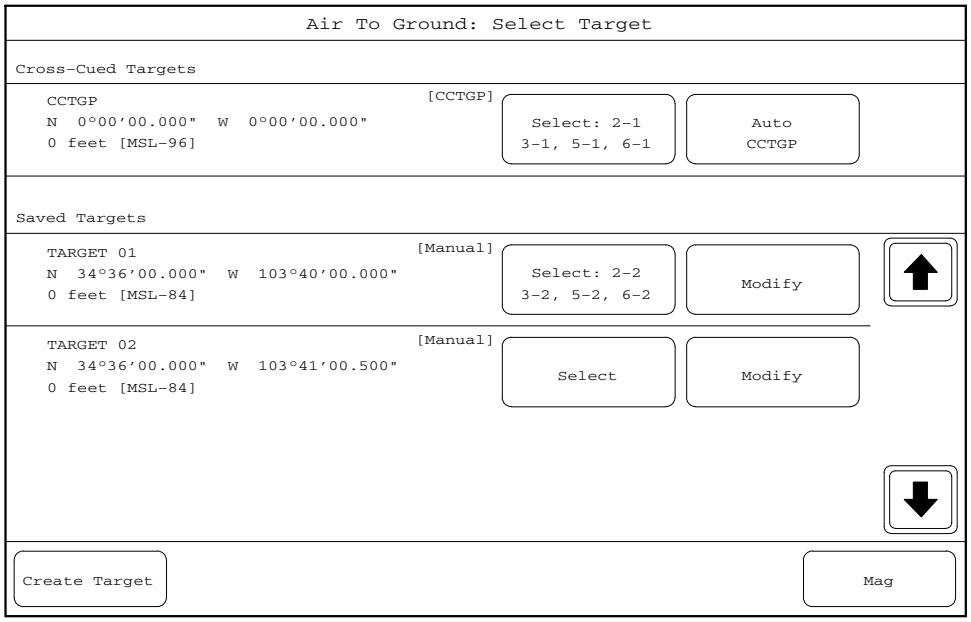


**Figure 1-83. Select Target Page with Individual Stores Selected.**

The saved targets list consists of three types of targets, manually created, TGP created, and SAR created targets. Manually created targets are targets that have been manually entered by the pilot through the Create Target button. TGP created targets are entered by the pilot with the TGP supplying the targeting information via an update. The pilot can then update the target name, as required. SAR created targets are provided by the Multi-function Workstation (MFW) in the GCS through the use of Control of Lynx and Analysis Workstation (CLAW)software and sent to the Saved Targets list. The pilot can update the target name, as required.

Each saved target will have Select and Modify button options. The Select button on a target's row is used to select that target.

The Modify button allows the user to modify some attributes of the targeting information and brings up the Modify Target screen (Figure 1-84). For manually created targets, all inputs are available for update except for the source option. To exit this screen once changes have been made or without making any changes, select the Modify & Exit button. To delete the target, select the Delete & Exit button.



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**NOTE**

- When modifying or creating a target, ensure the selected keyboard is fully populated prior to inputting coordinates or elevation. Failure to wait until the keyboard is fully populated may cause the GSMS to lock up and the GCS will need to be rebooted to regain GSMS operability.
- The hemisphere letter of the latitude and longitude variant formats (DD MM' SS.sss" and DD MM.mmmmm') and the zone number and zone letter in UTM or MGRS shall be auto-populated based on aircraft location when the user is entering a new coordinate location when the GSMS is connect to the aircraft. Care must be taken when entering coordinates as failure to do so may result in an unintended location entry due to the auto-population feature.
- The Hellfire does not use the target sent from the GSMS; it gets its targeting parameters directly from the MTS. The target icon on the tracker display and all release cues in the GCS use the target coordinates from the GSMS.
- Cross-cued target auto mode should be disabled prior to verifying weapons and not be enabled until verification has been completed. Not disabling auto mode may result in an "SCC Software Fault Detected" caution on the Warnings screen.

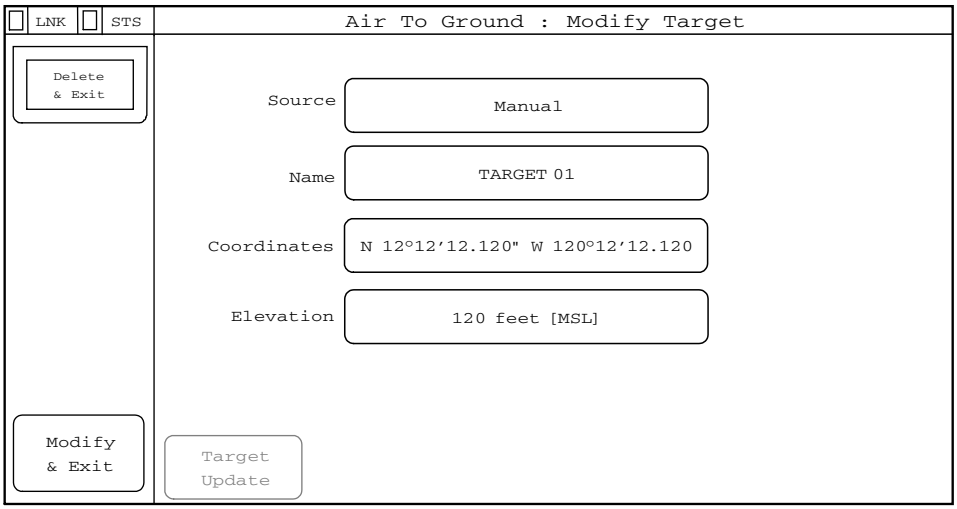


**Figure 1-84. Modify Target Screen.**

1.3.3.3.1 Create Target.

**NOTE**

When entering a UTM coordinate, verify that the entered coordinate is within the boundaries of the intended grid zone. If the coordinate exceeds the grid zone boundaries, and the coordinate format is changed to Lat/Long or MGRS and then back to UTM, then software may change the designation of the entered coordinate to a different zone, easting and northing than originally entered.

The Create Target screen is used to create new saved targets. It is displayed when the Create Target button, located on the Select Target screen, is selected. The Control Area of the Create Target screen (Figure 1-85) contains (from top-to-bottom) a guarded Delete & Exit button and a Create & Exit button. The Page Section of the Create Target screen contains a series of buttons for target attributes.

- Source
- Name
- Coordinates
- Elevation

The lower portion of the Page Section of the Create Target screen contains the Target Update button.

As each target attribute is set, its value will appear on the corresponding button. Before the target can be selected for weapons employment coordinates and elevation inputs must be entered. When all elements of Target Creation have been completed, touching the Control Area's Create & Exit button will add this target to the list of available targets. Touching the Control Area's Delete & Exit button will void any created information and take the user back to the Select Target Screen.



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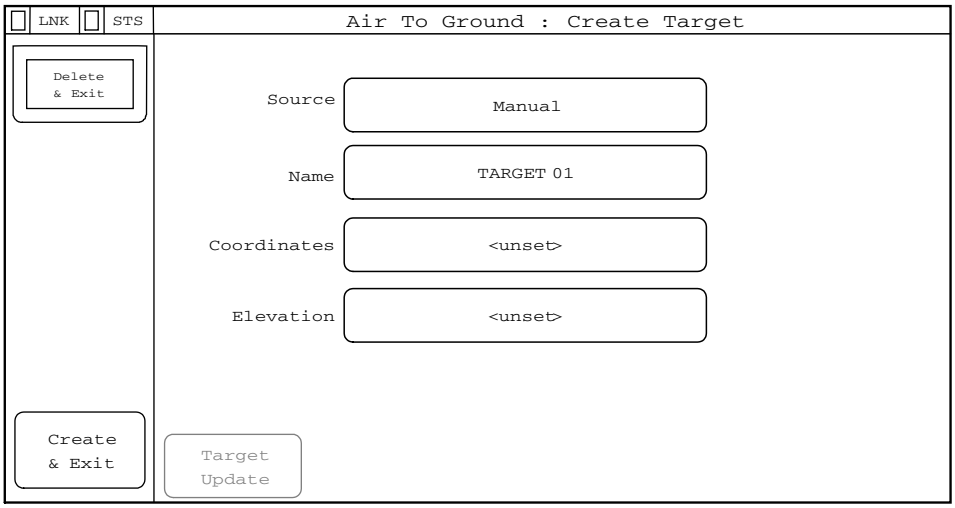


**Figure 1-85. Create Target Screen.**

1.3.3.3.2 Target Source and Name. The Source button allows the user to select the source of the target location information. Selecting the Source button will bring up the Choose Target Source screen (Figure 1-86) allowing the user to choose between Manual (default), or TGP. If TGP is selected, target location fields will be set to the current MTS target position once the Target Update button is pressed. The manual option will allow the manual input of target location information.

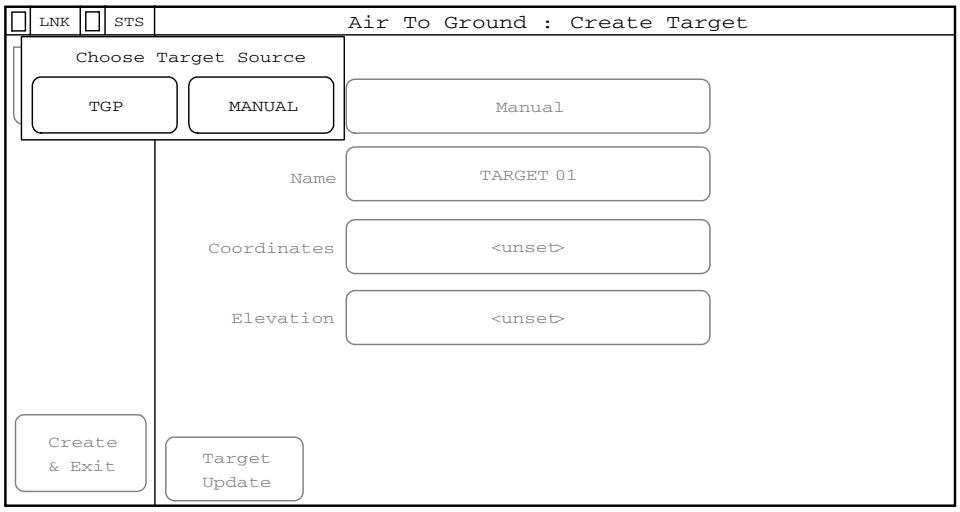


**Figure 1-86. Choose Target Source Screen.**

The Name button allows the user to name the target. The default value of this field is “Target 01” and subsequent. Selecting the Name button will bring up a keyboard taking up the Page and Control Area (Figure 1-87) allowing a desired target name to be set. Once a name has been input, selecting OK will close the keyboard. The system will not allow the Target name to be CCTGP, nor will it allow a blank name space to be saved. Of the targets inputted, the map tracker will only display the first eight targets.


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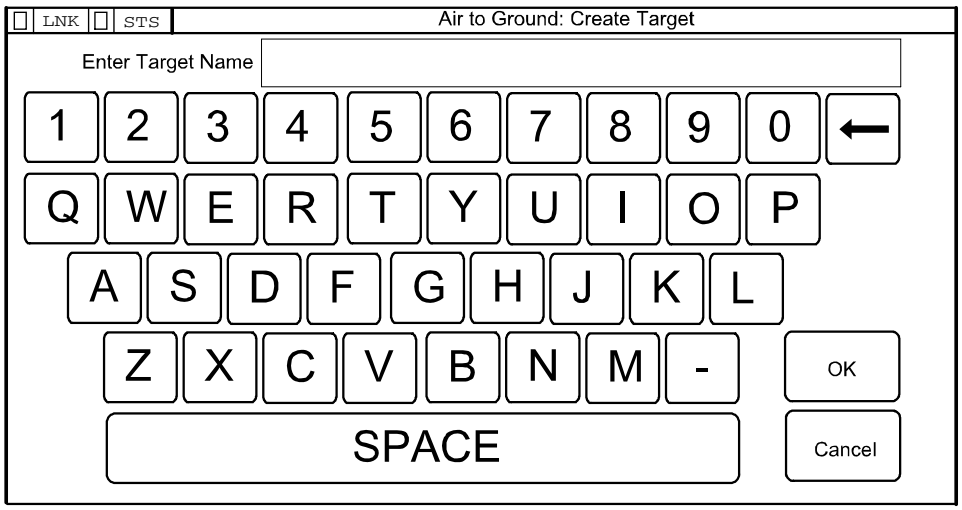


**Figure 1-87. Target Name Input.**

1.3.3.3.3 Target Coordinates and Elevation.

**NOTE**

- When entering a UTM coordinate, verify that the entered coordinate is within the boundaries of the intended grid zone. If the coordinate exceeds the grid zone boundaries, and the coordinate format is changed to Lat/Long or MGRS and then back to UTM, then software may change the designation of the entered coordinate to a different zone, easting and northing than originally entered.
- With an elevation value input, switching target elevation source between MSL-84, HAE-84, or MSL-96 will not automatically convert the input value. When switching between elevation sources, it is necessary to verify elevation value and adjust accordingly. Failure to do so may result in incorrect target elevation.
- MSL-84 uses the WGS 84/10 (EGM84) geoid as its reference, HAE-84 uses the WGS-84 ellipsoid as its reference, and MSL-96 uses the EGM-96 geoid as its reference. CCTGP targets have an automatic elevation reference selection of "MSL-96". When DTED is being utilized, the elevation is MSL-96. Manual and store targets have an elevation reference option of "MSL-84", "HAE-84", or "MSL-96".

The Coordinates and Elevation buttons allow the user to input target location information when Manual is selected as the target source. Selecting the Coordinates button will bring up the Coordinate Keyboard window (Figure 1-88). This keyboard is used to enter the desired target coordinate information in the following formats:

- Latitude and longitude coordinate system in DD° MM' SS.sss" format
- Latitude and longitude coordinate system in DD°MM.mmmmm" format
- Universal Transverse Mercator (UTM) coordinate system
- Military Grid Reference System (MGRS) coordinate system



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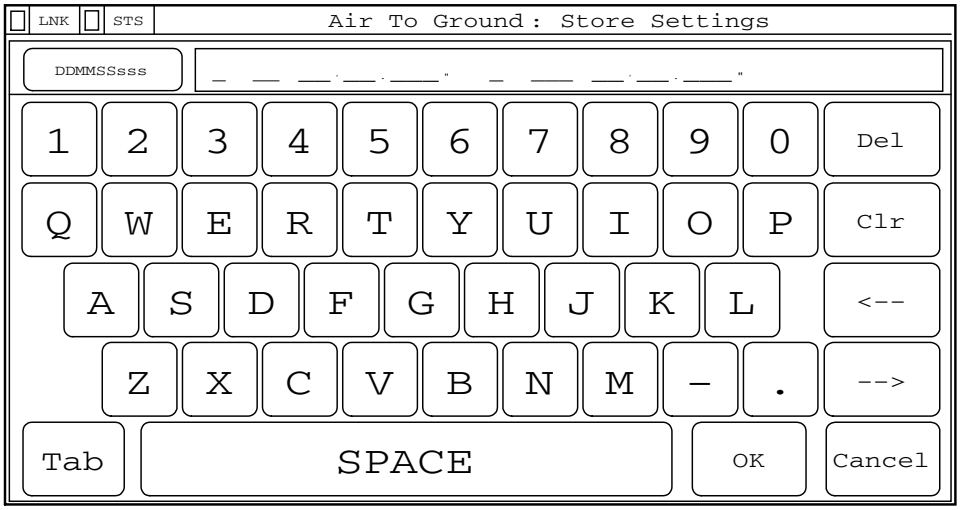


**Figure 1-88. Coordinate Keyboard Window.**

The information entered will be displayed in the coordinate bar at the top of the keyboard screen. Selecting the Coordinate System button in the top left corner of the keyboard screen will display the Choose Coordinate System window (Figure 1-89). This window allows the user to select the desired coordinate system format. At any time prior to, during, or after entering the desired coordinates, the coordinate system may be changed. When changing coordinate formats, the system will automatically convert the previously entered coordinates to the new format. If the entered coordinates are not valid or do not translate to another coordinate system, a Coordinate Validation Error window will populate the screen and the coordinates will have to be verified and adjusted.

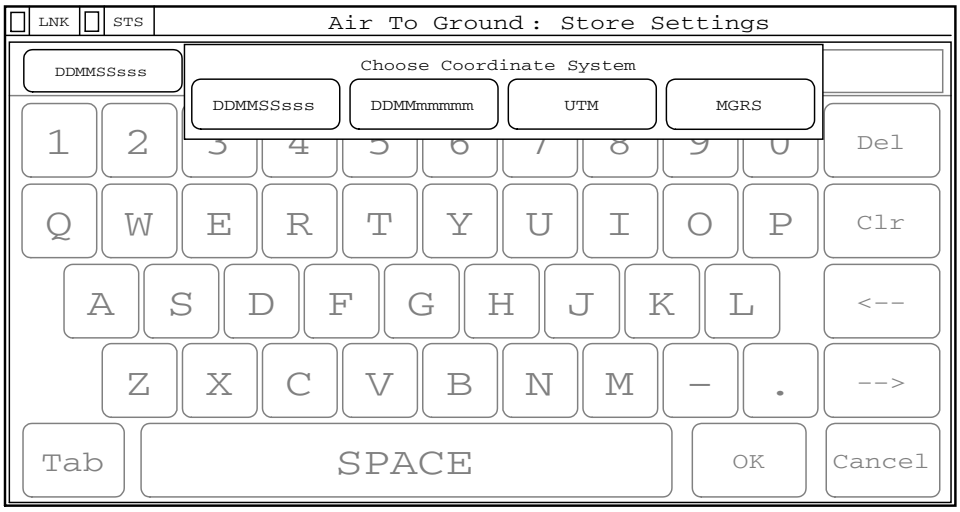


**Figure 1-89. Coordinate System Selection Window.**

Selecting the Elevation button will bring up the Elevation Keyboard window (Figure 1-90). This window is used to enter the desired target elevation. The information entered will be displayed in the input elevation bar at the top of the keyboard screen. The distance increment can be changed between feet and meters by selecting the Increment button to the right of the input elevation bar. The elevation source can be changed between MSL-84 and HAE-84.


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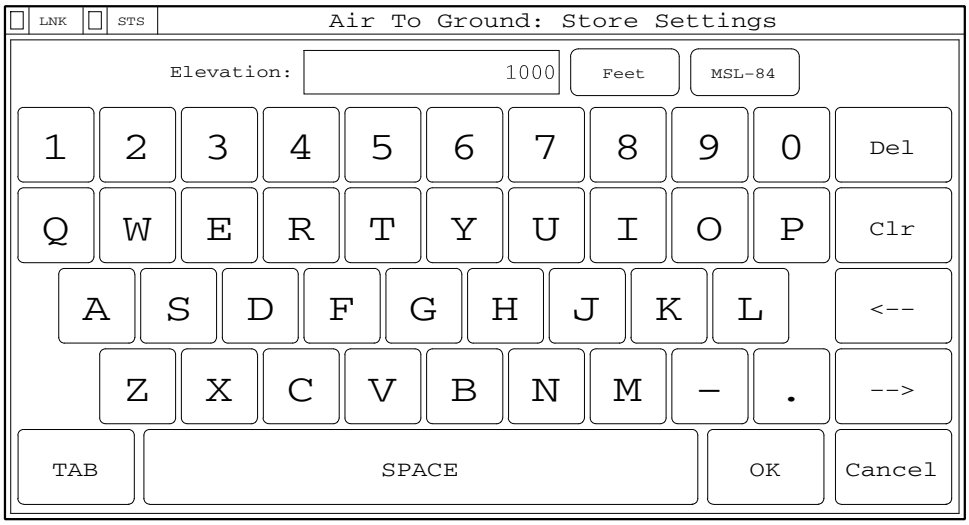


**Figure 1-90. Elevation Keyboard Window – Typical.**

1.3.3.3.4 SAR Target. SAR Targets are created through the use of specialized software working on three images taken during a three dimensional targeting scan. This results in a precision set of target coordinates and elevation. These coordinates and elevation are sent to the SMS target list and tagged with SAR as the target source with a default name, unless changed in the CLAW software prior to sending to the SMS. The pilot selects the target in the same manner as all other targets but can only modify the name and no other attributes. All SAR target elevation source information is depicted in the HAE elevation standard.

1.3.3.4 Release Settings.

**NOTE**

- GBU-38 and GBU-54 TOFs may vary even with similar settings applied. Use mission planning tools accordingly.
- OWT is not currently supported with UAI.

The Air to Ground mode Release Settings screen appears when the Release Settings Control button is selected in Air to Ground Mode (Figure 1-91). The Release Settings screen allows the pilot to configure the Tracker graphics and how the weapon will be released. Release Settings screen provides three separate sections listed below:

- Run-in Settings – Provides Mode, Course, and Time to Release (TTR) Type options.
- Tactical Settings – Release Cue Mode and Target Type options.
- Staple Settings – Contains Minimum Distance, Desired Distance, Maximum Distance, Time of Flight entry, Impact Heading, Range, Type and Staple Transition options.

Release Settings allow the user to choose between Manual Release Cue Mode or the WEZ, CCRP or LAR modes and will display the applicable weapon cues on the tracker and HUD. Selecting the TTR Type button will toggle the option between DPI and Tangent. In DPI mode (Figure 1-93 or Figure 1-94), the TTR displays the total time required to, if necessary, turn the aircraft for a course to intercept the DPI plus the travel time to intercept the RDL. In Tangent mode (Figure 1-93 or Figure 1-94), the TTR displays the total time required to, if necessary, turn the aircraft for a course to intercept the nearest tangent point on the circle defined by sweeping the RDL about the DPI plus the travel time to intercept. The TTR will display 0:00 when inside the RDL. The TTR will be calculated using the turn direction, if necessary, that will result in the lowest value.

For each UAI store type, a unique CDF determines what release settings are applicable. The setting can be a list option or a range value. These settings match the legacy weapon release settings page and apply to all UAI weapons. One exception is that the GBU12 Release Cue mode can be CCRP or Manual with a default of CCRP. Another exception is that the SDB max range is doubled to be 50 NM's instead of 25NM. The UAI Release Settings screen (Figure 1-92) provides the following:

- Run-in Settings – Provides Mode, Course and TTR type.
- Tactical Settings – Release Cue Mode.
- Staple Settings – Contains Minimum Distance, Desired Distance Enable, Desired Distance, Maximum Distance, Time of Flight, Staple Transition.

The following are differences for UAI weapons:

- GBU-12 Release Cue mode can be CCRP or Manual with a default of CCRP.
- The SDB maximum range is 50.0 NM.



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- Target type is selected on the weapon's store settings page.
- OWT is not currently supported with UAI.

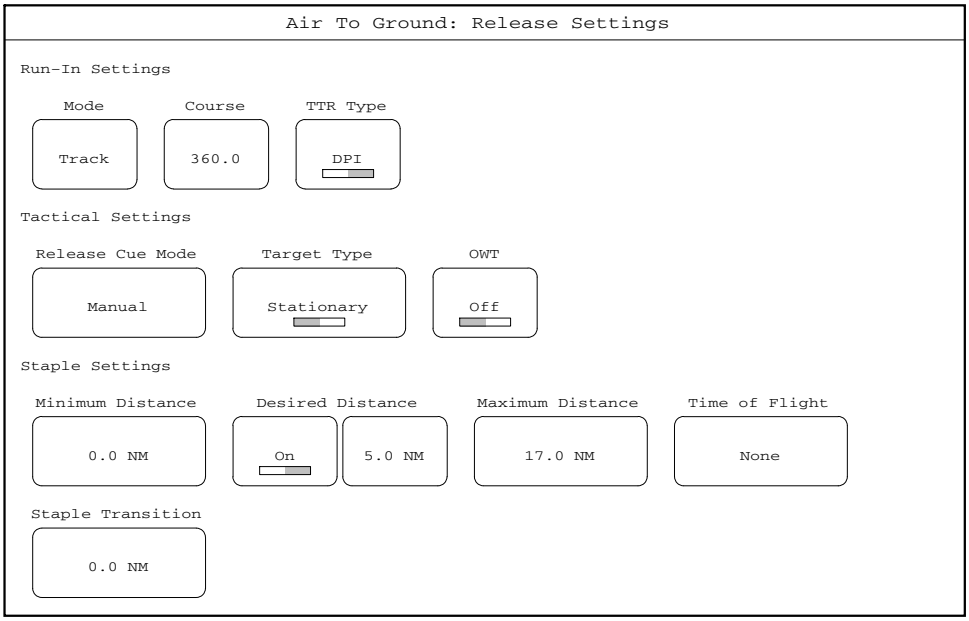


**Figure 1-91. Release Settings – Typical.**

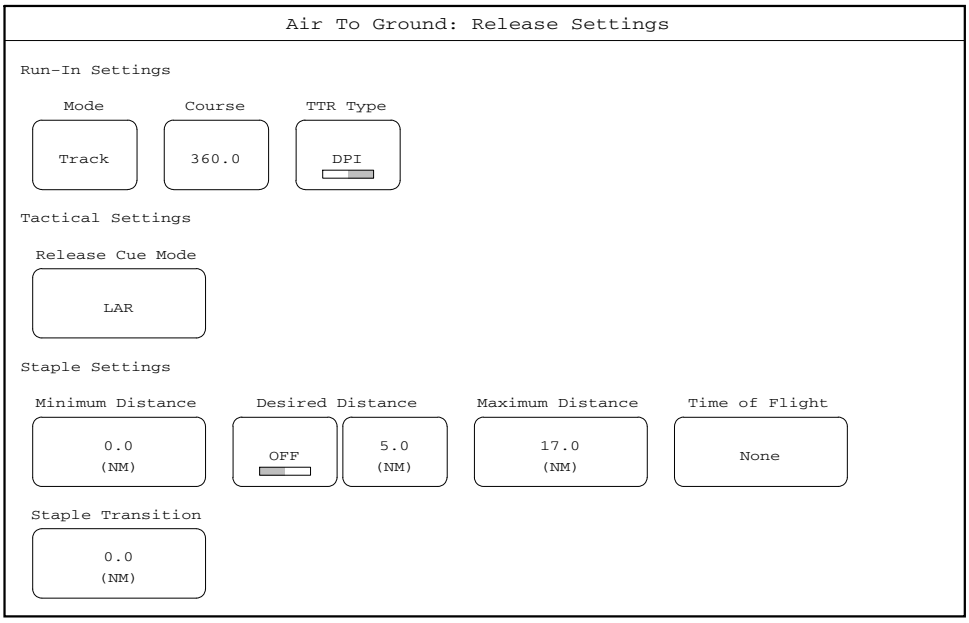


**Figure 1-92. Release Settings for UAI.**



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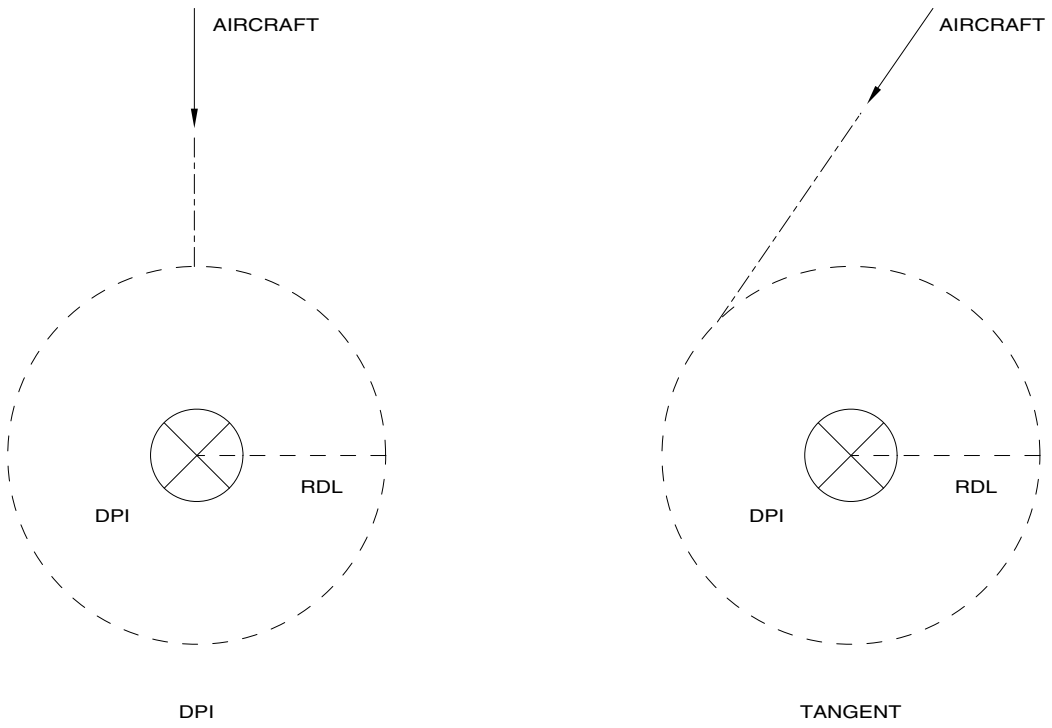


**Figure 1-93. DPI And Tangent, No Course Change.**

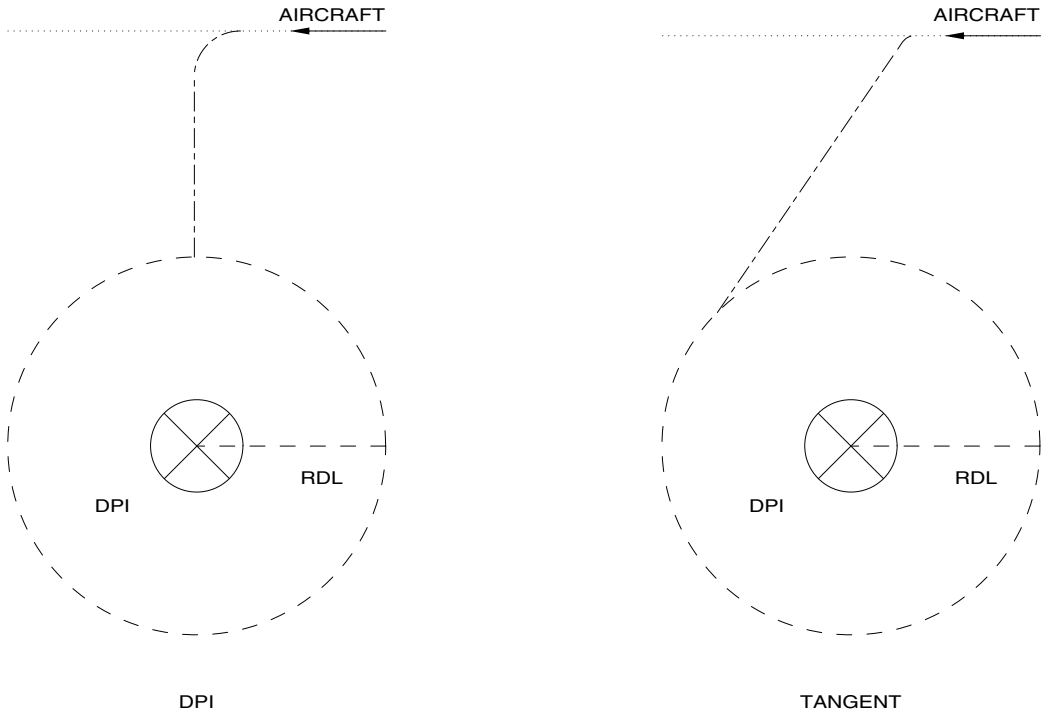


**Figure 1-94. DPI And Tangent, Course Change.**



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1.3.3.4.1 Run-In Settings. The Run-In Mode allows the pilot to decide if a run-line will be present on the Tracker display and what type of line will be present. Selecting the Mode button will bring up the Run-In Mode options window (Figure 1-95). There are three options available and are listed and described below:

- Track – Draws a Tracker screen run-in line connecting the aircraft and selected target icons. This is the default setting.
- Manual – Draws a Tracker screen run-in line with a specified bearing through the selected target.
- Off – No run-in line will be present on the tracker.

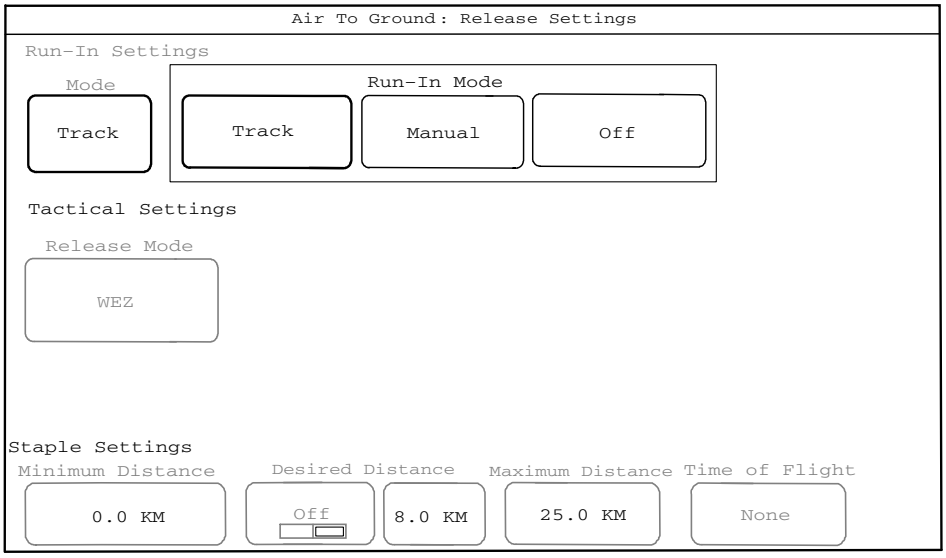


**Figure 1-95. Run-In Mode – Typical.**

When the Run-In Mode is set to Manual, the Course button becomes enabled. Selecting the Course button will bring up the Enter Run-In Course keypad (Figure 1-96 or Figure 1-97) allowing the pilot to set a bearing. The keypad allows a bearing to be set between 0° and 360°, referenced to true North. Once a bearing has been input and OK selected, the bearing line will be presented on the Tracker display. For all weapon types, when the user inputs a run-in course, the Enter Run-In Course keypad will display the current state of the global heading reference toggle button as either “True” or “Mag”.



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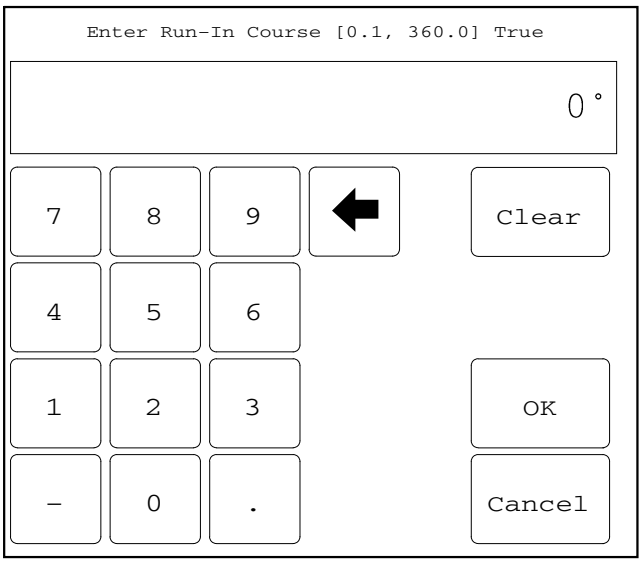


**Figure 1-96. Run-In Course – Global Heading Reference Set to True.**



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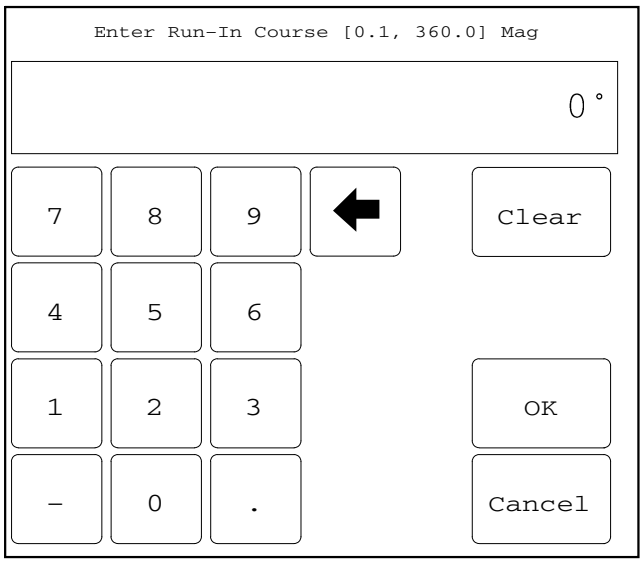


**Figure 1-97. Run-In Course – Global Heading Reference Set to Mag.**

1.3.3.4.2 Tactical Settings. Under Tactical Settings is the Release Cue Mode button. When the Release Cue Mode button is selected, the Release Cue Mode window (Figure 1-98) will offer Manual, LAR, or WEZ options. Manual mode allows the operator to input values under Staple Settings. WEZ/LAR mode will provide the operator with automatically updated Time of Flight (TOF) if the target is within the WEZ/LAR as well as minimum and maximum release distance values calculated by the CWL server. These values will continuously change with movement of the RPA. The Range of Desire Launch (RDL) value may be input by the operator if desired. WEZ/LAR mode is the default setting.


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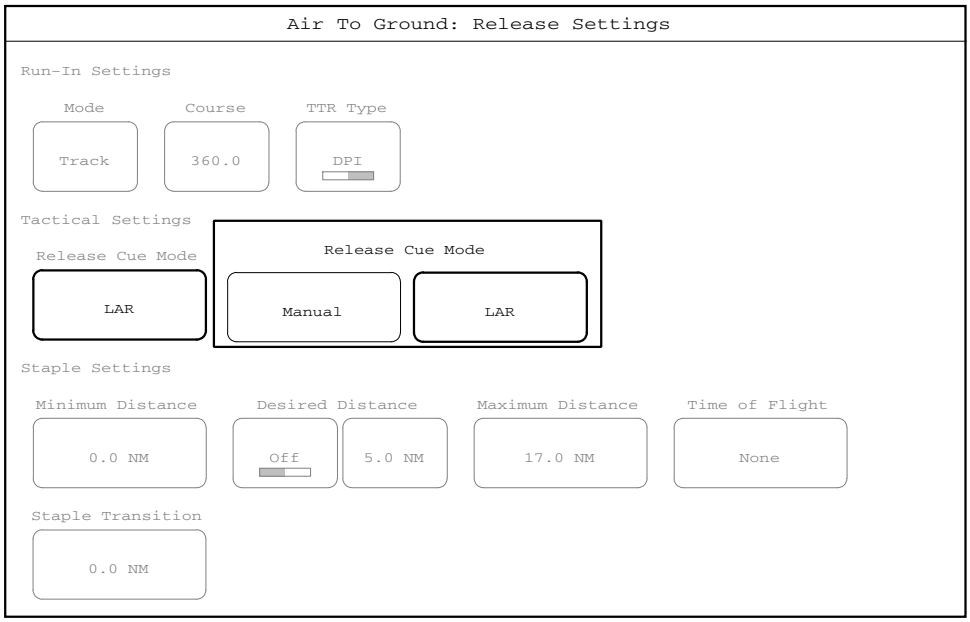


**Figure 1-98. Release Cue Mode.**

**NOTE**

End game calculations to include TOF may not display if missile performance data is not available.

The Release Cue Mode button allows the method of weapon release to be configured. Selecting the Release Cue Mode button will bring up the release cue mode options window. The two settings available are LAR and Manual, with LAR being the default setting.

The Manual option allows the bomb to be released at the moment the release command reaches the aircraft, but does not give the pilot cues to release the bomb. The release command is sent once the launch enable button and the control stick trigger are depressed in sequence. They must remain depressed until the weapon icon disappears from the HUD, indicating the weapon has been released.

The LAR option gives the pilot cues to release the bomb within a specified area. The release command is sent once the launch enable button and the control stick trigger are depressed in sequence. The launch enable button and control stick trigger must remain depressed until the weapon icon disappears from the HUD, indicating the weapon has been released and to allow the most effective weapon employment. The range LAR provides a visual reference to the area in which the store may be released for a successful weapon employment. It is determined based on aircraft altitude, wind speed, course, target elevation, wind direction, KU delay, and air speed. The zone LAR provides a visual depiction of the area in which the store will meet the terminal guidance requirements set in the store settings. It is determined based on all inputs used in the range LAR inputs as well as the desired impact angle.

Weapons with a desired impact angle are considered constrained, where those without are considered unconstrained. If weapons being dropped in a ripple are constrained, the CWL will calculate the zone LAR to be the intersection of the zone LAR of each weapon to be released. An overly constrained weapon in a ripple release will cause no zone LAR to be calculated as well as if no weapons in the release are constrained. If a mixture of constrained and unconstrained weapons are to be released, the CWL shall calculate the release's zone LAR as the intersection of each constrained weapon's zone LAR and each unconstrained weapon's range LAR.

The Manual option will display the minimum, maximum, desired ranges, and TOF on the run-in line. The release command is sent once the launch enable and the control stick trigger are depressed in sequence.

The second button is the Type toggle button and gives the user the ability to select Moving or Stationary for the target. The default setting is Stationary; while the Moving option will only be selectable if all weapons in the release support moving target guidance. If not, the target type selection will be disabled and default to Stationary. Additionally if all weapons support moving target guidance and Auto CCTGP is selected, the Type will automatically be set to Moving, though it can still be put back to Stationary.

Under Staple Settings, Desired Distance will always remain selectable. When Release Cue Mode is placed in Manual (Figure



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1-98), all Staple Settings will become selectable, allowing manual entry of values.

1.3.3.4.3 Staple Settings. Desired Distance will always remain selectable but when Release Cue Mode under Release Settings is placed in Manual (Figure 1-108), all of Staple Settings will become selectable allowing manual entry of values. For AGM-114 when Impact Heading is set to zero degrees, the Range button will be disabled. This will also disable Predictive Missile Impact Heading (PMIH) on the HUD. There are seven selections for AGM-114 under Staple Settings:

- Minimum Distance – The shortest ground range for weapon release at the intended release HAT.
- Desired Distance – The best ground range for weapon release at the intended release HAT.
- Maximum Distance – The longest ground range for weapon release at the intended release HAT.
- Time of Flight – The time of weapon flight before impact.
- Staple Transition – Determines when GBU steering cue will transition from Angular to Positional mode.
- Impact Heading – The desired missile heading for the PMIH
- Range – Acceptable range impact angles for PMIH
- Type – True or Magnetic North selection options when using PMIH.

Selecting Minimum Distance allows the user to enter the minimum release distance on the Enter Minimum Distance Keypad (Figure 1-99) between 0.0 through 25.0 Kilometers (KM) for Hellfire, and Nautical Miles (NM) for GBUs.

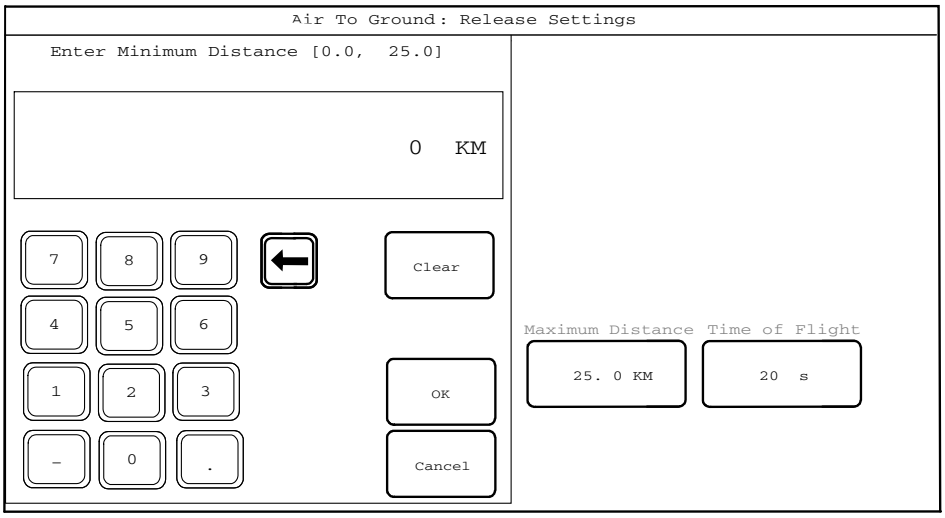


**Figure 1-99. Enter Minimum Release Distance Keypad.**

Setting Desired Distance to On enables the Distance button directly to the right. This button brings up the Enter Desired Distance Keypad (Figure 1-100) allowing entry of 0.0 through 25.0 KM for Hellfire, and NM for GBUs.



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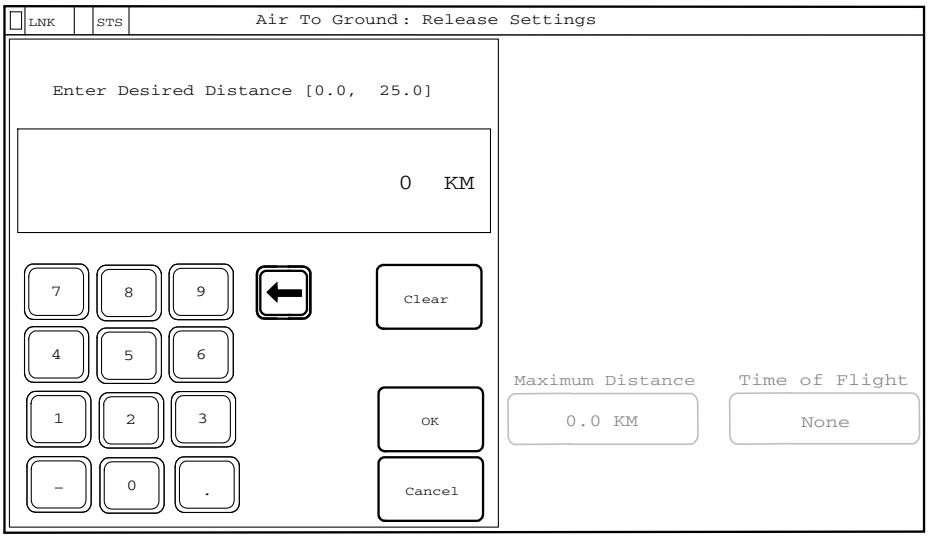


**Figure 1-100. Enter Desired Distance Keypad.**

Selecting Maximum Distance allows the user to enter the maximum release distance on the Enter Maximum Distance Keypad (Figure 1-101 ) between 0.0 through 25.0 KM for Hellfire, and NM for GBUs.

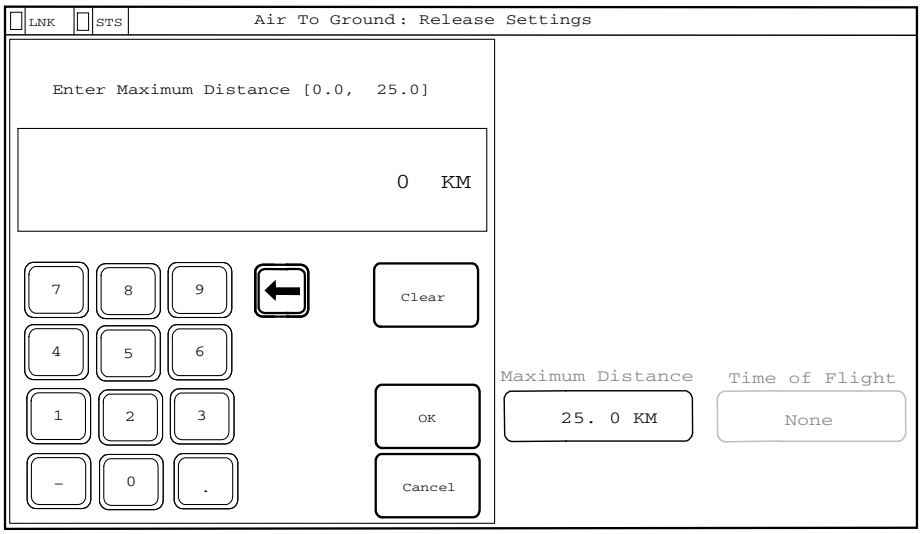


**Figure 1-101. Enter Maximum Distance Keypad.**

When the Release Cue Modes set to Manual, the Time of Flight button gives the operator the ability to enter a manual TOF into the Enter Time of Flight window (Figure 1-102). The entered values can only be entered between 0 – 99 seconds and are allowed in 1 second increments. An initial default value for manual TOF will be 0 seconds. When a desired value has been entered, the same value will be displayed on the HUD. Ensure the most current version of the R Missile Impact Tool is used to determine accurate release capabilities.



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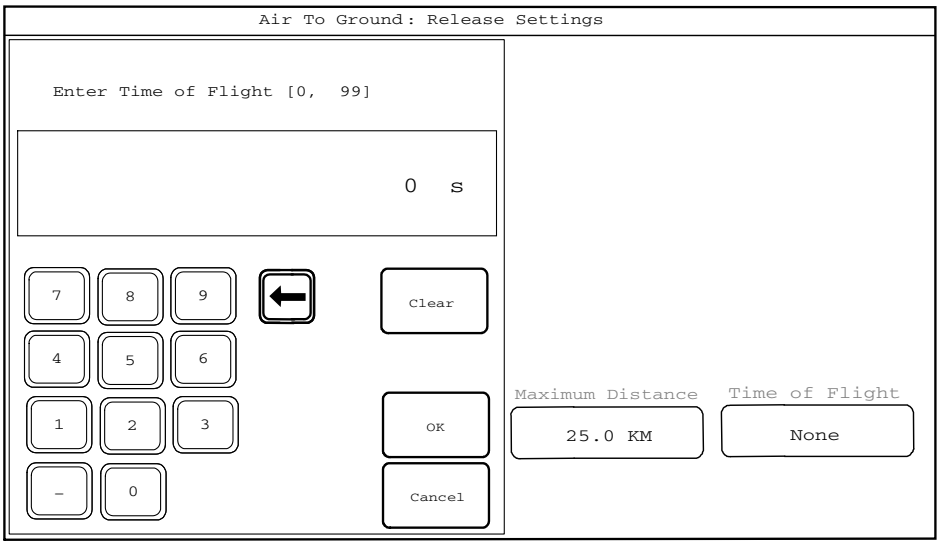


**Figure 1-102. Enter Time of Flight Window.**

For GBUs the staple transition button is displayed when release cue mode is set to LAR, the Staple Transition button will be selectable. Selecting the Staple Transition button brings up the Enter Staple Transition Keypad (Figure 1-103). This keypad allows the user to set the distance at which the GBUs steering cue will transition between angular and positional modes, in relation to the ground range from the outermost LAR intercept point. The user can select distances between 0.0 and 20.0, with the default transition distance being 0.0 NM. When the transition distance is 0.0, the steering cue will transition from angular to positional mode as the aircraft passes the outermost LAR intercept. When the transition distance is set to other values, the steering cue will transition from angular to positional mode at that distance from the outermost LAR intercept point. Once in positional mode, the steering cue will switch back to angular mode when the aircraft has flown the transition distance away from the outermost LAR intercept.

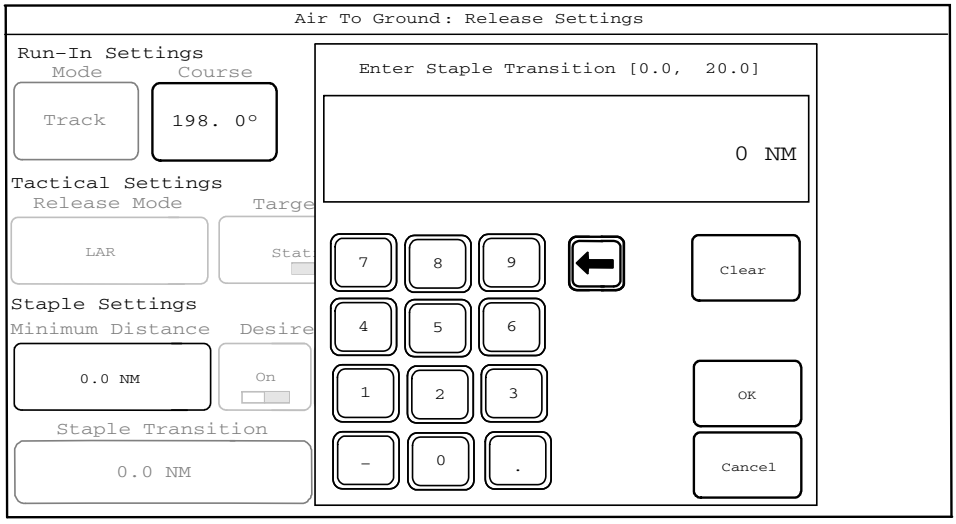


**Figure 1-103. Enter Staple Transition Keypad.**


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For AGM-114 PMIH, the Impact Heading button allows the operator to enter a desired missile impact headings from 0 – 360 degrees in one degree increments in the Impact Heading window (Figure 1-104). The default setting is zero and will set the desired impact heading to Invalid. PMIH will be displayed in green on the HUD and the Range button will be disabled. 360 degrees must be used instead of 0 degrees when setting the desired PMIH. To disable the desired PMIH, select Impact Heading and press OK. The range button will be disabled but will continue to display the range value.

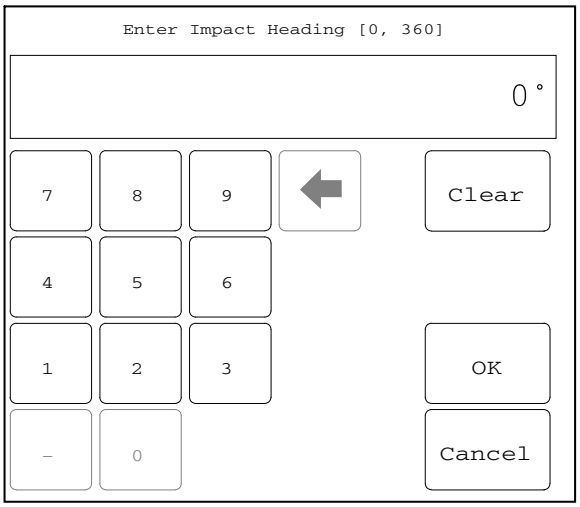


**Figure 1-104. Impact Heading.**

For AGM-114 the Range button is selectable when impact heading is set from 1 – 360 degrees. The Impact Heading Range window provides 0 – 180 degrees in one degree increments with zero being the default setting (Figure 1-105). The entered sum provides a range off acceptable angles for the Impact Heading, for example entering a value of 10 degrees to an impact heading of 270 would result in 270 +/- 10 degrees.

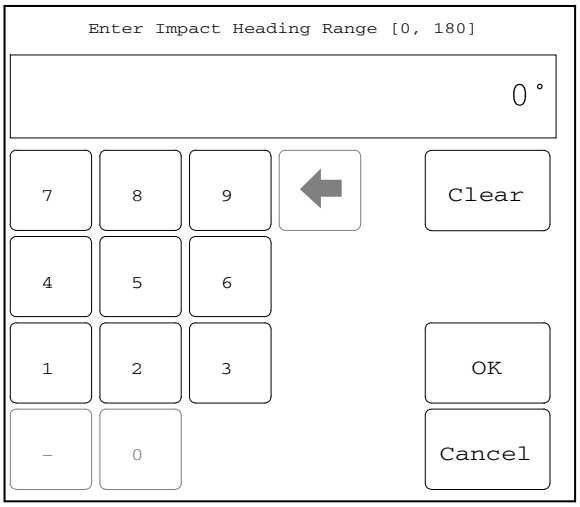


**Figure 1-105. Impact Heading Range.**

For AGM-114 the Type toggle switch located to the right of the Range button gives the option of selecting True or Mag. When True is selected the PMIH will be in true North which is also the default setting for Type. Mag when selected will place the PMIH in magnetic North. Magnetic North will not effect the display of the impact heading or range values and will only effect the values of the PMIH on the HUD.



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1.3.3.4.4 GBU-12/51 Release Settings. The GBU-12/51 has one additional setting available under the Release Settings column labeled Release Cue Mode. The Release Cue Mode button allows the method of weapon release to be configured. Selecting the Release Cue Mode button will bring up the Release Cue Mode options screen (Figure 1-106). The two settings available are Manual and Continuously Computed Release Point (CCRP).

When one or more GBU-12/51 munitions are selected, the GSMS uses the aircraft state information, selected target, wind speed and direction and munition capability to calculate a minimum release point , maximum release point and Ballistic Release Point (BRP). The BRP and aircraft HAT are used to create the munitions Time of Flight (TOF) displayed on the HUD. The minimum release point, maximum release point and BRP are displayed on the HUD and tracker Run-in line as the minimum, maximum and optimal release cues.

The release command is sent once the launch enable button and the control stick trigger are depressed in sequence. The Manual option allows the bomb to be released at the moment the release command reaches the aircraft. When the Release Cue Mode is set to CCRP the bomb can only be released at the BRP. The launch enable button and trigger must be depressed prior to the BRP. In either mode the launch enable button and trigger must remain depressed until the weapon icon disappears from the HUD indicating the weapon has been released.

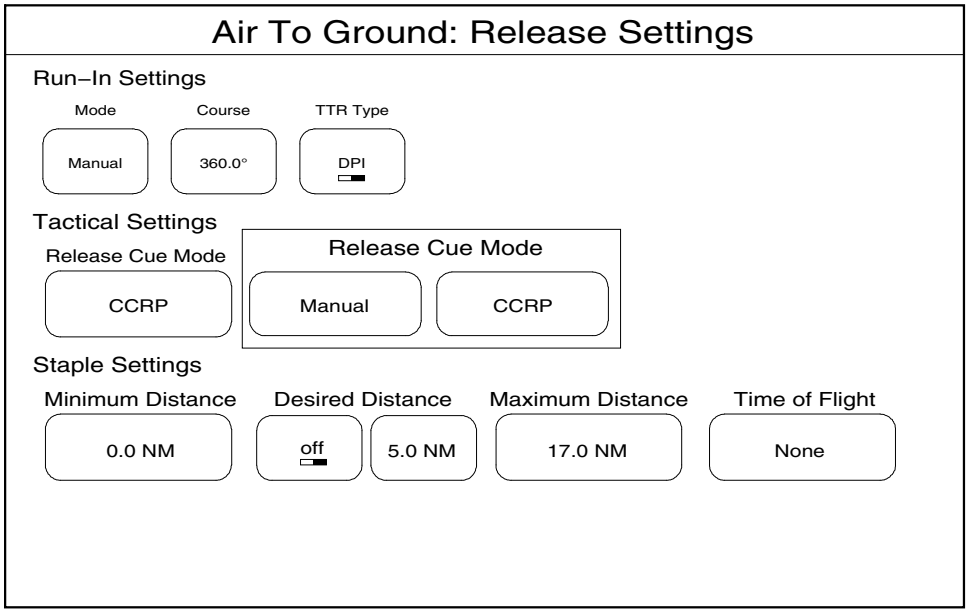


**Figure 1-106. Release Cue Mode for GBU-12/51.**

1.3.3.4.5 GBU-49/48 Release Settings.

**WARNING**

If a GBU-49/48 is released within the in-range LAR, the weapon will prioritize achieving terminal guidance requirements set in the store settings over hitting the target.

The GBU-49/48 has three different settings sections to the left of the screen under the Release Settings page (Figure 1-107). The three settings selections are:

- Run-in Settings – Mode and Course options.
- Tactical Settings – Release Cue Mode and Target Type options.
- Staple Settings – Minimum Distance, Desired Distance, Maximum Distance, Time of Flight entry, and Staple Transition options.



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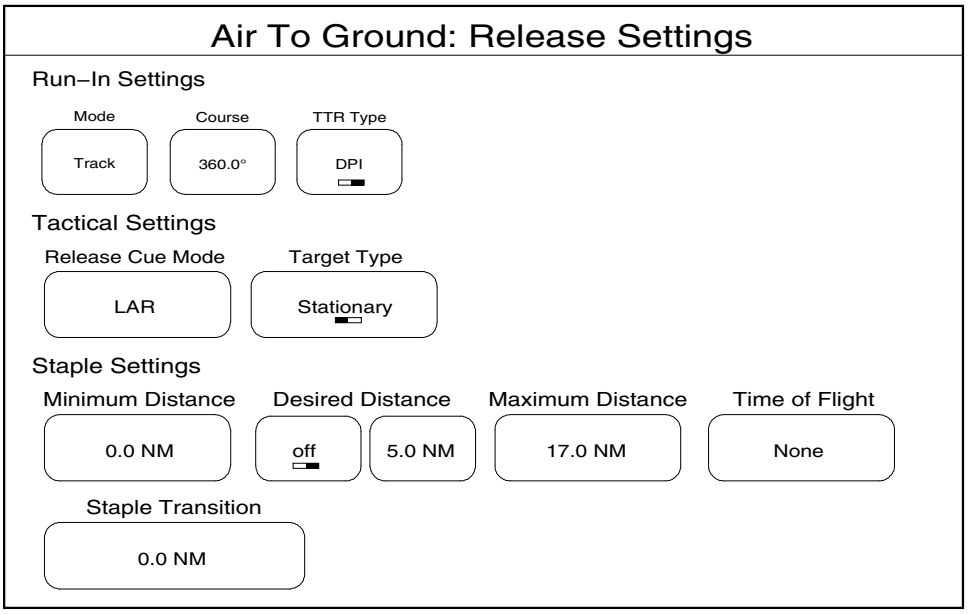


**Figure 1-107. Release Settings and Release Cue Mode.**

Run-in Settings allows the pilot to decide if a run-in-line will be present on the Tracker display and what type of line will be present. Refer to Release Settings for Run-In Mode and Course settings. A target type of Stationary should only be used on stationary targets, a target type of Moving should only be used on moving targets. Currently only Lot 5 ECCG GBU-49 supports moving target capabilities.

Block 1 The SCC shall receive target updates directly from the MTS-B for moving targets and the SCC shall receive target data via MTS to DFCS to DGCS to GSMS for stationary targets.

Block 5 The SCC shall receive target data directly from the PCC for moving targets and the SCC shall receive target data via PCC to DFCS to DGCS to GSMS for stationary target. When tracking a moving target, if the speed of the target drops below 5 mph, course and speed will show 0 for both.

1.3.3.5 JDAM Release Settings. On the JDAM Release Settings page, there are two options to the right of Release Cue Mode, Target Type and Non-UAI On-Wing-Tracking (OWT). When Release Cue mode is set to Manual, Target Type will be disabled. When Auto CCTGP is not selected as the target, Target Type will default to Stationary. OWT can be set to On or Off. OWT allows the JDAM to track, but not acquire satellites prior to release. OWT should be turned on prior to entering a denied environment to give the weapons a better chance of acquiring satellites post release. When OWT is enabled the weapon may intermittently become degraded. If the degraded status does not clear, check detailed status for root cause.

- NOT COMMANDED – Indicates that OWT is not commanded.
- INIT – Indicates that JDAM is not attempting to track. The JDAM performs a normal GPS satellite acquisition if released.
- ACQ – Indicates the JDAM is attempting to track. The JDAM reverts to a normal GPS satellite acquisition if released.
- MARGINAL – Indicates that the JDAM is tracking some but not all of the satellites required for GPS tracking causing the weapon to become degraded. The JDAM will continue OWT through the release sequence.
- GOOD – Indicates that the JDAM is in track. The JDAM will continue OWT through the release sequence.



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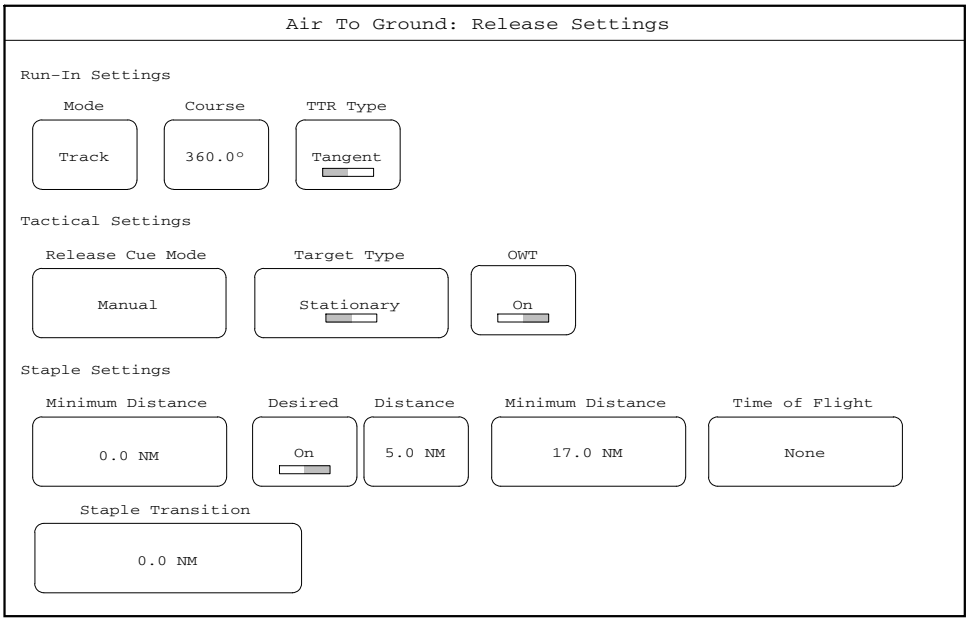


**Figure 1-108. JDAM Release Settings – Typical.**

A target type of Stationary should only be used on stationary targets, a target type of Moving should only be used on moving targets.

Block 1 The SCC shall receive target updates directly from the MTS-B for moving targets and the SCC shall receive target data via MTS to DFCS to DGCS to GSMS for stationary targets.

Block 5 The SCC shall receive target data directly from the PCC for moving targets and the SCC shall receive target data via PCC to DFCS to DGCS to GSMS for stationary target. When tracking a moving target, if the speed of the target drops below 5 mph, course and speed will show 0 for both.

1.3.3.6 LSDB Release Settings. The LSDB has three different settings sections to the left of the screen under the Release Settings page Figure 1-109:

- Run-in Settings – Mode and Course options
- Tactical Settings – Release Mode
- Staple Settings – Minimum Distance, Desired Distance, Maximum Distance, Time of Flight, and Staple Transition options



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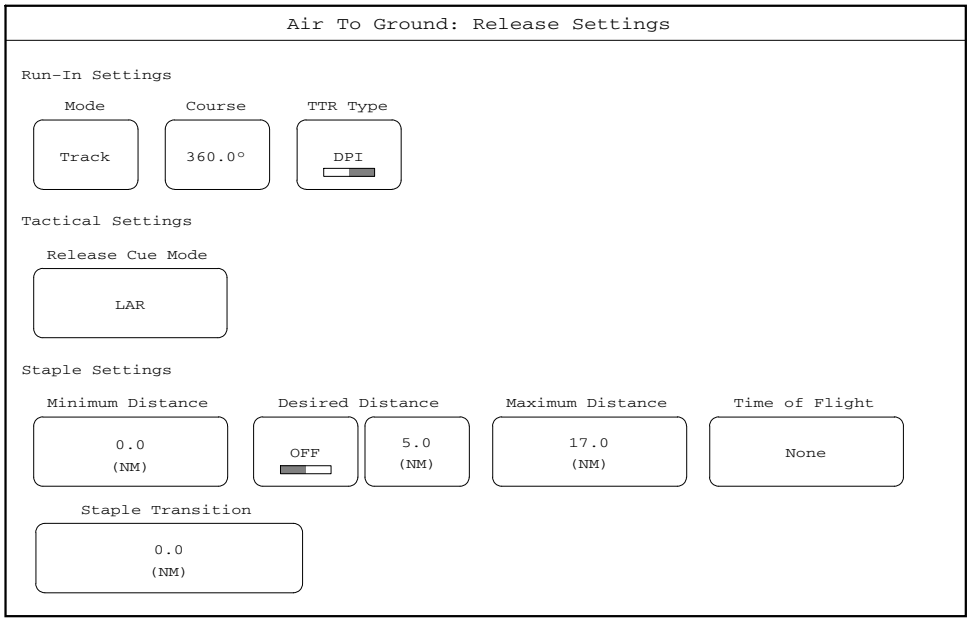


**Figure 1-109. Release Settings for LSDB.**

Run-in Settings allows the pilot to decide if a run-in-line will be present on the tracker display and what type of line will be present. Refer to Release Settings for Run-In Mode and Course settings. Tactical Settings section offers the Release Cue Mode button and allows the method of weapon release to be configured. Selecting the Release Cue Mode button will bring up the Release Cue Mode options screen (Figure 1-110). The two settings available are LAR and Manual, with Manual being the default setting.

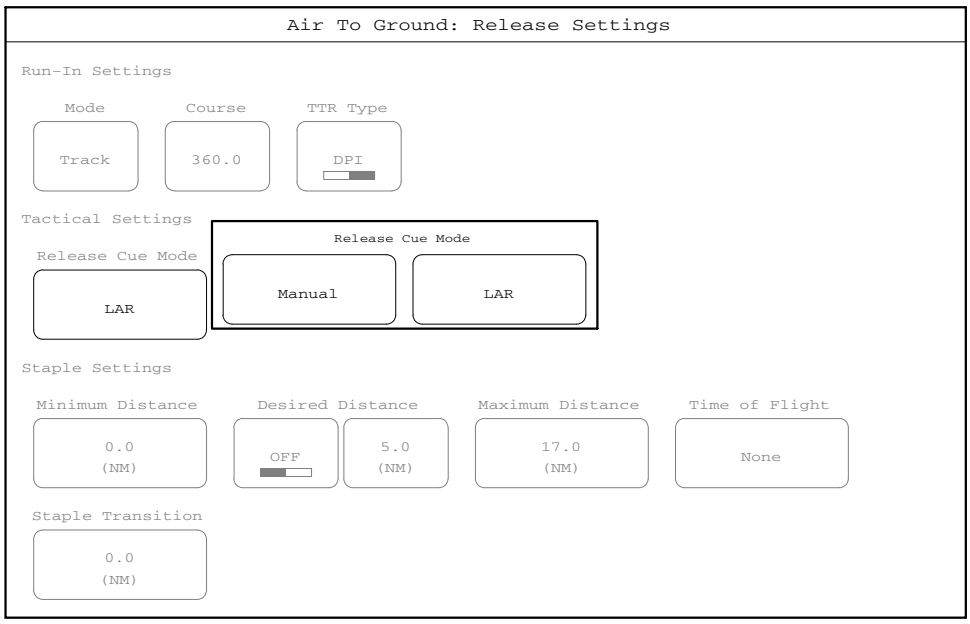


**Figure 1-110. Release Cue Mode for LSDB.**



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The Manual option allows the bomb to be released whether the aircraft is inside or outside the LAR. LAR release cues will remain on the HUD and map tracker. The release command is sent once the launch enable button and the control stick trigger are depressed in sequence regardless of whether the UAV is inside or outside the LAR. Under Staple Settings, Desired Distance will always remain selectable but when Release Cue Mode under Release Settings is placed in Manual, all of Staple Settings will become selectable allowing manual entry of values (Figure 1-110) although Staple Transition will not be functional as LAR is not available.

1.3.3.7 Launch Status.

**WARNING**

When a Lot 3A/Lot 5 ECCG Manual IBIT is commanded, a Laser Seeker BIT is performed automatically. If the Laser Seeker BIT fails, the laser seeker may not guide to laser energy. If a reattempt of IBIT is unsuccessful power cycle the weapon and again reattempt IBIT. If the Laser Seeker BIT still fails, the weapon can only be dropped in GPS guidance mode.

The Launch Status Page shows the full target list for all selected weapons which allows the use of a up and down arrow scroll button so the user can verify all selected targets prior to launch. It provides the ability for the user to switch any store assigned to a manual target type to the CCTGP target type.

The Page Section will contain a status list with a corresponding indicator on the left hand side. To the right, will be a box containing PRF code information for the MTS and weapons selected for a release if the weapon is laser guided. The target data is at the bottom of the screen.

The GBU IBIT button is in the lower right hand corner next to target information. The button will be present and selectable if a Lot 3A/Lot 5 ECCG is in the inventory and any GBU-49 type is selected.

Displayed to the right of the PRF codes are the fuze function settings for selected fuze function weapons. To the right of Fuze are the LOAL setting and Interval times (Figure 1-111).

The Launch Status Page Layout after the UAI system integration is slightly larger. The Store status area size has been increased to accommodate up to eight weapons. The profile preview group boxes have be shrunk down in size to allow eight weapons to be displayed. The Profile Preview group box for the on-deck and in-the-hole profiles show the following items: Number of weapons selected, weapon type, and profile name. For each weapon, the store status area items that get displayed are determined by the CDF file for that weapon type.

2411.0.0 The Launch Status page Target Display button(s) have been increased in size to accommodate up to eight munitions getting displayed on the button after assignment to a target ( 2411.0.0 Figure 1-114).

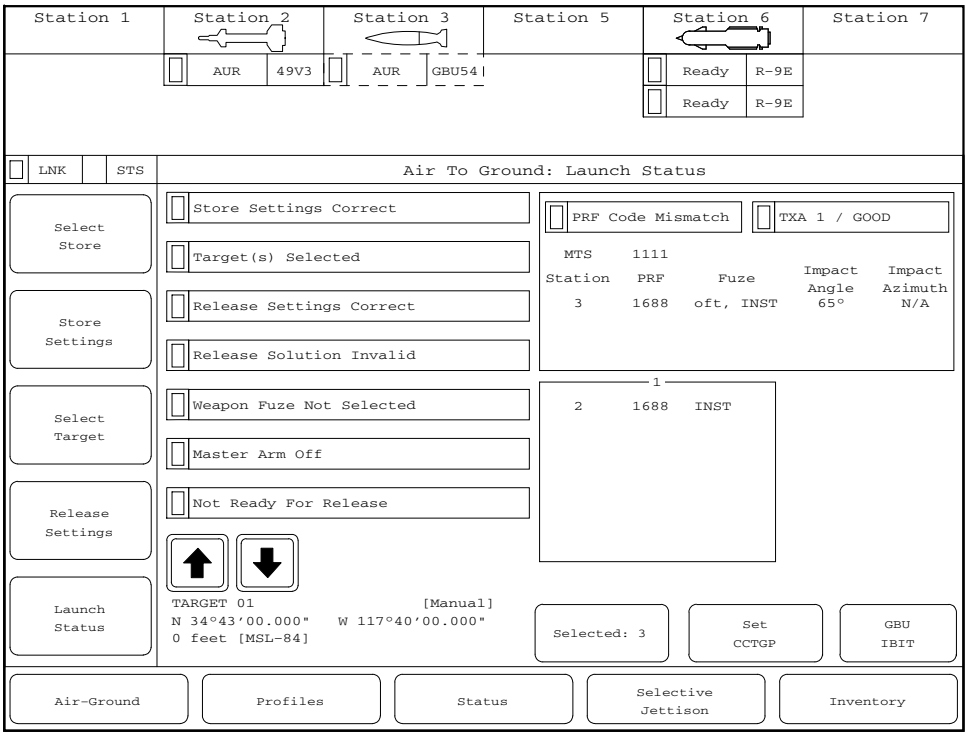


**Figure 1-111. Launch Status for AGM-114 with Active Profile Setting.**



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The Launch Status indications differ slightly between weapons, but some common indications are present for all. The common launch status indications are as follows:

- Store Not Selected
- Store Settings Correct or Incorrect
- Target Selected or Not Selected
- Release Solution Unknown, Invalid, or Valid
- Master Arm Off or On
- Ready for Release or Not Ready for Release

If a Store or Target is not synchronized, or when Master Arm transitions between safe and arm, the LNK indicator will blink green.

Most indications provide a reference to a specific setting or group of settings without the need for further explanation. The Release Solution indication applies to all weapons but functions slightly different for each weapon and will be explained as each weapon is explained.

Launch status indications specific to the AGM-114 are as follows:

- Store Powered or Not Powered
- System Ready or Not Ready For Launch

Release Solution for the AGM-114 applies for all missiles in a ripple train. The indication allows the pilot to identify if, with the current settings, the weapon will be released from the aircraft within the minimum and maximum range release points. Release solution is also checking that the predicted missile time of flight allows for 16 seconds of uninterrupted laser time during flight. Launch status indications (Figure 1-112) specific to the GBU-12/51 are as follows:

- Release Settings Correct or Incorrect
- Weapon Fuze Armed, Not Selected, or BRU-71/A Center Fuze Not Selected

The GBU-12/51 Release Solution only applies when more than one weapon is selected. The indication allows the pilot to identify if, with the current settings, the weapon will be able to hit the target. This is determined by known weapon performance abilities and restrictions.

BRU-71/A A Center Fuze Not Selected caution will be displayed on the Launch Status screen if the Center Fuze is not selected and any combination of Nose/Tail Fuze is selected.

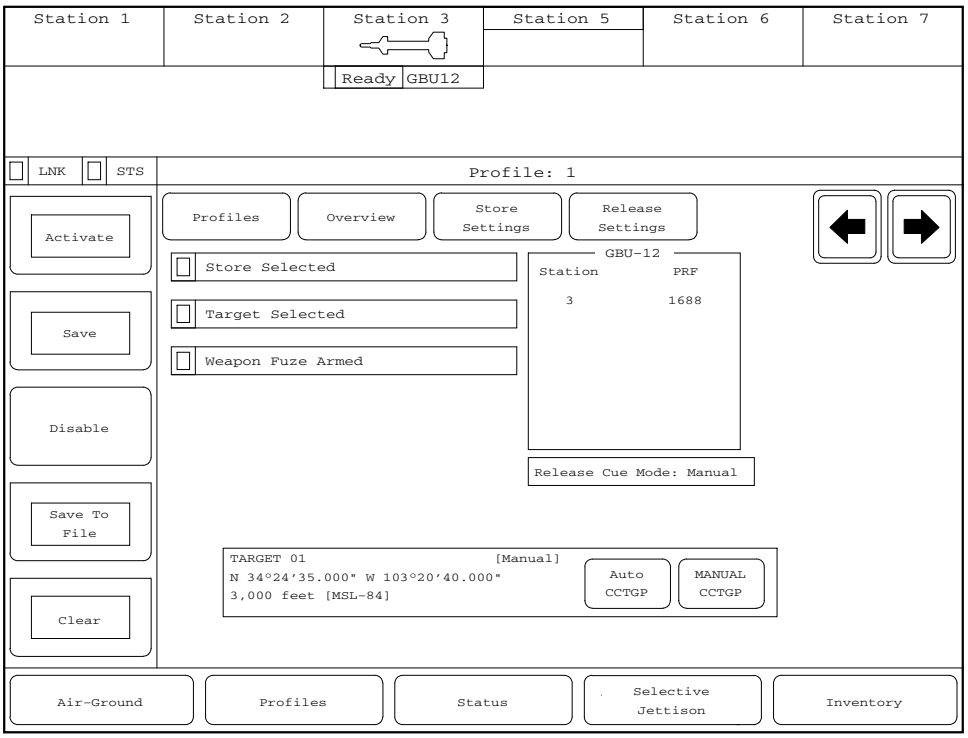


**Figure 1-112. Profile Overview Page for GBU-12/51 with No Active Profile Setting – Typical.**

Launch status indications (Figure 1-116) specific to the GBU-49 are as follows:

- Store Settings Correct, or Incorrect
- Target Selected, or Not Selected
- Release Settings Correct, or Incorrect



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- Release Solution Valid, or Invalid
- Weapon Fuze Armed, Not Selected, or BRU-71/A Center Fuze Not Selected
- Master Arm On, or Off
- Ready, or Not Ready For Release
- TXA and Nav Quality

The GBU-49 Release Solution only applies when more than one weapon is selected. The indication allows the pilot to identify if, with the current settings, all the weapons will be released from the aircraft prior to departing the LAR or within 3 seconds of the CCRP.

BRU-71/A A Center Fuze Not Selected caution will be displayed on the Launch Status screen if the Center Fuze is not selected and any combination of Nose/Tail Fuze is selected.

The GBU IBIT button provides the ability to manually command a GBU-49 IBIT from the Launch Status page. When the GBU IBIT button is selected, store status indicates IBIT for approximately 10 seconds until IBIT is complete. Under PRF Code Mismatch, displayed to the right of the Station and PRF codes, are Mode, Fuze, Impact Angle, and Interval. This ability is only available when the bomb is equipped with a Lot 5 ECCG and shall not be commanded during a power-up sequence or release sequence. When the GBU-49 is equipped with a Lot 3 or Lot 3A ECCG, IBIT runs automatically during warmup and cannot be manually commanded.

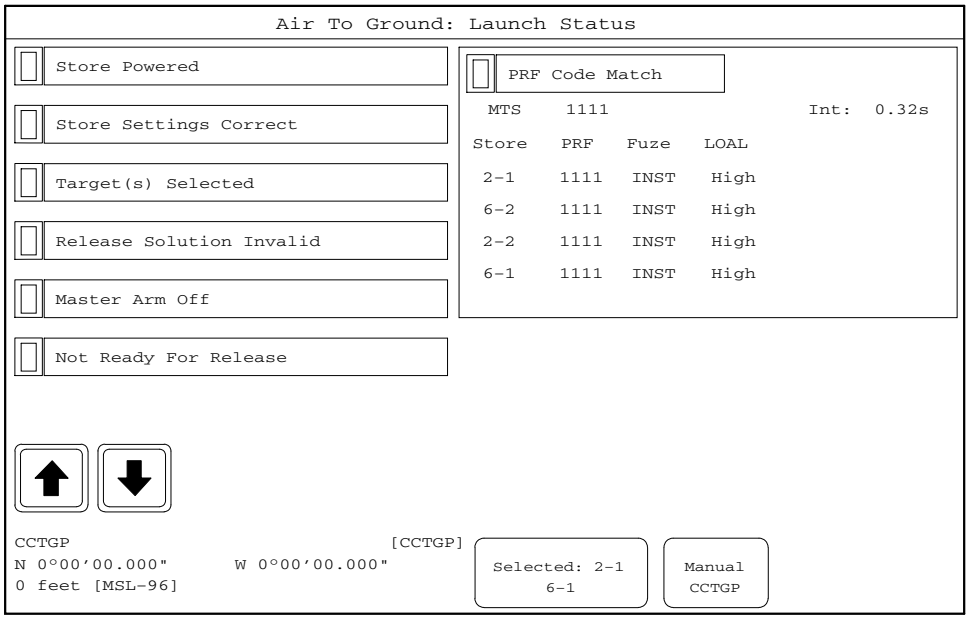


**Figure 1-113.** 2410.0.0 **Launch Status for AGM-114.**



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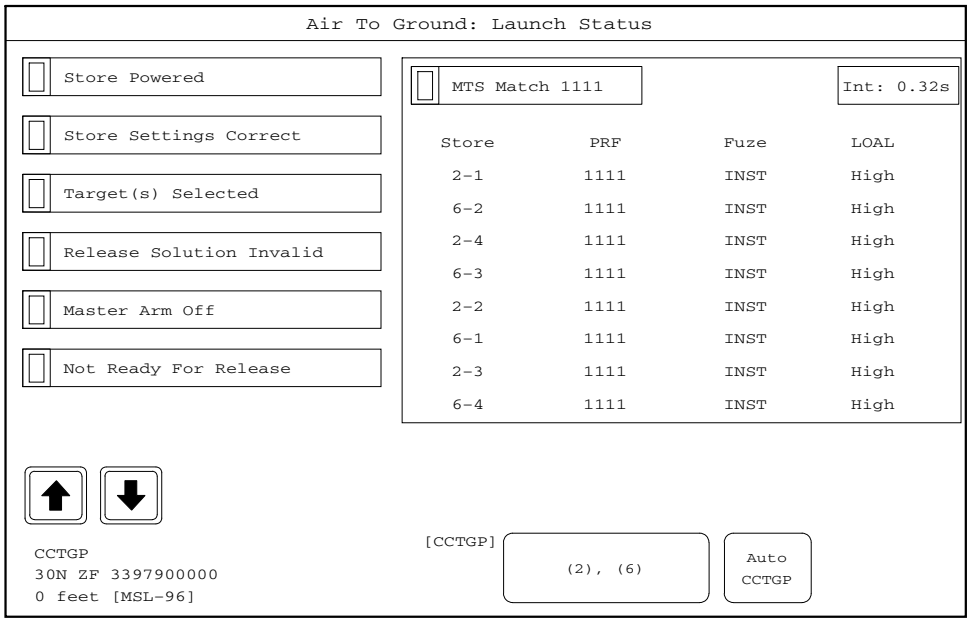


**Figure 1-114.** 2411.0.0 **Launch Status for AGM-114.**

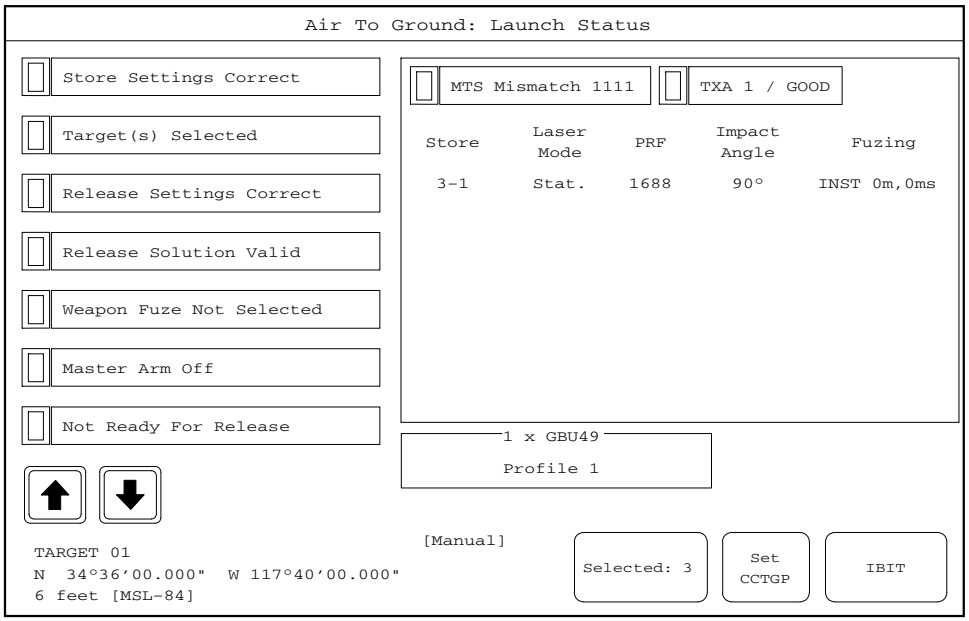


**Figure 1-115. Launch Status for GBU-48.**



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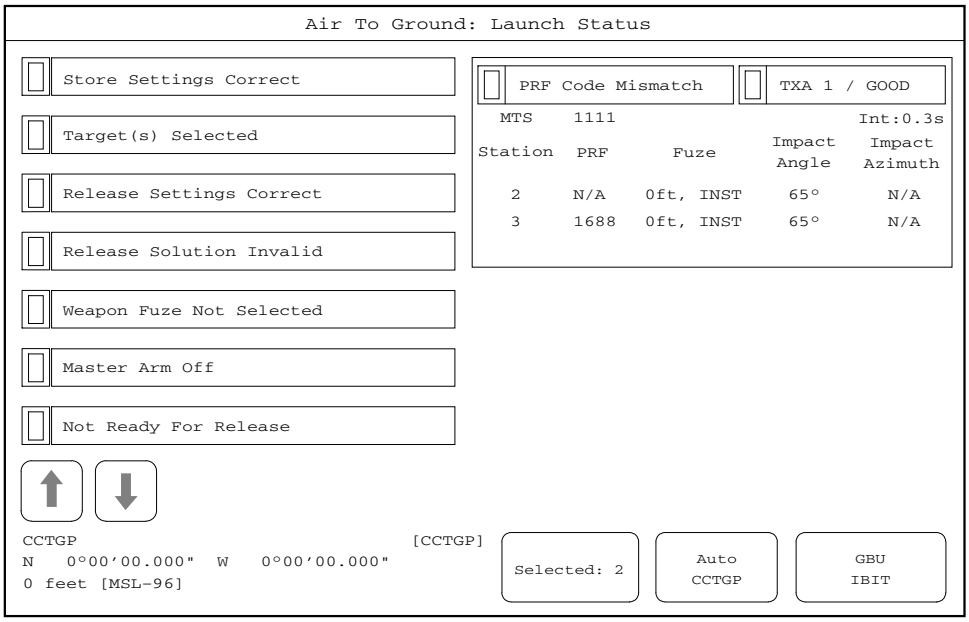


**Figure 1-116. Launch Status for JDAM/GBU-49.**

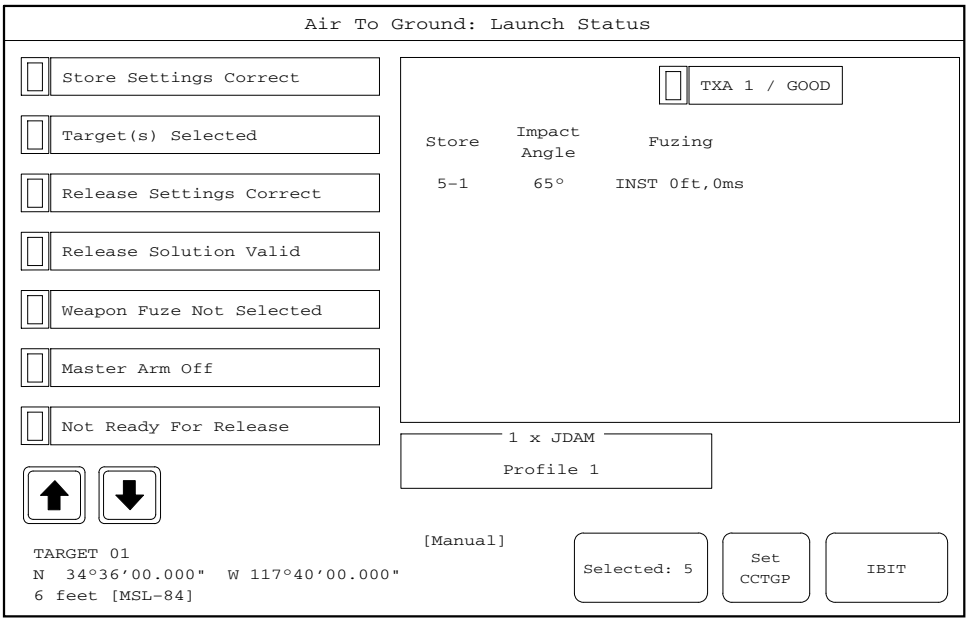


**Figure 1-117.** 2410.0.0 **Launch Status for GBU-38.**



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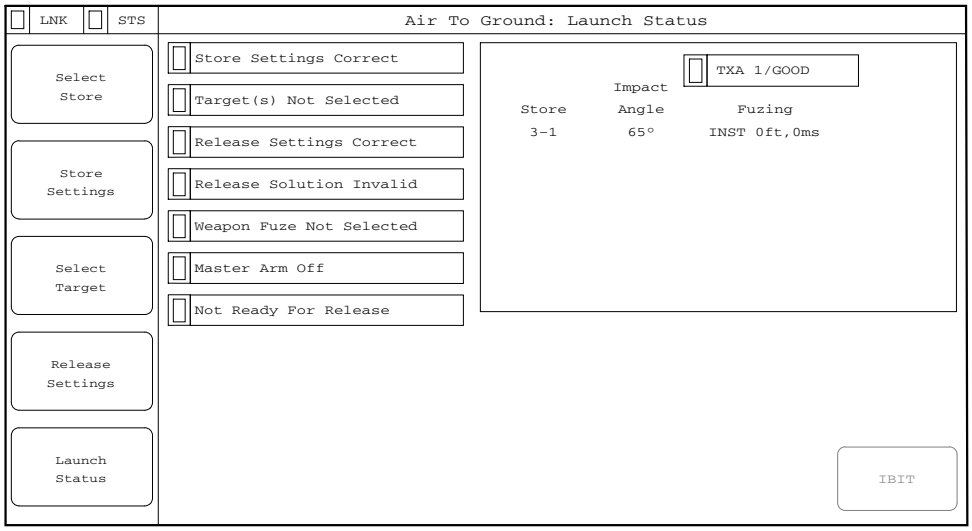


**Figure 1-118.** 2411.0.0 **Launch Status for GBU-38.**

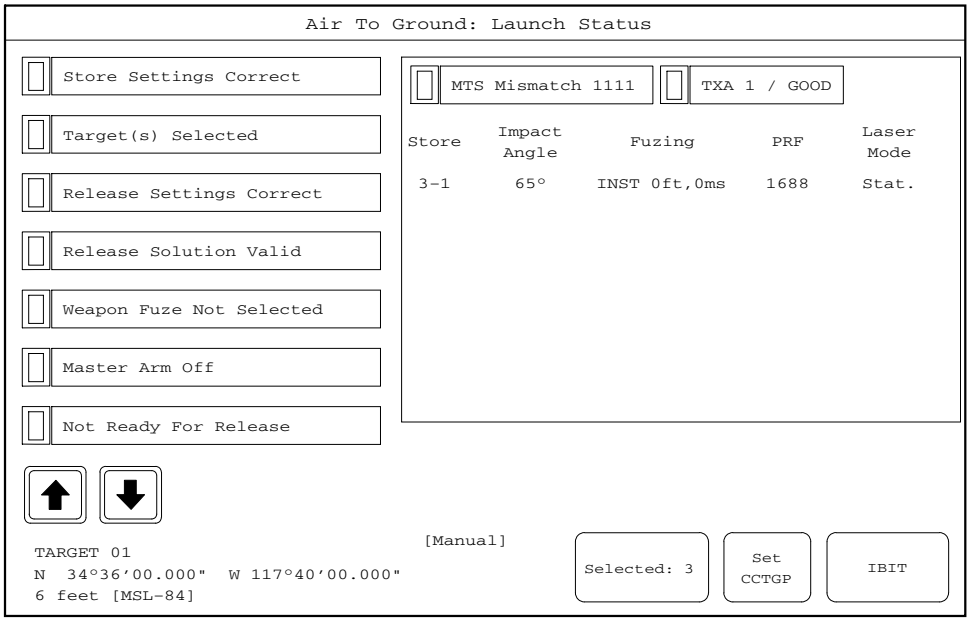


**Figure 1-119.** 2410.0.0 **Launch Status for GBU-54.**



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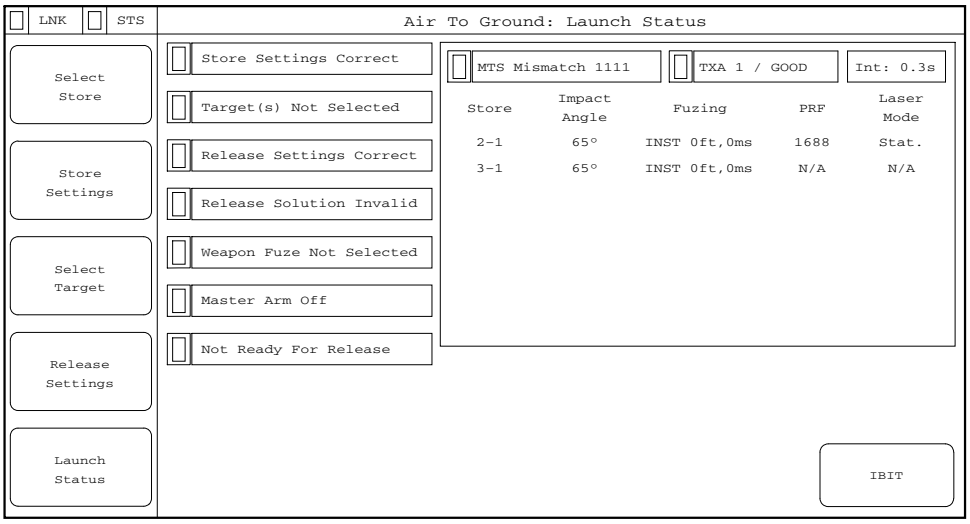


**Figure 1-120.** 2411.0.0 **Launch Status for GBU-54.**

JDAM Launch Status page indications (Figure 1-117, Figure 1-119 or Figure 1-121) for JDAM weapons are as follow:

- Store Settings Correct, or Incorrect
- Target Selected, or Not Selected
- Release Settings Correct or Incorrect
- Release Solution Valid, or Invalid
- Weapon Fuze Armed, Not Selected, or BRU-71/A Center Fuze Not Selected
- Master Arm On, or Off
- Ready or Not Ready For Release

The JDAM weapon type will display the following store status items: Store Number, Laser Mode, PRF code, Impact Angle, Fuze Delays and Fuze Heights (Fuze settings display only when a fuze is connected). When only GBU-38 is selected Laser Mode and PRF code will not be displayed. When a GBU-38 and GBU-54 and selected for a ripple, the GBU-38 Laser Mode and PRF code will be set to N/A.

TXA and Navigation Quality are displayed in the Store Settings Summary Group box (Figure 1-116 or Figure 1-121). The GBU IBIT button is located in the bottom right corner of the JDAM Launch Status page. When this button is selected, an IBIT will be performed on a selected JDAM weapon.



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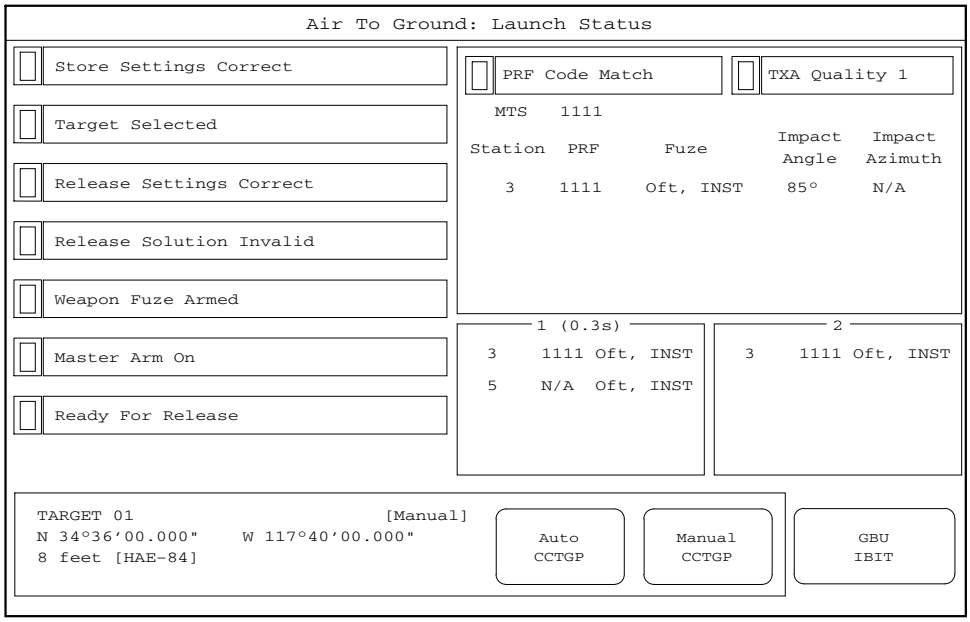


**Figure 1-121. Launch Status for JDAM with Saved Profile.**

Each indication will have a color associated with its current state. These colors and meanings are described below:

- Green – satisfactory
- Red – unsatisfactory
- Gray – unknown or not applicable
- Yellow – caution
- Blinking green – in-process

A Yellow status indication will not prevent release of a weapon.

LSDB Launch Status page indications (Figure 1-123 or Figure 1-124) are as follows:

- Store Settings, Correct, or Incorrect
- Target Selected, or Not Selected
- Release Settings, Correct or Incorrect
- Release Solution, Valid or Invalid
- Weapon Fuze Armed, or Weapon Fuze Not Selected
- Master Arm, On or Off
- Ready or Not Ready For Release

The Store status area size can accommodate up to eight weapons. The profile preview group boxes can allow eight weapons to be displayed. The Profile Preview group box for the on-deck and in-the-hole profiles show the following items: Number of weapons selected, weapon type, and profile name. For each weapon, the store status area items that get displayed are determined by the CDF file for that weapon type. TXA Quality is displayed in the Store Settings Summary Group box (Figure 1-124). In the bottom right corner of the Launch Status page is the IBIT button. When this button is selected, an IBIT will be performed on all powered IBIT capable stores.



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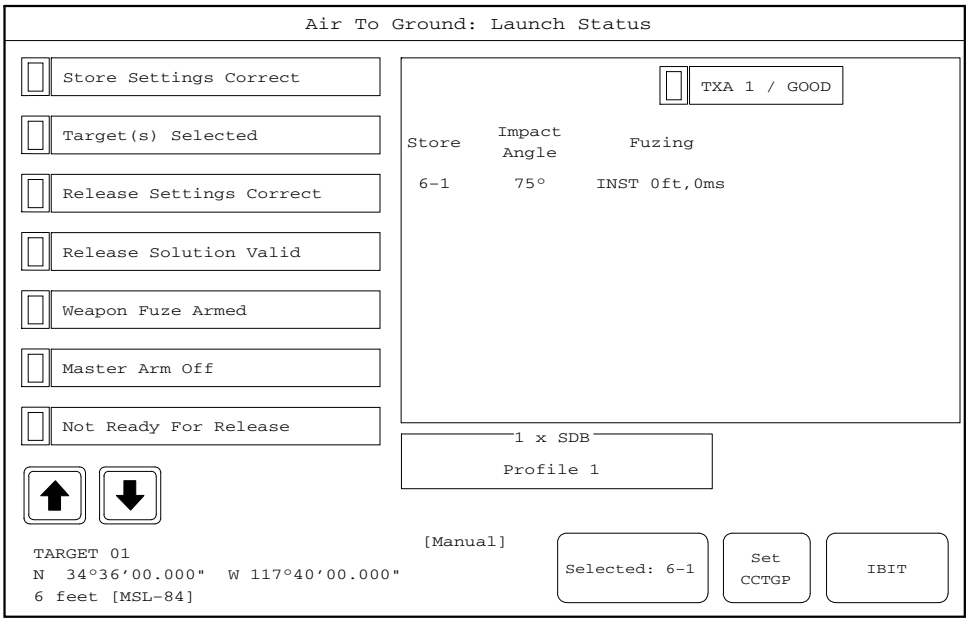


**Figure 1-122. Launch Status Page for SDB.**

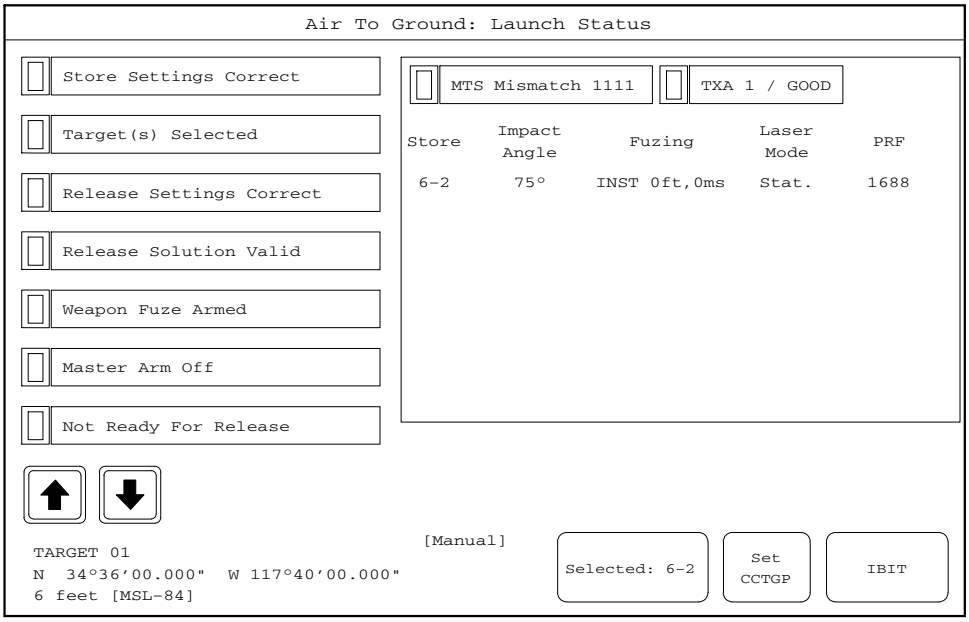


**Figure 1-123. Launch Status Page for LSDB.**


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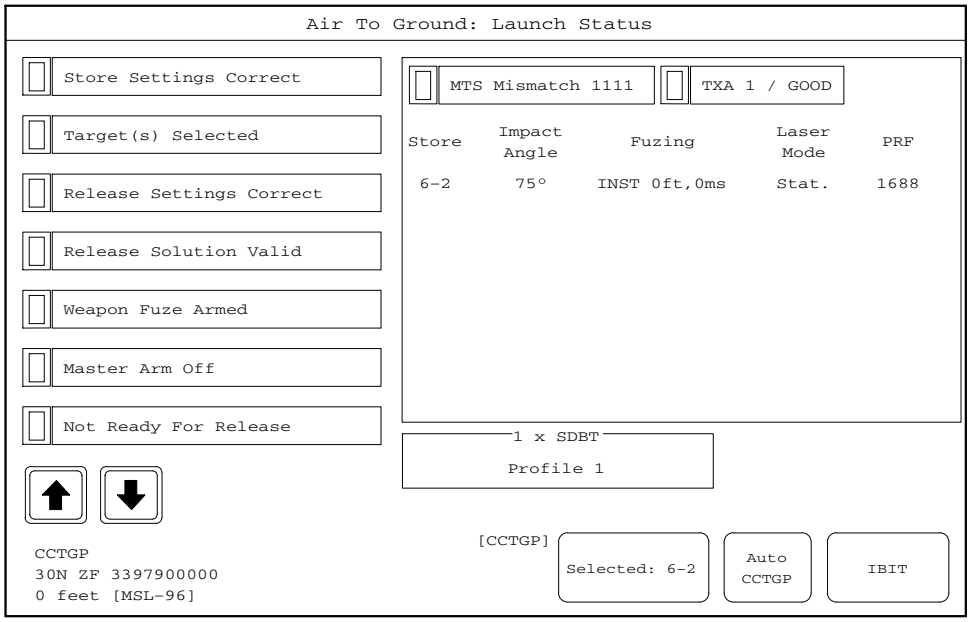


**Figure 1-124. Launch Status for LSDB with Saved Profile.**

2411.0.0 The TXA and Navigation Solution Quality Display widget on the Launch Status page will display the "TXA Good" flag instead of the TXA Quality numerical value whenever a Store on a DCS was selected for release. The store status message received from the DCS doesn't provide the TXA Quality as a numerical value. It provides the "TXA Good" flag which will be set to true whenever the TXA Quality is a value of 1 or 2. When the TXA Quality value is greater than 2, the TXA Good flag will be set to false. For GBU-12 variant bombs, the widget will be entirely decluttered, as TXA and Navigation Solution Quality do not pertain to such stores (2411.0.0 Table 1-8, 2411.0.0 Figure 1-125 and 2411.0.0 Figure 1-126).

**Table 1-8. 2411.0.0 Widget Status Text and Light Color.**

| TXA Good Flag | Navigation Solution Quality | Status Text | Light Color |
|---|---|---|---|
| True | Good | "TXA GOOD/GOOD" | Green |
| True | Marginal | "TXA GOOD/MARG" | Yellow |
| True | Unsatisfactory | "TXA GOOD/UNSAT" | Yellow |
| False | Good | "TXA DEGRD/GOOD" | Yellow |
| False | Marginal | "TXA DEGRD/MARG" | Yellow |
| False | Unsatisfactory | "TXA DEGRD/UNSAT" | Yellow |



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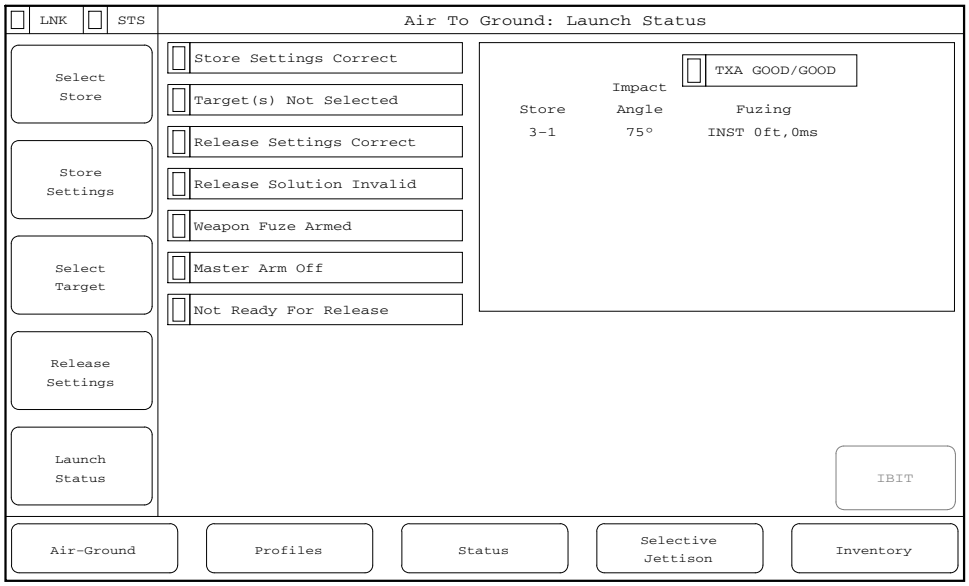


**Figure 1-125. 2411.0.0 Launch Status Page.**

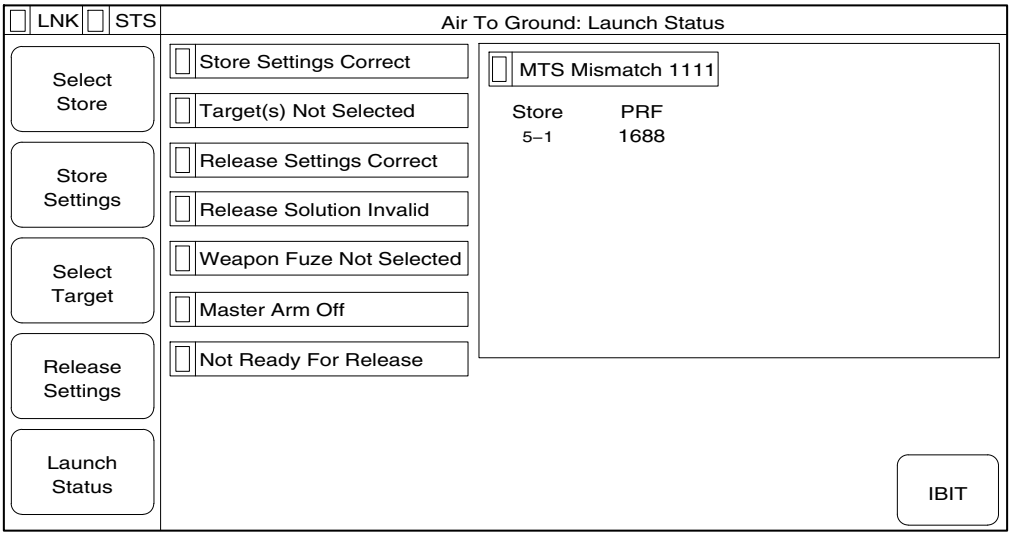


**Figure 1-126. 2411.0.0 Launch Status Page – Decluttered TXA Display Widget.**

1.3.3.7.1 Target Information. Displayed in the lower section of the Page Section is the selected target information. Next to the target information are Auto CCTGP and Manual CCTGP buttons.

Above the selected target information are up and down arrow scroll buttons that provide the user the ability to scroll through the target list and verify all selected targets prior to launch. To the right of the target information is the Select button that displays the selected stores associated with the target. It provides the ability for the user to switch any store assigned as a manual target type to a CCTGP target type.

If the cross-cued target is selected, the current update mode will be identified. Selecting the other button will change the update mode to that selection. The selected update mode button will be identified with blue. The mode descriptions are described in Select Target.

If a saved target is selected, that target information will be displayed. The target update mode buttons will be available for se-



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lection but both will be gray. Selecting either of the target update buttons will switch the target to the cross-cued target. The target information and buttons will function as described previously.

If no target is selected, no information will be displayed. The target mode buttons will be displayed and function as described above.

1.3.4 Profiles Mode. Profiles Mode allows the user to enter and save different weapons configurations for faster weapons setup. When the Profiles Mode selection button is pressed, it will display the Profiles Page Area (Figure 1-127). The left and right traversal arrow buttons located at the top right corner of the Profiles Page Area, allows the user to scroll through different profiles that have been entered and saved. The Profile Page work area contains four buttons:

- Profiles
- Overview
- Store Settings
- Release Settings

The Weapon Profiles pages show a Target list. The Active page will show the current selected target list, and the saved profiles will show the list of targets that are saved to that profile.

The Profiles Page supports the following features: up to eight weapons selected for release, and the ability to rename a profile. The display items within the store settings status are determined by that weapon type's CDF file. When a profile is selected and the rename button is pressed, the following dialog appears that allows the user to rename that profile to a string of choice (Figure 1-127). The name is then displayed on the launch status page within the profile preview group boxes.

2411.0.0 The Profiles page Target Display button(s) have been increased in size to accommodate up to eight munitions getting displayed on the button after assignment to a target.

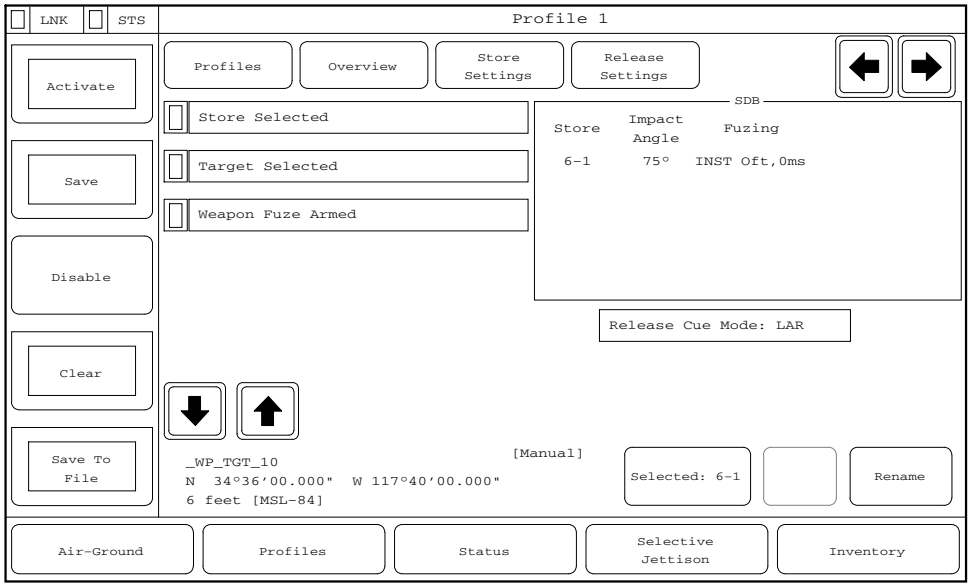


**Figure 1-127. Profiles Page Area.**

On the left side of the Profiles Page Area are four Profile Control buttons (Figure 1-127). These four guarded buttons can activate, deactivate, save, and clear Profiles that have been entered and saved in GSMS. When all of the Profile Control buttons have been disabled, all of the buttons will be greyed out and unselectable. When all four Profile Control buttons have been enabled, all four buttons will be selectable. Refer to Table 1-9 for various states that Profile Control buttons can display.

**Table 1-9. Profile Control Button Status.**

| Control Button | When in Disabled State | When in Enabled State |
|---|---|---|
| Activate | No data saved to the selected profile or selected profile is disabled. | Data saved to the selected profile and selected profile is not disabled. |
| Save | Store not selected and/or target not selected. | Store and Target selected. |



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**Table 1-9. Profile Control Button Status – Continued.**

| Control Button | When in Disabled State | When in Enabled State |
|---|---|---|
| Enable/Disable | No data saved to the selected profile. | Data saved to the selected profile. |
| Clear | No data saved to the selected profile. | Data saved to the selected profile. |

When a Profile is Enabled, the Enable/Disable button text on the left side of the Profiles Page Area ( Figure 1-128) will display "Disabled."

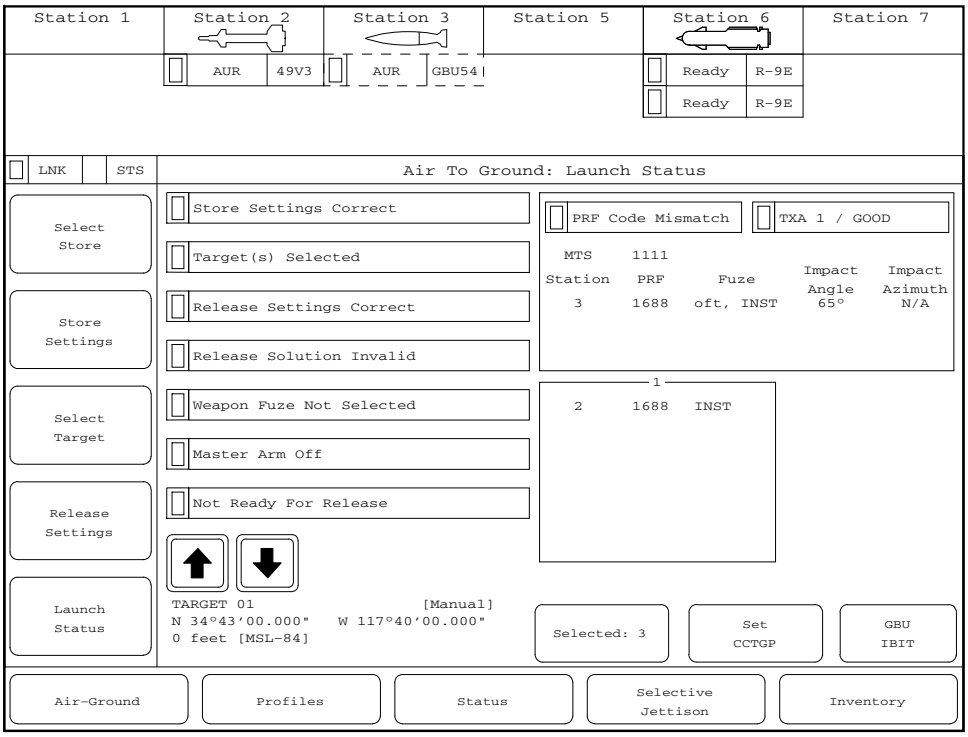


**Figure 1-128. Enabled Profile Page – Typical.**

When a Profile is disabled, the Enable/Disable button text on the left side of the Profiles Page Area (Figure 1-128) will display "Enabled." On the right side of the Profile Page Area, when a Profile is deactivated, "Disabled" will be displayed in yellow (Figure 1-129).



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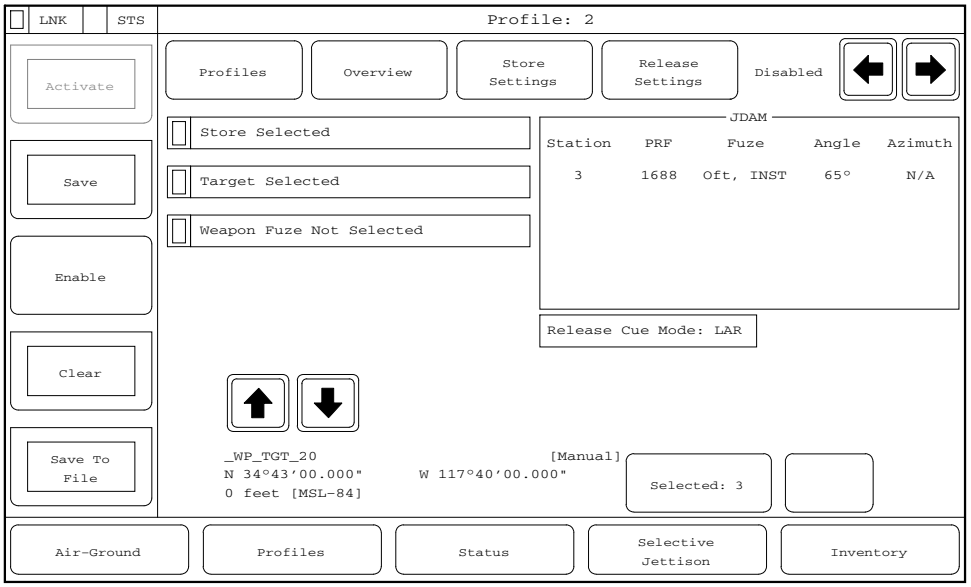


**Figure 1-129. Disabled Profile Page.**

1.3.4.1 Profiles. When the Profiles button is selected on the Profile Area screen (Figure 1-127), it will display the Profile Selector screen (Figure 1-130). This screen is used to switch, activate, or deactivate different profiles that have been entered. Up to eight different weapons profiles can be entered and saved. There are four conditions that can exist on the Profiles button page:

- Saved and Enabled – Active button will be highlighted in blue with white border, and selected Profile button will be highlighted with a white border
- Saved and Disabled – Active button will be highlighted in blue with white border, and disabled Profile button with only associated Profile number highlighted in white
- Selected – Selected Profile button will be highlighted in blue with a white border
- Empty – Only the Active button will be highlighted in blue with a white border



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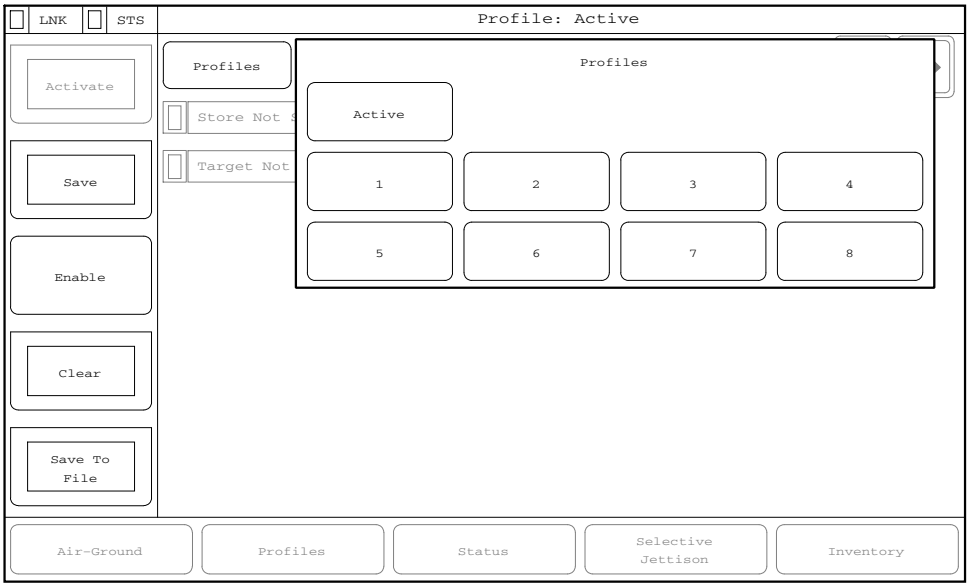


**Figure 1-130. Profiles Active Page.**

1.3.4.2 Overview. The default selection on the Profiles Page Area (Figure 1-127), is the Overview button. The Overview Page Area (Figure 1-131), displays the following four indications:

- Status Light Indicators
- Target Settings
- Store Settings Summary
- Release Cue Mode

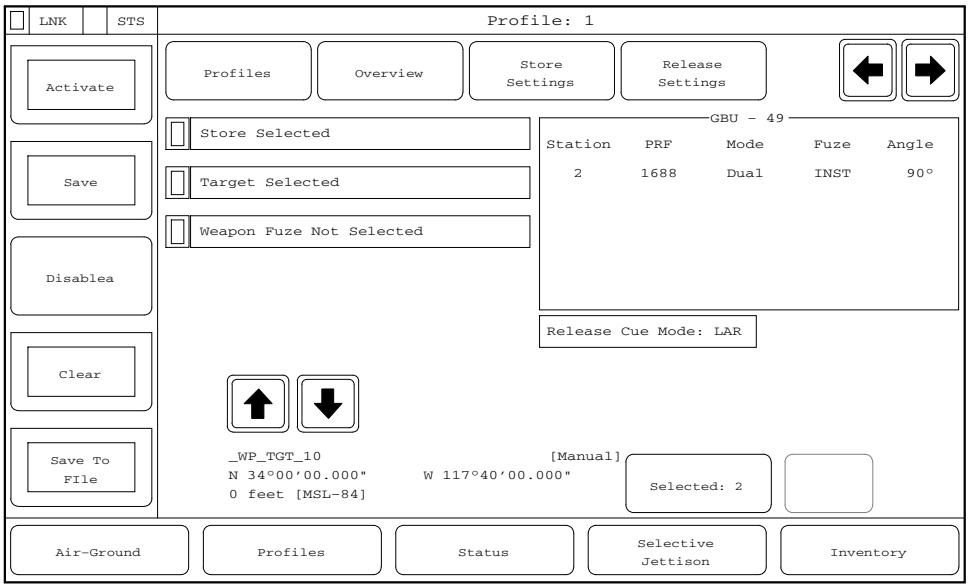


**Figure 1-131. Overview Page Area.**

1.3.4.3 Overview: Status Light Indicators. The Status Lights indicate the status of what has been currently selected. Refer to Table 1-10 for further explanation.



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